

Bus 220 Successful Mistakes-In Business

Numbered Transcript

BUS 220 • Derick Cooper & Dr. Weng Tao

Contents

Lecture 1. The Power of Mistakes

Lecture 2. The Growth Mindset

Lecture 3. Information and Decisions

Lecture 4. Personality and Perception

Lecture 5. Fundamentals of Statistics I

Lecture 6. Fundamentals of Statistics II

Lecture 7. Creating Resilient Organizations

Lecture 8. Leading with Integrity

Lecture 1: The Power of Mistakes

At a glance — Cooper on the power of mistakes.

- 1.1 Good morning. I'm Derick Cooper. I run a company called QOL Medical. It's a specialty biopharmaceutical company. Over the next couple of days, we're going to be talking about successful mistakes, which is really a concept about how the analysis of mistakes ultimately is the mechanism that leads to success. You need to incorporate the idea of making mistakes and that errors are a natural part of the human condition into your thinking and how to analyze that from a people perspective, a data perspective, and that kind of thing. How to ultimately be successful by understanding how to be wrong.
- 1.2 QOL is a mid-size biopharmaceutical company. We're focused on rare diseases. We make drugs for diseases that are generally genetic in terms of their nature. Very few people have them. They're highly nuanced in terms of a lot of the technical challenges that patients with rare diseases face. We're not going to go through technical biopharmaceutical things much today, but we're going to talk about the systems and the architecture and the leadership structure that I and my COO, our COO, Dr. Weng Tao, have developed over roughly the last 10 or 12 years in terms of just considering how to consider people and data and structures in terms of good decision making.
- 1.3 We have a good size company. We're fully integrated. We have sales, manufacturing, R&D, clinical development, etc. We're roughly one of the top 10 largest private biopharmaceutical companies in the US, maybe top five. It's a little hard to tell with private companies. But the first lesson that I want to introduce at the start here is that, that is not the most important thing to understand. Another way of saying it is the fact that we have a successful company, a successful large company, that we do complex things with polymorphisms and the periplasmic space.
- 1.4 That's not what you're looking for, when you're looking for someone who is going to be helpful in offering you good advice. You don't want to look to authority. You don't want to look to their history, their track record of success. It can be a contributing factor, something that you want to take into account. But really what you want to think about is this. So a fairly smart guy here, Dr. Feynman. I'm just going to read this quote. Why should you listen to not just me, but Dr. Feynman or anyone? So first, "have no respect whatsoever for authority." That's a really good starting point.
- 1.5 Be a little cynical. Question things. If somebody says, "I am the expert," and they beat their chest, that is a good starting point for understanding that they're trying to convince you that their point of view is right because of who they are rather than because of what they know. So what you want to understand is what people know and why, which is a function of how they explain things. So again, have no respect for authority. Forget who said it. And instead, look at what he starts with, where he ends up, and ask yourself, "Is it reasonable?"

- 1.6 You're looking for the essence of the message. Dr. Peterson says this well: "That which you most need will be found where you least want to look." What you're actually looking for is the information that you don't like, that you don't want to hear, that your personality has a negative reaction to, because it's telling you something that's actually valuable. So the best thing you could do is understand how to gather the broadest spectrum of information, including the information that you don't like, that bothers you.
- 1.7 We all know this. We have friends who tell us things. They're very good—you can tell a really good friend because they'll tell you something that they and you know you don't want to hear, but you know if you think about it, it's actually really good deep advice. They're telling you something very important. We're going to go through how to understand whether or not somebody's giving you good advice. I call it an authority trap where they appeal to the authority, and you want to focus on the why and not the —you want to focus on the why and not the who.
- 1.8 You want to be careful with jargon. A whole lot of fancy word salad is a good indication of somebody who's trying to tell you something because of their worldview or because they're trying to convince you of something, not necessarily because they actually have a good perspective on things. We're going to talk about how to differentiate between those things. How can you tell somebody who's really truly an expert versus just trying to razzle-dazzle you with a bunch of fancy words.
- 1.9 A good start to this is: success does not necessarily always mean competence. There are plenty of examples of people who have gotten to very successful positions in life through some good luck, random sets of circumstances, etc. So you still want to make judgments about people in terms of what they're saying and how they're relaying information to you, not just because of the position that they're in.
- 1.10 My partner here in this lecture is Dr. Weng Tao. She's got an MD and PhD. You'll learn real soon she's kind of smart. She's got experience across the entire spectrum of all of the biopharmaceutical world from early stage biologic development, gene therapies, which we've all learned a whole lot more the last couple or three years. We all know what a gene therapy is now. She's very good at assessing a new situation in a new area where she does not have direct, explicit background or experience and figuring out how to solve the puzzle in that new place. She's our COO. She runs R&D, manufacturing, our regulatory group, etc. She's got a strong background in immunology, so if we have a chance to get into that at some point, she'll befuddle us all with her expertise.
- 1.11 All right, I promise I won't quote Dr. Peterson too much, but this is a good place to start. I think this may be his most well-known quote: "In order to be able to think, you have to risk being offensive." I'm going to tweak that quote a little bit. This is a pretty offensive thing to do. I'm going to modify Dr. Peterson's quote: "In order to be able to think, you have to risk making mistakes." You have to be comfortable with the

possibility of mistakes. You have to understand how to analyze the likelihood of mistakes, and you have to incorporate that into your decision-making process.

- 1.12 Mistakes are not something to be avoided. They are something to learn from because they are going to happen no matter what you do. And so what you want to do is you want to pay attention when they do. And we'll go through a lot of the way that mistakes cause the human ego to compound problems here in in a little bit.
- 1.13 So ultimately, there's a human paradox, right? The human condition is imperfection. We all recognize that we don't navigate the world perfectly. We make mistakes every day in all kinds of different areas of our lives. And ultimately, to be a successful human being, is to emerge from that sequence of mistakes. You have to understand them and how the sequence of how you got to a failure gives you the ability to avoid that picture, what that failure picture looks like in the future, because you want to incorporate it. You want to integrate it into who you are. Success doesn't come from avoiding mistakes. It comes from recognizing and correcting the patterns of mistakes. We're going to talk about how to do that.
- 1.14 So there's three sources of information that teach us about the world. And when we're analyzing a new situation, we're trying to develop a new product, we're expanding our business into a new market, we're developing a new piece of software, whatever it is. There are three ways to think about how to make that decision and how to do that well. One is yourself, right? You're perceiving the world. You're making judgments. Another is gathering information from people that you know and you trust, other experts, and the third is data. We're just going to talk about how each of those impacts the way that we ultimately view the world.
- 1.15 Our agenda over the next couple of days, we'll start with the path to good decisions. We're going to talk about mistakes themselves and how to fall in love with them. We're going to talk about how to consider, conceptualize, information and how it impacts the way that we make decisions. We're going to talk about our personalities and how different types of personalities judge information, others, and decision-making processes differently, and they're biased in different ways. We're going to talk about a few real-world examples of mistakes that Weng and I have made and how we've learned from them. Then we're going to integrate all of that into how to communicate, how to listen to others, and how to ultimately discern the decision-making process in running a business, hopefully pretty well.
- 1.16 So the path to good decisions. We're going to start with this guy who, I don't know that he said anything that isn't very valuable, but this may be the most valuable. So this is Socrates. This may be the most valuable advice that he started 2,800 years ago when he gave us, which is: "All that I know is that I know nothing." What does he mean by this? He doesn't mean that we're all idiots. What he means is that you want to be cautious about what you think you know. You want to be careful. You want to adroitly

second-guess your own perspective. And you want to get good at that habit. It's really a boundary of epistemic humility, which we're going to go into here in a minute.

- 1.17** Here's how our own brains get us in trouble. The human brain is binary. It wants to know the truth, like with certainty. Like, I know this is the answer to the question. The world, however, is probabilistic. It's a distribution of— it's a foggy distribution of many different possibilities that could happen. We all know this. You go to the casino and roll the dice. What are the odds that—you have two dice—what are the odds that you're going to get a seven? You may not know that if you're not a gambler, but it's 16.7%. Okay, does that mean you're going to get a seven? It's the reason why the house wins on the seven: it's the best odds. The seven with two dice is the most likely outcome. So you don't know. There are 36 different possibilities. So that type of perspective needs to be pervasive in terms of all of the decisions that you make, that you want to understand how to rank order the different possibilities.
- 1.18** So the world's foggy. Academia makes this worse because part of what happens when you go to school is you get multiple choice tests. This is a funny example. If anybody can figure out the answer to this one, please let me know. Most multiple choice tests are like, here's a question, here are four different answers, pick the best one. Have fun in school doing that because nothing else in life works like that, ever, in any possibilities at all, in any circumstances. There are no multiple choice tests in the world of business. Everything is fuzzy and foggy. Everything's fuzzy and foggy in terms of your kids, who you're going to marry. All of these things are complex. It's not like, yes/no. It's rarely yes/no.
- 1.19** If you pick an answer to this question at random, what is the chance that you will be correct? There are four different answers. On the surface, it's 25%, right? Okay. But now there are two 25% in the answer group. That means it's 50%, right? But 50% is one of the answers. But if you pick 50%, you've just now not done it randomly. So most multiple choice questions, this is a good example. What I've done here is try to find a multiple choice question that actually doesn't have an answer, and that's more like life.
- 1.20** We're taught, as we grow up, to decide on things when we're certain—particularly in the educational world—but we live in a world of uncertainty, where you don't know what the outcome is going to be. The consequence of that is it becomes hard to discern exactly what it is that leads to failure. Because you're seeking certainty, and then something goes wrong, it's like, well, I was certain. Why did something go wrong? Well, is it because you made a mistake? Is it because there's random bad luck in the world? So you need to try to understand how the certainty causes you to perceive what it is that led to a mistake.
- 1.21** So the consequence of this dichotomy between knowing something for sure and the real world being a fuzzy realm of possibilities, is the ego fills in the gaps. Because the ego doesn't like to not be certain. It wants to know. So what it does is it grasps at distal, low probability possibilities, and it assigns more certainty to those possibilities than it

should. We all do this, and we do it at a subconscious level. So you need to understand that your brain is doing this in order to transcend it.

- 1.22** So the ego overestimates the knowledge in order to seek the certainty, because the ego wants certainty in terms of the decision making. And this begs a question, right? The natural question will be, well, how else are you going to make decisions in the world other than trying to be as certain as you possibly can? It's not "possibly" can. I mistakenly threw in a word there that I shouldn't have. You're seeking certainty, well, how else are you going to make decisions other than to seek certainty?
- 1.23** So the problem is the pursuit of the certainty itself. It's an unrealistic goal. The ego hates the possibility that it could be wrong, and it associates uncertainty with the likelihood of being wrong. That's actually the nuance that gets us. It's that if I'm not certain, it means I'm more likely going to be wrong. So you need to understand that about yourself. Consequently, because it doesn't want to think about it might be wrong or there could be a possibility of failure, it imposes, it forces certainty and assuredness onto a world that isn't like that at all.
- 1.24** And so the result, when that happens, when the ego sneaks in and says, this is the right decision, when it may or may not be, and it may or may not be naturally, right? It's like rolling the dice, and the ego says, "a six is the answer." Well, you might not get a six, and then the ego freaks out. You might not get a six because the law of the dice roll. The result of this is inconsistencies, and it's an inability to second guess the process of decision making and moving through the world because the ego keeps coming back to the assuredness. It assigns a certainty and preconceived truth to that level of assuredness that it creates for itself.
- 1.25** Then anything that challenges that, any person in your organization or a friend or your wife or husband or somebody comes in and says, "hey, I think you may be making a bad decision here. I think you don't have this just right." The ego goes, "no, I can't re-conceptualize this path that I'm on because if I do, that means I might be wrong." And it associates "might be wrong" with being wrong. And the consequence of that is ultimately a sequence of catastrophic failures because we keep insisting to ourselves that we're right, no matter what the information. There's new information that comes in that says, well, "hey, you're working on this drug, and you ran this clinical trial, and guess what? It didn't work. The drug failed. There's no efficacy whatsoever." I have seen this over and over and over again.
- 1.26** The CEO goes, "well, no. If we redo this and we change the structure of the statistics or the patients that we enroll." They go, this data tells me this drug just failed, but I know it's actually going to work. And so what do you do? If you're successful at that failure sequence and your ego is able to continue to take you down that path, what ultimately happens is that you run another trial. You take a failed drug and you run another study and it fails again. Well, now, instead of wasting 10 or 20 million dollars, now, you've done it twice. You've done it twice because you did not pay close enough attention to

what the information was challenging your ego with the first time that you failed. So the ego doesn't like second-guessing. Series of continuous failure is the consequence of that. The ego wants to avoid mistakes.

- 1.27 Here's a good example of the space shuttle launch from 1987, I think it was, when it blew up. And how did that happen? It was cold. It was really cold that morning in Florida. I think 42 degrees. It was the coldest space shuttle launch ever. Maybe it was even freezing. There was an engineer in Utah who had worked on the O-rings of the rocket boosters with all the fuel in them. The boosters, these big white things on the side, are actually just full of rocket fuel. They're bombs about to go off, right? So everything has to go perfectly in order for those things to work.
- 1.28 This engineer called the command control center in NASA and said I don't think you should launch this thing. We don't know what's going to happen. He literally predicted exactly what the problem was. These O-rings are not, we don't know—they're going to contract in this temperature—we don't know if this is going to work. We should not launch the space shuttle. All right.
- 1.29 Now, you're head of NASA, and you've only launched a couple of space shuttles. You're just getting the program going, and you're excited about the possibility, and you want to get back to space, and all of the motions of the entire system, you've got thousands and thousands of people around you and the whole world watching on TV are built to see the space shuttle launch that morning. And one person comes along and says, "Hey, I don't think this is going to work. This is a big risk." And he was ignored, and it blew up. And that's a very good example of: ignore the information that you don't like because it tells you something that deep down, you secretly know it's true, but your ego doesn't want it to be true.
- 1.30 So this iterates to more failure. We've covered this. There's ultimately a way to assess the sequence of mistakes, which is: What went wrong? Were the chips stacked against me? I had just bad luck with a roll of the dice, or did I actually do something wrong? Did I do something that led to this problem? We all know what the right answer is. It was me, especially if you're the CEO. This is the best part of the job, if you're CEO, is you have to take responsibility for everything. If you're a good CEO, you take responsibility for all of the mistakes of the organization.
- 1.31 If you're a good leader, you take responsibility for mistakes and you dole out responsibility for successes. People will follow you anywhere if you do that because it's not scary. You create a situation that isn't scary, where they're not scared of the person in charge because you're actually giving them the possibility of making mistakes when you say the mistake is my fault. The right answer to this, it doesn't matter if you're CEO or where you are in the world, the right answer is always: What did you do—I as an individual—to contribute to this sequence that resulted in the space shuttle blowing up? Hopefully, you guys don't blow up any space shuttles.

- 1.32** You want to understand what it is—how did you, even if you think it's completely not your fault and you didn't do anything wrong—what is it that you possibly did that contributed to the ultimate failure? But what do most of us do? We all know this, because we're human beings. We've done it ourselves and we've seen lots of other people do it. "But it wasn't my fault. I didn't do anything wrong. It was Joe. I didn't mess it up." It's unbelievably common, especially in the hierarchies of the world and business, etc., where people don't want to—the fundamental problem here is ultimately the ego. It associates its level of—its rank in the hierarchy—with more successes and fewer failures. So the ego wants to disassociate itself with failures and associate itself with successes because it perceives the world around it based on where you fit with other people, based on your rank of successes and failures.
- 1.33** We want to try to learn to do is the exact opposite of that. This is the reason, this is the fundamental reason why you want to do the opposite. It's because if it's not your fault, if it's not my fault when something goes wrong, then really there's no reason to dig in and analyze why it went wrong. Like, "I didn't do it. Why should I try to understand what happened?"
- 1.34** The default protection mechanism when something does go wrong, especially when you disassociate yourself from it, is to not understand it and learn from what led to the mistake. I refer to this as "I did my best-ness," which is: it's not my fault. I did my best. I did my best. What else could I do? When you believe you did your best, but the project failed anyway, I can't do any better. I did the best I possibly could. It failed. If you're talking to your boss, I don't know what else I could possibly do. How could it be my fault when I did the very best. And I think what you want to do here is be wary of the I did my best-ness tendency.
- 1.35** Don't think of your best as not failing. Think of your best as understanding the sequence of mistakes that can lead to failures. You want to constantly be judging information and thinking about how your own inadequacies, you want to understand what it is about yourself that could possibly contribute to an ultimate failure and realize that those trapped doors are there.
- 1.36** Because, look, if you understand a gap between where you are and where you need to be in order to be successful, if you can truly perceive that gap, then you can fill it. Part of filling it is understanding the inadequacies of your own personality. Like, what are you good at? If you're extremely conscientious, you get everything done. You're unbelievably good at getting things done, but maybe you're not that creative. And so you're going to be focused on making sure that everything gets done. Well, in the middle of that process of making sure that everything gets done, you may need to change direction.
- 1.37** The new information comes in and says, well, this drug was a failure. But you're like, I'm going to get this done. I'm going to run another trial. You're really, really focused on the sequence of steps rather than the bigger picture. And so you need to take a step

back and say, maybe my tendency to focus on the sequence of steps is missing something. But the ego doesn't like to recognize its own inadequacies. And this denial is really the genesis of the repeat pattern of failure because it refuses to acknowledge its own contribution to the sequence that led to a mistake. You can't learn from not analyzing mistakes that you think you didn't make.

- 1.38 So Epictetus, you're going to learn here soon that I like Epictetus, "It is impossible to begin to learn that which we think we already know." If you think you know it. You think you understand. You think you understand yourself. You think you understand a particular situation. You're much less open to new information that could contribute to better decision making.
- 1.39 Even if you think you're the world's expert in a certain area, you want to continue to be a learner. You want to continue to be receptive to new information because, well, technology is evolving faster and faster and faster, and it's going to iteratively teach us new things. You want to stay on top of your allowing new information to come in. If you get good at this, you understand how your decisions, your emotions, these human factors contribute to ultimate decision making and the possibility of failure, you get good at recognizing the pattern that leads to failure.
- 1.40 Now, this is going to vary depending on circumstances, certainly wildly different in the pharmaceutical world. But if you run a gym and you want to open a new gym and your first new gym that you open doesn't go very well, the best thing to do is really try hard to understand what it was that led to that. So the next time you open a new gym, you do it well.
- 1.41 So ultimately, understanding the sequence that led to the mistake is the mechanism of success. So our greatest failures are a consequence of believing that which we most want to be true, or which our ego most wants to be true, that isn't actually true. It's hanging on to something for too long that we shouldn't.
- 1.42 Our lecture goal over the next few days, we're going to learn to identify the blind spots in ourselves, the disguised firmly held beliefs, and question them so we can transcend beliefs that lead us down a path that is not the right path. And the reason you want to do this is because, ultimately, you want to second guess yourself. Because your best source of information, the person that you need to best understand is you. If you can understand yourself well, it's actually a path to understanding other people better. Because if you can properly judge what you're good at, what you're not good at, where you need to fill in gaps, that'll be helpful in terms of the way you navigate the world, then it's a guide. It's a foundation. It's a background step that teaches you how to understand other people better.
- 1.43 And really, this is a leadership piece here. Because if you don't question yourself, you don't create an open environment that encourages questioning. If you, as a leader, if you're always, "I'm the leader, I'm certain." We all know these overtly authoritarian

types who just say "this is the answer. It's not open for discussion." When you do that, the farther you can go on the spectrum of questioning yourself and thinking about things carefully and acknowledging legitimate doubts, the more open you'll be in terms of a discussion with your employees, your friends, etc., in terms of expanding your worldview and your perspective.

- 1.44 Look, if you can understand the way another person sees the world, it expands your vision. Because now you have not only access to the way that you see the world, which is one person. But imagine you're a smart person, you're capable, and you add another person to that equation who's also smart and also capable, but who sees things differently than you do. Well, you can go two ways with that.
- 1.45 One, you can say, well, I don't like what they have to say because they're argumentative. This happens routinely with me and Weng. We debate things vociferously, which is simultaneously very fun and frustrating. But if you can listen to another point of view, you actually gain a whole, it's a whole new window on the world. You want to encourage others to think about their own perspective. You want to create an environment that encourages others to think about their individual perspective and yours and how these things dovetail together.
- 1.46 If you don't second guess, if you go down a path, the ego says, "we're going here." It sets a target. We're not going to let anything get in the way between here and there. That's the target that we're aimed for. If you don't have a compass, if you don't second guess, well, the wind has blown me off course. There's new information that I need to take into account that's going to change, that should change, the decision-making process, then you get off course and the ego just keeps pointing the ship at the wrong port. You got to second guess. The way to do that is with free speech, open dialog, debate, open inquiry, encouraging challenge.
- 1.47 So the secret to success: learn to correct mistakes by identifying the pattern of mistakes earlier, get the ego out of the way, and you'll improve substantially. The ego is ultimately the enemy here in terms of good leadership and good decision-making processes. Because it does associate itself with your position in a hierarchy, which is success = good; mistakes/failure = bad. It will continue to construe successes as a mechanism of climbing in an organization. That's not actually what you want to do.
- 1.48 You don't want to get promoted. What you want to do is demonstrate to senior executives that you're good at decision making. Sometimes that can be giving senior executives information that they don't want to hear. And because if you are good at identifying the right decision-making process, part of which is second-guessing yourself, you will string together a sequence of more successes. The probability of success is higher when you can identify the pattern of failure. And that will result in promotions and a lot more success your life.

- 1.49** "When proven wrong, the wise man will correct himself; the ignorant will just keep right on arguing." This is the gist of what we're trying to cover here. Really, here's a telltale sign: if there's a mistake made in an organization, somebody comes to me and says, "it wasn't my fault. I didn't do anything wrong. I did all right." Well, that tells me a couple of different things. One, I know I have an actual problem I've got to fix: the manufacturing isn't working, something has gone wrong with the FDA, whatever it is. I know there's a problem. When somebody says there's not really a problem. I didn't do anything wrong. I also know that that person can't fix the problem. Because they're actually explicitly telling me "I do not acknowledge that there's a problem." If you're not acknowledging the problem that we're having, how are you going to fix it?
- 1.50** Back to Socrates. Make good decisions. You have to get to know yourself, so you can outsmart yourself. Part of that is taking into account information from other people. Want to know why and when you're likely wrong, so you can become right, which is a factor of understanding your blind spots.
- 1.51** And here's the challenge. A blind spot is a really big difficulty because a blind spot is, by definition, something that you cannot see. The only way to see a blind spot, and there's a tendency as individuals—especially competent, assertive, smart, young people—there's a tendency to want to figure out how to see your own blind spots. Okay, maybe I'll admit that I've got some blind spots. I'm not perfect, so I'll just figure out how to see them. Well, you can't. That's the definition of a blind spot.
- 1.52** And so the only way to see a blind spot, the only way to see something that's causing you to go down a path to an ultimate error is from somebody else. It's literally the only mechanism. Because, again, by definition, the way that you're wired, the way that your brain perceives the world, won't do it. So you got to get good at listening to other people.
- 1.53** So ultimately, this is about epistemic humility. It's the path to being a superhuman decision-maker. The path of epistemic humility is to be: "If you want to improve, be content to be thought foolish and stupid." You have to let that into your personality, because if you're going to learn something that you don't know, you have to admit that you don't know it.
- 1.54** The first part of epistemic humility is epistemology. It's a fancy Greek-derived word that means the knowledge of knowledge. You need to know what it is that you know. You need to know your own boundary of ability to perceive the world accurately. Humility is the path to understanding where that boundary is. You won't overestimate it. The mistake is the overestimation of your own knowledge, the overconfidence problem.
- 1.55** Humility is a funny word. Because humility tends to be construed as weakness or meekness or cowering in the corner. I should make sure and clarify that. That's not the kind of humility that I'm talking about here. What I'm talking about is adroit doubt and second-guessing of the self, like confident second-guessing. That's what I mean by

humility, not being scared of the world or intimidated by navigating the world and making decisions or shying away from making decisions.

- 1.56** So how are we going to be stupid fools? First of all, you need to understand there's almost always a gap between what you know and what you need to know in order to make a good decision and ultimately achieve success. In order to manage that gap, you need to see where the potential errors are and the mistakes between you and where you need to be. You need to understand what it is that you likely will need to correct along the path to a goal, in order to perceive where you're wrong. And then once you can acknowledge where you likely will be wrong, which is going to upset your ego, you can improve it. You can course correct.
- 1.57** So think of a mistake as like a treasure map. It's actually really valuable. It's something that you can learn from and put it in your panoply of mistake history. It's a library. Think of navigating life, making good decisions, good leadership, as a library of mistakes that you can reference. Because every mistake that you experience in life ultimately will teach you something about how to avoid it in the future, if you openly acknowledge it and learn from it and take it into account.
- 1.58** Just real quick side note here. I'm not talking about neurotic flogging. Acknowledging mistakes is not an excuse to beat yourself up. That's not what we want to do. It's more of an analytical approach to mistakes. If you can figure out how to separate the decision making process as its own entity and put it on the table, and it's not you that is making the decision. It is a decision that needs to be made. And the different contributing factors that can influence that decision as a objective external thing that's outside of your ego and outside of your individual personality. Gather information from other people, data, whatnot, all the different contributing factors, and then make that independent decision outside of the self. Put it on the table and consider it as a literal thing outside of you, then it depersonalizes the process. And that depersonalization helps remove your ego from the process because it's not my decision. It is a decision that we're trying to make well.
- 1.59** We'll go a little bit into antiquity here, maybe a little after antiquity. But Dante's Inferno, he talks about this as he descends into Hell. And he talks about doubt, mistakes, and second-guessing. When he first encounters doubt, he thinks of it in simpler terms. "All taint of cowardice must here be dead." Just get over the doubt. Be strong. Be brave. Make a decision. Keep moving. Then he learns from that as he continues to descend.
- 1.60** Later, he understands that doubt is actually a valuable tool. It's not something to just overcome and keep going. It is, "Born of that will, there rise up, like fresh shoots, pure doubts. These flourish at the foot of truth." What he's saying in a fancy way here is: incorporate, integrate the doubts, pay attention to them. It's something in your intuition. It's something in your personality. It's something from your friends and colleagues that is informing you of information that you need to be aware of in order to

get back on the right path. It's not something to just overcome and set aside. It's something to integrate.

- 1.61** So to doubt is to find the true source of error. Which is: something I did to contribute; I misread the situation; I listened to other people who gave me bad advice that I shouldn't have I listened to; or is it just a natural vagary of life? These are the things that ultimately contribute to the sequence of error. It's one of those three things. It's you. It's others. Or it's information. You want to try to be able to discern between those three things and how each of them contributed. And obviously, the most important part is you. So you being: you just misread the situation, you misread others' knowledge—we're going to go through how to understand whether or not somebody else knows what they're talking about in a little bit—or you did not understand the probabilities that led to the dice roll. And you made a bad bet.
- 1.62** So there are really basically two aspects of this. You need to understand yourself in terms of understanding how you perceive information and how others perceive it, and then how data ultimately impacts your decision making process. With that, I think we actually have time for some questions.
- 1.63** The ego has a very bad reputation, and I mean, understandably so for some things. But for yourself personally, when it comes to scaling a business, do you think that there is some benefit to an ego, especially in the early days? You're going to have inevitable obstacles, things you got to overcome. And can that ego serve somewhat of a good purpose in building up? I get when you get to a certain size, the humility and with the staff, but is there any benefit to an ego in building?
- 1.64** No. It's an impediment at every step of life. And I would say that there is a subtle difference between ego-driven overconfidence and insecurity, which I don't think is helpful at all. Because it actually continues to defend itself no matter what the situation is, and it continues to say, I'm right no matter what the world is telling me that I'm wrong.
- 1.65** Now, there's a subtle undertone to your question, which is really important, which is as you're building a business, you're going to have lots of pieces of advice from people that are contrary to what it is that you're trying to do. So you need to be able to differentiate between good advice and bad advice. That you definitely want to do. We're going to touch on that over the next day or two. Because it's key. Because you do need to, if you don't have the confidence to override advice that is wrong, you're going to make just as many mistakes.
- 1.66** So it's a balance between assertiveness and willingness to continue to pursue the goal no matter what other people are telling you while simultaneously adroitly second-guessing and taking into account new information. And the ego gets that equation wrong because it can't take this. It just does the assertiveness piece. It just charges

ahead no matter what. And it can't take into the account the second-guessing piece. The ego has no capacity for that.

- 1.67** You say humility is confidence, second-guessing?
- 1.68** Yeah.
- 1.69** Were there anything in your path that you didn't have that ability?
- 1.70** Yeah, I learned this by failing. Everything that I've been, this whole entire thing, I conceptualized as: if I could go to when I was 30 years old and give myself advice about all the things that I've learned over the last 22 years, it would be— learn to second-guess yourself carefully. Because a level of overconfidence in your judgments can too easily sustain itself for too long, and to the point you were just making, continue to compound the error. That's the thing you're really trying to avoid is the compounding of the error.
- 1.71** And so if you can recognize the signal of the error earlier, which requires, "I might not be on the right path." Ultimately, this is truth. Truth will tell you the thing that you most don't want to hear and that you don't like, but that is most necessary to get you back on the right path. You go run a clinical trial. It fails. What's the truth? It failed. I don't want to spend a bunch of money on a clinical trial that fails, but it gets a whole lot worse if I start trying to convince myself that it didn't really fail. That's when it just goes kablooeey.
- 1.72** I was taught that there's no such thing as failure. There's only feedback. My ego identifies with that. When I'm triggered by something, my ego automatically goes like, okay, there's something wrong there, and I have to be better or look at it, so I can be better. And I feel like a lot of it has to do with my ego and the identity that I have with that.
- 1.73** There are those of us who are wired to be bigger risk-takers and those of us who are not. And I think what you're saying is when you sense risk and that something may go wrong, the ego is helpful in that process because it gets you over the bubble.
- 1.74** There's risk in not taking risk.
- 1.75** Yes. It's a really good point. I think it's a good correction to the overall message that I'm trying to convey here. Listen to the intuition because it's telling you something that you need to be hearing. So definitely listen. But then if the ego is the part of you that just says, well, I'm just going to be brave and overcome that, regardless of what's going on in the world, that's a danger on the other side of the equation. So you want to be in the middle.
- 1.76** You want to listen to your intuitive self and what it's telling you about taking a risk. Maybe it's saying you really need to be taking more risk, and you need to recruit the ego in order to be able to do that. I think that's the core of your real question. I think that's okay. As long as you don't let it get carried away—to damn the torpedoes, and

I'm going to charge this hill no matter what. Yeah. So my problem is probably different than yours. I'm more of a damn the torpedoes guy. I can stay on a path no matter what.

- 1.77** And so what I need in terms of second guessing myself is a little bit more on the too aggressive side of the equation. You're asking a question, I think, on the not aggressive enough side of the equation. And so, yeah, it's yin and yang. It's a balance. It's like you can't just overcome your intuition and charge ahead with something that doesn't make sense. You need to continue to second-guess. Because if you do that, you'll end up making the same mistake that somebody who's more aggressive and entrepreneurial and assertive, which is the damn the torpedoes mistake.
- 1.78** How do you give your team the confidence to explain to you how they best take information? I had one example with an incredible staff member I have who we communicated largely through text. Then at the water-cooler we're having a conversation one day. She told me about her childhood and how she was extremely dyslexic, and she was an avid reader, but only listened to audiobooks. And so much of our dialog was predicated on this one form of communication. Then as an individual who's maybe apprehensive to ask for that, how do you encourage that out of your team to tell you what they need from you? If that makes sense?
- 1.79** I think she did. I think she told you pretty well, and I think you know what it is. It's built into the question. I think with somebody like that, you, as a leader, you can't be so diffuse that you modify your approach with every single person in the organization. It'll pull you apart. You won't be able to just, you have to be able to send emails. I'm sorry that she doesn't like to read emails. She'd rather hear it from you directly. But get her to get an audio program that will read the email to her, if that's what it takes.
- 1.80** But I think some modification is necessary. You have to be willing to adapt to the way that she perceives the world, but not so much that, again, there's a yin and yang balance here. Not so much that you're like, well, I'm going to speak French with you. I'm going to speak English with you, and I'm going straight to Canada here, and I'm going to speak ancient Greek with you. That's not actually going to work.
- 1.81** But look, I'll tell you, I very intentionally communicate with our scientists and operational people completely differently than I communicate with the sales force and the marketing team and the more extroverted people. Because the extroverted people are much more relationship-oriented. The scientists and the operational people are like, just tell me what you want me to do. And I'm super direct, straight to the point. Do this. Do this. Do this. If I did that with the salespeople, they'd be like, what's wrong with Derick? Why is he upset with me? You do have to manage this, but I think she told you what it is that she needs from you, which is more verbal communication. That's fine. If you're okay with that and it doesn't disrupt you too much, just do it.
- 1.82** If you invite open dialog and discussion in every aspect of your organization all the time, you're going to have the biggest mess you could possibly imagine in your entire

life. Well, you don't want an opinion from your personal trainers on which bank you should be using or which accounting firm you should be using. You need to draw a line here. And people that work for you need to understand that there's a line. But you can do it in a way that's not mean or rude. You can just say, look, this window is your responsibility. Within that window, we can have an open dialog about whatever it is that you want to talk about. But you wander over here, we're both going to get in trouble because you don't actually have expertise in accounting. We've got a problem.

- 1.83** So you do need to define the boundary of where that openness and creativity can start and end. What you really want in an organization is a willingness to tell the boss something that's wrong. Because I can't fix a problem that I don't know about. So you want to encourage people to tell you something that they perceive as wrong, but not open the doors to every single person having an opinion on everything all the time. You're speaking of emails, you won't be able to get through the emails every day.

Lecture 2: The Growth Mindset

At a glance — The growth mindset.

- 2.1 This morning, Derick introduced the concept that mistakes are an essential part of success. My session will focus on how to not only accept mistakes, also embrace mistakes as opportunities for growth. There's a quote from Winston Churchill, "Success is not final, failure is not fatal: it is the courage to continue that counts."
- 2.2 I'm Weng Tao, Chief Operating Officer at QOL Medical. I read the book called Being Wrong 10 years ago. This is the author, Kathryn Schulz. This book literally changed my life. I credit the principle in this book as an important role to lead the growth of QOL into a successful company. So Derick and I would like to share what we learned with you.
- 2.3 Mistakes are inherently useful. All of us make mistakes because we're human. I'm going to get into some of the root causes of mistakes and why it is normal and natural for us to make mistakes and what we can learn from our own mistakes. Because mistakes can help us to identify areas of improvement— most importantly, to enhance self-awareness. It is so important because we saw this all the time when people do not understand themselves, and that is really a biggest obstacle to their success.
- 2.4 Once we understand why we're human and we make mistakes, that will help us to practice resilience. We're going to also practice, in this session, to identify the mistakes we've made and to develop a strategy, how to analyze the mistakes and how to grow from our mistakes. That will help us to build the problem-solving skills. Most importantly, we actually discuss at our break, is to develop humility. Without it, we will not ultimately be successful. That altogether can help us to develop the habit of critical thinking.
- 2.5 First, "everybody makes mistakes, and nobody is perfect." Even the most successful people is not spared. We have the quote from Bill Gates, the quote from Albert Einstein —the genius—and most successful business people, and the famous author—and Steve Jobs. I think we all have his products. But all of them, through their career, had made mistakes, and they were able to learn from the mistakes and get up and keep going. This is what we would like to share with you today.
- 2.6 However, with that insight, why is it so difficult for people to acknowledge that we make mistakes? Because being wrong feels bad emotionally. Like Derick mentioned earlier about ego, what ego is that when you realize you made a mistake, it feels horrible. It's an emotional response to the mistake. Here, just to protect the ego, as you can see, what's associated with a mistake: embarrassment, shame, frustration, and anger. We've all been there. The difference of how we want to be and how we actually are. There is a self-image, and there's a reality.

- 2.7** There's a fear of the consequences of the mistake, and they're real. Like Derick mentioned, some organization will punish their employees if they make a mistake or they admit they make a mistake. So the fear is real, and sometimes it's justified. There's also cognitive dissonance. It's what you think and believe and what you actually behave in reality. There's regret, vulnerability, and social judgment from our peers, our families, and society. So all of these are real and true.
- 2.8** Certainty, on the other hand, feels good. It feels great, right? Because if everything is certain, we will always make the right decision. There'll be no risk. In this case, being right is confirmation the world works the way we expected. And that's natural, from our desire to be right. It reduces anxiety and uncertainty. It gives us more confidence in decision making. We feel in control. There's a reduction of confusion, ambiguity, and I feel a sense of identity and validation. And, ultimately, we feel safe.
- 2.9** But unfortunately, certainty is a feeling, it's not a fact. In the real-world situations, very few things are certain. The reason is that human perceptions are inherently fallible. Because we're human, there are built-in mechanisms that make us not perfect. Like Derick mentioned earlier, we have our blind spot. We have our inherent limitation. If we don't realize that, we usually will end up making the wrong decision. To realize that, to understand ourselves, our own limitation, also understand those who work with us, surround us, and their blind spot limitation is a very, very powerful tool to help us make the right decision.
- 2.10** Just to give you a few examples of our sensory. You see there are three lines. Isn't that immediate reaction if they are not the same length? But if you really measure them carefully, they are exactly the same length. Our visual perceptions can mislead us. There's another example, is that the illusion, when you look at this, it can really sometimes confuse you.
- 2.11** But this is one of my favorite examples. Let me just take a poll. How many people see this shoe as a gray background with a green border and a lace? Anybody? Yes? One, two, three. And how many of them seeing this, the background is rather pink and with a white lace? Most of you. Okay. There are only three, and plus me, four of the audience seeing this one way, and the majority of you see in a different way. This is why we challenge the notion "seeing is believing." Because you can see this, but we have a completely different perception of the same thing. This is how different people see with the same shoe. Isn't that amazing?
- 2.12** Now, here is another example. [tones play] Okay, let's poll the audience again. How many of you hear the first tone as the first tone was higher than the second one? One, two, three. So almost half. How many of you hear the tone, the first tone, is lower than the second one? Interesting. Again, this is exactly the same tone, but because we're wired differently, we actually hear different sounds.

- 2.13** So those are examples just to demonstrate that because we are human and we have a different background, different biological background, genetic background, environment, that shape us in such a way, we have preconceived notions. Even with the same shoe, the same sound, we see different color, we hear different sound.
- 2.14** To understand that is very, very important because when we make a decision with the same thing, we know different people bring different perspectives. Not because they're bad, not because they're dumb, not because they're difficult, just because they're different. Now, in order to make the right decision, it is very critical to allow yourself to have all different perspectives available. Then you will make the best decision possible. Otherwise, you will make half the decision.
- 2.15** As we can see, if you insist the shoe is gray and green, then you dismiss the pink and the white. You only make this decision for half the population. The same thing goes with the tone, with the sound and music. Because our sensory perception is prone to error and that's built-in in our system, it's very important to recognize that in any given situation, we have a blind spot. Not because we chose, just because we're human.
- 2.16** And how to overcome that? You cannot train yourself to see the green become pink; you cannot train yourself to listen to a tone. It's very difficult. Maybe it's possible. But I think the most effective way is to identify those people who could bring complementary skills, to bring you their point of view for you to have a broader understanding what actually is going on. That will allow you to make a better decision.
- 2.17** Because of this, a few examples I showed you that humans have quite a few biases here. I'm not going to go through all of it. I think throughout the day that Derick will provide examples of those biases that will give you a much better understanding that what is actually going on. So we, hopefully, move forward with that awareness, then we can make a better decision.
- 2.18** A lot of times, this is just a few examples, you probably have the experience as well, is that after something happens, "oh yeah, that's what I thought." That's called the confirmation bias. And sometimes, like Derick's example earlier, if you did one clinical trial, it failed, but already invested 5 to 10 million dollars, might as well keep going. That's sunk cost fallacy. Just because you invested in the past, it doesn't mean it's the right thing to do in the future, but we need to be aware of that.
- 2.19** Sometimes when things happen, I knew it. I knew it. I told you so, right? But actually, you didn't. That's hindsight bias. The availability bias is: I keep hearing about the plane crashes. They're really unsafe. Just because it has a lot of press, it doesn't mean it's actually unsafe. In fact, airline industry is one of the safest industries. Those are just a few examples, but I think over the next two and a half days, we're going to learn more examples of the various fallacies and biases.
- 2.20** So just to summarize this portion that being wrong is fundamental to human experience. The important thing is how to acknowledge, accept, and endorse, and now

to try our best to avoid it. Because we're human, we have all of this biological bias built in us. We will be wrong from time to time and we will make mistakes. It is okay to make mistakes, but it is not okay not to acknowledge it and keep making the same mistake over and over again.

- 2.21** I'd really like to take this opportunity to practice what being wrong is like. Hopefully, together, we can develop the strategy in the future when we realize we made a mistake, how we deal with it, and how we learn from it.
- 2.22** At this point, I would encourage you to think about one time that you made a mistake, the most painful mistake, the most humiliating mistake you've made. Think about that to yourself, and write down what happened. The more crushing, the better; the more painful, the better. One of the worst mistakes you ever made, write that down. After you record that event, ask yourself: when that happened, when you realized the mistake was made, what was your reaction? How did you feel? Write down the negative sensations you experienced when you realized you made a mistake.
- 2.23** Now, ask yourself: is your emotional reaction in line with the mistake? How critical was that mistake? Did anyone die? Did anyone get hurt? Would that mistake kill you? Remember, what doesn't kill you makes you stronger. Will I remember it in one to five years? Did I lose a lot of money? Is that mistake easily fixable? Most important, what can I learn? Does it really matter in the grand scheme of things?
- 2.24** Now, let's challenge your reaction. Let's engage our analytical brain to challenge our emotional reaction to our mistake, and ask those questions again. How awful that mistake was? Anybody died or got hurt? Did I lose a lot of money? Does it matter? Does it really matter in the grand scheme of things? If I may, I would like to have the audience volunteer to share one of these mistakes. Who would like to be the one to share?
- 2.25** I got arrested for drugs, and I was facing a felony and two misdemeanors. And I felt a lot of shame and embarrassment and disappointment and fear. I guess when I asked the questions: was it in line with the mistake what I was feeling? I felt at first, yes. Then when you took us to the other questions, like, was it easily fixable? I guess at first I thought it wasn't, but it kinda was. It did take me, at first, I had to process those emotions and take responsibility for the choices that I had made for the experiences that I was living in my life.
- 2.26** Thank you. That is very helpful. Yes. Anyone else want to share?
- 2.27** When I was 21 or 22, I had some money saved for school, and I used some of the money to go out and party and do all the fun stuff, I would say. And later in life, I couldn't really finish what I was studying because of that. So I had to find other ways to find that money and finish that. And it took some years, and I felt frustration, regret, and sadness. Did anyone die? Not that I remember. Nobody got hurt. But did it matter in the great scheme of things? I would say yes because if I would have finished that

earlier, I could have helped a lot of people, especially family. And I didn't have—that took a lot of courage for me to admit and to fix the problem. So I had to find all the ways to finish it and to finish that license. YYeah. Thank you for sharing. Yes. Really appreciate it. I think this is very helpful. The reason people, or the ego, try to avoid a mistake or admit they made a mistake is exactly the initial emotional reaction. Because it's so humiliating, and so hurtful, and so devastating. But I know all of us are human. We will make mistakes. The next time when something like this happens, you'll let the emotion, you don't try to suppress it. You'll let it out. You let it process.

- 2.28** But once that's out, do not make a decision then and do not really react. Let your analytical brain to engage and then ask those questions. How critical that is? Anyone died? Anyone hurt? Did we lose a lot? All of these simple questions. Very quickly, you will realize in the scheme of things, it's not that important. But it's more important to accept it, acknowledge it, learn from it, and not to make the same mistake again. You're allowed to make different mistakes, but if you keep making the same mistake, there's something wrong. Okay?
- 2.29** Along the same line is that we need to have compassion to those around us if they make mistakes. We also need compassion for ourselves. We need to forgive ourselves for being human and make mistakes, and I think it's a very important part of the healing. Identify the lesson, as mentioned, that all of you, when you recall some of the biggest mistakes, but you all end up in an area of learning. Some of this is more immediate, some of this is annually after. Even after all this time, when you recall that mistake, you still learn something. This is why mistakes are so valuable to us. Because once you acknowledge it, it becomes a source of a learning opportunity. And that is the course you want to be on to ultimately achieve success.
- 2.30** Reflect is important. How can your life be better now that you've learned the lesson? And some of this is the homework for the rest of your life: it's every time when you make a mistake, you want to make that as a part of you, a valuable experience that you have that you can carry with you for the rest of your life. Because of that, we already discussed this, is that from the mistake we recognize and from the learning that we now benefit from, and that is the source for our growth.
- 2.31** Forgiveness is important, right? Because I think one thing, this is something I learned as well, and Derick knows, that I have very high standards for myself. If I make a mistake, I just beat myself so hard. So Derick had to call me to let me forgive myself. So this is something I'd like to share with you, is that we are human, accept it, don't be too hard on yourself. From time to time, it's okay to forgive yourself.
- 2.32** I think from there, you can be a more compassionate person toward others. I'm sure that all of you had the experience that other people made mistakes. It's so frustrating. Why does people doing this? Only if they can behave differently. But now we learn the lesson that we are human, we make mistakes. Other people are human. They also make mistakes. Let's extend that forgiveness and grace to others as well. By doing so,

you will really cultivate the loyalty to others, and other people will not have the fear to come forward with their mistake. Because in my line of work, to involve a lot of people, their mistakes making left and right all the time.

- 2.33** What I don't like is people hide it and not to tell me. But sometimes they call me up, out of blue, and tell me all the things they just screwed up. They are expecting me—they say, "now you can yell at me now." I was like, "no, I appreciate your transparency, and you let me know all these mistakes were made. Let's focus on the future. What can we do to fix the mistake, so it will not happen again." They were like, such a relief. So next time, if they make mistakes, rather than hide it, they will communicate with you, and together you can find a solution to solve the problem and move forward together.
- 2.34** You're probably wondering what made me start to read the book, *Being Wrong*. This is a place I'm going to share my story. I think Derick heard it before, but anyway, I think you will also learn how we met. This is a part of my life, I would say, talking about most painful, humiliating experience, and that summarizes all of that.
- 2.35** It was about 2000, 24 years ago, against all odds, that I started my own company. Nobody believed that's possible for a woman scientist from China—no money, can't speak English well—want to start a company: no chance. Everybody said it's not possible. I don't know if that's ego, or that's passion, or that's confidence. I was like all of you naysayers, you watch. I was able to: successfully start this company, assemble an amazing team, and with the management, also raise money for the next, I would say, 13 years; led the effort of developing a very complex intraocular implant to treat blindness.
- 2.36** My goal is to build an ophthalmology company into the best company in the world. That's the time I hardly could spell ophthalmology. I was in a different field. I was trained as an immunologist. So along the way, that was quite successful. But one thing that I learned I did not know back then is that I believed decisions will be made by people with expertise and experience. But the structure of the company is that decisions will be made by whoever invested the most money.
- 2.37** Over the years, until about 11-12 years ago, there was a disagreement between me and our investors. They're also on our board. They want to apply this technology to compete with Genentech, Amgen, Regeneron, large markets. But our technology is only suited to genetic rare diseases. The market is not huge, but it's a reliable market. But because of the vision difference, I was so confident that if I just show them data with the logic, they're going to be swayed to my view of thinking. But little did I know, they have decided they've had enough of me, and they fired me. It was 2013, I still remember, June 10th. Completely surprised. The company I founded myself, I've been working there for 17 years. My career just ended like that.
- 2.38** I think with that shocking, humiliating experience, that's when the book, *Being Wrong*, was introduced to me. I really, really wanted to understand why, because I didn't

understand what happened. Through that book, I learned about myself. I learned about other people, how they make decisions. Rather than being bitter and fighting and suing, because I had a lot of leverage that way, I decided, I'm going to be a better person. I started to look at different project to work on. It was through the different project that I met Derick.

- 2.39** Derick was an investor of that project. I was an operator of that project. I went to New York City to make a presentation for our investors; that's how we met. Together, we worked as a team. Because of this book, I realized that Derick and I come from a very different background. Derick has a business-finance background. I have a medical science-researcher background, but that is a very powerful combination. Together, we were able to get the first project off the ground and had a successful outcome. The project was sold to a large company, and Derick and I worked together to discuss, negotiate a deal. But after that transaction happened, my job went, left.
- 2.40** Sometimes, the success—I basically lost the job, and then Derick said, "Weng, why don't you come work for me?" So ever since then, Derick and I have been working together, and we understand this aspect of human nature, and we sometimes argue a lot. Not because there's no respect. Just because his background, he looks at the problem from one angle. My background, I look at the problem from different angle, we come back together, ultimately, reach consensus. But it's not easy. Sometimes it can be very painful.
- 2.41** But what we learned, if we have the same viewpoint, that's the problem. There are times that we completely disagree, we negotiate and reach the same consensus. There are sometimes—at the get-go, we agree. Immediately, both of us say that's not a good sign. Let's bring in a different point of view. We actually intentionally bring a different point of view to argue with us because we can be both right or both wrong. We don't know what it is. Now we have a different perspective, and that allows us to make a better decision.
- 2.42** Here, what we learned is to understand our mistakes, the nature of mistakes, and that hopefully allows us not only to have compassion to ourselves, but we could also support others. This is actually not easy. How do you deal with someone who is wrong? If they don't have the understanding that you have here, they think they're right, but they actually wrong. This is something that Derick and I have to deal with this quite a bit. It really required understanding and compassion. But you really need to understand that if you just use logic alone, you're not going to convince anybody to be behind you.
- 2.43** The thing is that if you're the only one right, you cannot lead. You need people to actually see your vision. There's a part of the human factor that I did not realize before, and I'm learning now, and it's a challenge on a daily basis, is that people don't necessarily make decision purely based on scientific rationale or data. Because humans are emotional beings, there's emotion involved. There's an ego involved. There's social

pressure. There's politics. We need to take all of this into consideration when we try to persuade others to be on board with the decision we make.

- 2.44** Again, this is a very similar theme, and this we're going to go through the next few days as well, is that we need to normalize mistakes. We need to make mistakes part of the journey. It's not like people say, "We start from here, there's a success, and there's success all the way." Actually, it's exactly the opposite. Right? So at the beginning, there's a success, there's a mistake along the way. We need to learn, make a correction, learn. So by the time you reach success, then you are expert in dealing with mistakes.
- 2.45** There are some lessons here is that we need to live by setting examples. I think earlier someone asked a question, how you encourage open dialog, how you encourage people to share with you if they're wrong and they don't feel embarrassed. I think it's not only you need to actively seek their feedback, you need to create an open environment. You allow different point of view to come in and encourage them, even though sometimes they could be wrong, but not to punish them. It's very, very important because people take that reward/punish behavior very seriously. If you want your organization to have a healthy culture, you need to truly encourage open dialog.
- 2.46** Here, just some of the statistics showing that if you work in a company with a growth mindset, you will have people feel more strong sense of belonging. They're likely to see their company provide risk-taking. They're not in fear, and they're more likely to see their colleagues are trustworthy. So there's trust there. They don't feel mistrust. They're more likely to say that their organization encourages innovation. If you want new ideas to come from your employees, I think it's very important to have an open growth mindset.
- 2.47** Just to conclude: our emotional reaction, being wrong, gets in the way of learning. Again, we all have the experience that when you first realize you're wrong, it's a very strong emotion. I think biologically, all hormones and neurotransmitters probably just flash into your system and you feel overwhelmed. But I think let's engage our analytical brain soon after to really sort out what exactly is going on. I think that's the best way to really learn, analyze, and grow from our mistakes.
- 2.48** Now, I know certainty is comfortable. Now, I realize that there are people seek comfort in their life, but those people will never be successful. I think all the successful people I know, they seek out discomfort. They challenge themselves. They surround themselves with people who are not, "yes, sir." If you surround yourself with people who are confident enough to challenge you, that's when you benefit the most. Just basically to understand what a mistake is and understand that's part of our own journey, I think that will be the starting point of your success. There's no doubt in my mind, if you have this philosophy, you will be ultimately successful. Any questions?

- 2.49** Is there something from that book specifically that stood out to you that changed your mind about using the legal leverage that I'm sure you had against the company and go a different path and start anew?
- 2.50** Yes. I would say the question I asked myself is: why these very successful investors— I thought they were, knowingly, making decisions against their own best interest? Because they don't have the expertise, but why do they think that they're the one who should make decision? I didn't understand that. The second one is that I know I have the knowledge and the experience and the data. What have I done that failed to communicate effectively to the investors so they did not make the right decision?
- 2.51** There are two parts. At the time, I thought it's more important for me to understand myself, and I realize it's a completely different skill-set. One is that: you know the stuff, but in order for you to communicate that to other people effectively, that is on me. If I have a person communicating—this is including Derick as well at the beginning. If I keep saying this, he's not getting it. It's his fault? Or it's my fault? He's an intelligent person. Why he didn't get it? What I learned is that I need to find a different way to communicate, and that is so important.
- 2.52** The second one is that: why those successful people that put in 50 million dollars into a trial I know is going to fail—I told them they will fail if they do it—they did it anyway. Why is that? I did not understand that. But the book, *Being Wrong*, actually showing that the blind spot—just because people had achieved success financially in a different area, it doesn't make them the expert for this area, but they do not know. So, there's a blind spot on their part. Basically, when things didn't work out, they were multi-component.
- 2.53** Now, for me, I have the previous many years experience. I have the remaining years. Do I want to spend the time to go back to history to right a wrong in the past, or do I want to use my time moving into the future? In this case, I know people say, "Oh, that's the worst thing happened, but now it turned out to be the best thing." I literally can tell you I'm so thankful they fired me because I would never have met Derick if they didn't fire me. I would be still in this company, fighting, butting my head with the investors. Every day, it becomes excruciating, but I allow myself to acknowledge the error and then acknowledge the learning opportunity. I move on, and I'm very grateful that Derick gave me another opportunity. Very few people have that.
- 2.54** I start to practice the principle of this book, not only just myself, I think of Derick as well, and then the people we work with, and then we built this remarkable company together for the last 10 years. I would never have the opportunity for this if I wasn't fired. To me, when I think back, every day, I said, "I am so lucky that I got fired." Even Derick said that, "I'm glad that happened."
- 2.55** Because the first day they let me go, every single second I was obsessed. I want to go back because I lived, breathed, and work 24/7 on the technology for 17 years. I know

every single physician. I know every single of my patients. I was invited speaker all over the world. I have a lot of visibility. Overnight, I have none, zero. I have to build my career from scratch, from zero. But there's no regret because the last 10 years, it is the best 10 years of my life. I even said to Derick, I said, I wish I met you 10 years ago. Now I'm looking forward to the next 10 years.

- 2.56** Do you find emotions more as a hindrance or something that you work with to maybe improve yourself, improve business, the science, etc.?
- 2.57** Well, I think even when I was young, I always felt a sense of almost a mission to make an impact as an individual. I think that made every decision I made from that perspective, because I understand—this is something I think a lot of, quite a bit, I don't know about you—because each one of us has only limited time on Earth. It's finite. And how we're going to utilize this time effectively to make the most positive impact.
- 2.58** I realized very early on that I am limited and I have a blind spot—definitely not the most intelligent person. If I want to make a bigger impact than just me individually, I need to collaborate with people who can have this one-to-one, one-plus-one-equal-to-five, effect. This is something that I find that to partner with Derick, definitely, it's consistent with what I believe. I think together, we definitely make much bigger impact than us working individually.
- 2.59** I wanted to know if the investors gave you feedback, directly, as to why they decided to take this decision. And, had you not agreed with that decision or the opinions of these investors, would you have decided to take another path in the sense of not looking at yourself and the book, *Being Wrong*, and making it about yourself, but rather maybe not agreeing with the investor's decision about your abilities? And if so, what would you have done in that case, if you had not agreed with their decision or their opinions about why they let you go?
- 2.60** It is pretty obvious. For a start-up biotech company, the scientific founder usually survives three management changes just because the scientific founders usually are associated with the earlier development. In my case, I actually saw like six management changes. That's why I thought I was pretty solid with the investors, but the investors had a certain life expectancy. When they put money in, they want their reward back very quickly. In their mind, they believed if they opened up an indication that was in direct competition with a billion-dollar drug, that would increase the value so they can exit at a high value. That's their objective.
- 2.61** Now, my objective is different. My objective is: solve real problem for real patient in the long haul. It's not like a right and wrong. I understand the business incentive of it, but I also know the technology that I developed will not be able to compete with a billion-dollar drug because the mechanism of action is different. But at that time, the investors, they were so driven. I think early on, we talk about a public company. It's

very similar to that scenario, is that they are under the pressure to build up the value and then just get rid of the technology. They can move on.

- 2.62** But I told them that's not going to happen, just because the technology doesn't work that way. If they did not let me go, ultimately, I will tell you what happened with the company. They went ahead with that trial and failed. They lost their 50 million dollars for that particular trial. They put in hundreds of millions of dollars. Ultimately, one of my patients who is a multi-billionaire took over the company and went ahead with the technology that I thought will be a benefit to the indication. They went ahead with that indication that I had designed; they were successful.
- 2.63** They're waiting for FDA approval any minute now. I know in a way that to me, it's a validation. It's not like I really needed it, but it's good because that's something I worked all my career. It validated the technical decision I made was correct. But it still did not solve the problem. I was ineffective in communicating that to the investors. To me, that's on me. If that happened again, I probably would hopefully do a better job, but I am so grateful they let me go because I wouldn't have this new career, the second career I have with Derick, if that didn't happen.
- 2.64** When you asked us to reflect on our own mistakes, I found one of the things that was quite difficult was: based on my past experience, I started to instantly re-frame all of these anecdotes as learning experiences. Or I was like, was this really a mistake because of all of these secondary benefits? And I wonder if that could be a consequence of maturity or age, because as you reflect, maybe you protect your self-concept by not admitting that those are mistakes. So I guess as a person in your position with this level of experience, how do you have the composure to label those experiences as just mistakes? Not like, oh, that was a pivot, or that was a learning experience. What kind of mental models do you use to be able to hold yourself accountable?
- 2.65** Great question. I think just by asking that question, it tells me you have the awareness, right? Otherwise, you wouldn't have asked that question. To me, again, the book, *Being Wrong*, is for me also to realize that we are all great at revising history. Because I think that is an evolutionary trait: to self-protect. Because we want to make sure that our well-being is taken care of. And then if we totally screwed up, and then you will really face through it, it's pretty depressing. We have a mechanism to revise the history to protect ourselves.
- 2.66** But I think to recognize that allows us to use our analytical brain more effectively. Because we are all humans. We all have our emotions. When the emotion takes over, it's so overwhelming that you can't really sort out right and wrong. In that case, I would say it doesn't matter if you are 7 years old, 17, or even 70. It's just part of the norm. But I think when you are 70 versus 7, at 7, you probably wouldn't know. Just think of whatever you experience is the real world. But when you're—later on, you realize that your experience is your transient emotional reaction.

2.67 You let it go, let it process itself. Then you ask yourself tough analytical questions, and to really deal with it. Because I think to be successful, you need to have a fine line. One is to really make yourself more capable than you are. Otherwise, you will never take on the challenge because the chance to be successful is so low. People say someone like you, your background, to start your own company? Are you kidding? Nobody will be successful. But you got to have that belief in yourself to carry you through the hard times.

2.68 But in the meantime, you got to understand: you don't have the humility to understand your own limitations and your own blind spots and to really assess the situation in such a way that you will make, I would say, ideally, will make only the right decisions the rest of our life. But that's not going to happen. We're going to make more right decisions than wrong decisions. If we can do that, we'll be successful. I would just end here to celebrate the journey of being wrong.

Lecture 3: Information and Decisions

At a glance — Information and decisions.

- 3.1 So there are really two, I think, superordinate aspects of understanding information that comes into your worldview, that impacts the way that you ultimately perceive the world and then the consequent decisions that you make as a result of the way that you perceive the world. One is—and they're both manipulative in a way. They're not explicit. You're getting inexact information that comes into your brain. You want to understand the two reasons why that happens.
- 3.2 One is, especially in today's world with social media and Google and everything else, the information that you get to perceive as being tailored and custom-tuned and algorithmized to present you with information that those things think you will find important. The second is the way that your brain actually works and what it can and cannot perceive in terms of blind spots, etc. You want to understand what it is that you understand, the meaning behind what it is that you're saying, so you can then both understand the information and then speak it to an organization in an eloquent way so that people can follow what it is that you're doing and why.
- 3.3 How do you know if you know? This is really epistemology. One of the tell-tale signs of whether or not—you want to get good at separating the people that advise you into two different categories. One is, do they really have expertise and knowledge? Are they actual experts in this area that are giving me information because they know that the result of that information will help inform my decision-making process in a way that results in the best decision I can possibly make? How do you discern who those people are?
- 3.4 The other type of information flow, I call them "fake-sperts," are people who want to impress you. They want to beat their chest and demonstrate to you that they know something about the world, when they actually don't. And so you want to try to understand, you want to try to differentiate between what those two things are. So what's a really genuinely knowledgeable person versus who's just telling me something because they want to flatter their own ego, basically.
- 3.5 OK, so what's the difference between the two? Here's what you're looking for. In the entire world of information, there's a lot of jargon. In every area of expertise, you're going to hear words that none of us are going to understand in any respect. If it's in statistics, it's sigma and p-values. If it's in taking a cell apart, it's going to be the periplasmic space versus the cytoplasm. They're going to be all of these words that are unique jargon to one particular area that are just— it's like a fire hose. How in the world do you differentiate whether or not somebody knows what they're talking about when the experts and the "fake-sperts" are giving you a lot of jargon?

- 3.6** One of the best ways to do it is to ask for the advisor to give you the information in a more interpretable fashion. This requires you set aside your ego in order to do this. We've been talking about the ego a lot. You have to say, "I don't understand this. I don't know what this means. I don't have expertise in every single area in the world. I need you to explain it to me. OK? Explain this to me like I'm five, literally." And if they can, then they understand it. Right?
- 3.7** If someone, so I'm going to challenge myself with this, with you guys tomorrow. We're going to go through p-values, and we're going to go through t-distributions, and if your eyes glaze over, then I'm not really an expert because it tells me that I have not done a good job of explaining something pretty complex because—I call it de-jargoning. You want to take a complex idea. You're looking for somebody who takes a complex idea and distills it down into something that any human being can interpret.
- 3.8** Because, ultimately, all of the complex things are really not that complex, maybe particle physics or something. But everything else can be broken down into relatively simple concepts. So explain it like a five-year-old. Give me metaphors. In terms of you understanding it yourself, you want to do the same thing. So conceptualize a kindergarten version of yourself and go, "I'm going to try to understand all this jargon and all these fancy words. I want to understand them well enough and then translate them into something simple, into a basic concept that my five-year-old self can understand."
- 3.9** Can somebody else do that? And then can you do that for yourself? Basic ideas along that line would be five-year-old, clear metaphors. Metaphors really can help with this thing. Do you really know the background information and the variables that are involved in the situation that tell you whether something is right or wrong? Do you know the trunk of the tree all the way out through the limbs to the trees? Do you understand the whole thing? Or is somebody just giving you the leaves and saying, "Well, here's the answer." It's like the multiple choice test in academia. Here's the answer. You memorize the answer, but you don't actually understand the whole construct.
- 3.10** OK, so you have to understand the whole construct, from the trunk to the leaves, if you're going to make the right decision. If you don't, and many CEOs—they have so many things going on. They don't take the time to understand every aspect of accounting or understand it well enough to understand what the information is telling them, the gist of it. Do you really understand the basic background of this thing and whether the idea is right or wrong? Can you simplify it to yourself enough so that you can do that? Then can you extrapolate from that?
- 3.11** OK, well, what is the trunk of the tree telling me about where the tree is growing? What is this information really telling me about where we're going as an organization? How should I be processing this, etc.? Ultimately, the big thing is, am I focusing on information that I want to be true because I have a preconceived notion about what it

is that I think is the right answer? Or am I focusing on information that's genuinely, objectively true, even if I don't like it? And what you want to see is the stuff that you don't like. You want to train your brain to be able to see things that you don't like.

- 3.12** Again, it's just some basic patterns of "fake-spertise." We all fall in these traps. By the way, we're all "fake-sperts" to some degree. Just memorize the formula. We know, you guys, if you took fairly advanced math in college, you just memorize the basics of the way the problem works and you know how to work it, and you do enough homework examples, but you don't actually totally understand what it means, but you can just get through the quiz. Well, that's not understanding. You memorize it, you can do the basic math, you can say the words, but you don't actually have a depth of understanding.
- 3.13** That gets translated into a lot of jargon in terms of the way that you interact with other people. You want to see that in other people when they're doing it to you. The bedazzling aspect of using a whole bunch of real fancy words in sequence, knowing full well that if I start saying a sequence of really complex things in statistics or whatever it is, without a background in statistics, you're not going to know what I'm talking about. But if I'm trying to impress you with my words, rather than my simplicity of underlying knowledge, I'm not what you're looking for. So ultimately, it's really simplification. Simplification is the real discernment that you want to figure out here.
- 3.14** You're in a room of people, and you're trying to figure out where you fit. You're trying to figure out where other people fit. You want to know who actually knows what they're talking about and who does not. We've all met people who have the smartest person in the room syndrome, and they use a lot of jargon, and they want to make sure that everybody understands how smart they are. That's got ego built into it. That proclivity to impress others is something that you want to be aware of, and you want to be aware of it in yourself, and you want to be aware of it in others.
- 3.15** But also, going back to the ego, some people are driven by, "I want to make sure I impress everybody," and others are driven by, "I want to make sure that the things that I say don't imply that I don't know what I'm talking about." And that's actually the secret— to get over the fear of appearing dumb. If you want to be a true gatherer of knowledge, you have to be comfortable appearing dumb and foolish because there are so many areas in the world that—we go straight back to Socrates, the start of this whole thing. "The only thing I know is that I know nothing."
- 3.16** There are so many areas in the world that you don't understand. Well, of course not. How's any one individual human being going to be educated on everything? So really, at the core of that is to know whether or not you know. And then once you figure out that you don't know, find somebody that can help you know, and you want to get good at identifying that person. So be the dumbest. Humble, stupid questions, honestly. Just ask them. Get over yourself, get over your ego. Just ask the question.

- 3.17** This is harder the higher you get up in an organization, because if you're a vice president of whatever it is, you're not supposed to be naive. You're supposed to know what you're talking about. Well, what if you don't? Well, that's a really dangerous situation. You want to impress everybody with your knowledge and make sure that they know that you're the vice president, and you're really smart. At the same time, you're presented with a problem in a situation that you don't understand, and you don't know what to do, and you're scared to death that somebody's going to find out.
- 3.18** The secret to that is to do your best to not let that happen. The way you keep that from happening is to be willing to expose your naiveté. You just be open and honest with it. And this is one of the biggest challenges for human beings. It is extraordinarily rare, but if you can figure it out, if you can transcend that, it opens the door to every single avenue of potential knowledge because you are admitting the things that you don't know.
- 3.19** So if your greatest concerns in impressing others, you're not going to be a transcendent decision-maker. You're not. You're constraining yourself. Because you will constantly be placing a boundary on your own ability to learn because you're so concerned that you'll expose yourself to somebody else who will go, this guy doesn't, I don't know what I'm talking about when you don't. You don't know what you're talking about, and you need to admit it. Like, I'm CEO. I'm supposed to know everything. Let me assure you, I do not know everything. So I constantly have to be saying, "Tell me more about that. Tell me how that works. I don't understand it. Explain it to me. Make it simple."
- 3.20** And it's not an admission of a lack of intelligence. It's exactly the opposite. What it says to other people is that I acknowledge my own boundary of what I know and what I don't know. If you could do that, people will tell you things because you're inviting it. And you're making them feel comfortable. So academia is one of the biggest obstacles to the dumb questions, frankly, because it's got such a construct of certainty built into the multiple choice question that you go through academia and you do end up in a position where it's expected that intelligent people in academia know the answers.
- 3.21** You're supposed to have done your homework. You're supposed to know everything. I'm not saying, don't do your homework, but you go into a class on Shakespeare. Even if you've read Shakespeare, you don't necessarily know the nuances of what Hamlet's soliloquy really is about at a very deep level. You want to just expose yourself to, "Hey, teach me. Let's talk about this." In the real world, there's not that academic multiple choice answer. It's a fuzzy distribution of probabilities.
- 3.22** And I'll tell you honestly, and this is particularly true in the scientific field, the smartest kid in school who made straight As, who never really was exposed to the humility of, "You don't know what you're doing," struggles with this unbelievably. Because the real world puts you in positions where you don't know and you have to understand that and admit it. So there are a couple of different tricks. You can softly ask the question by saying, "Well, what do you think maybe could go wrong here?" Then you can talk about

the things that may actually lead you to a mistake, and that opens up the door to your understanding. It opens up the door to a discussion.

- 3.23** If you say to folks, "Well, don't tell me the answer." And it's actually a good trick to discerning whether or not somebody knows the whole situation well, is because if you say, "Don't tell me the answer, tell me the things that you think may lead to a bad decision," then they can give you the trunk of the tree. They can paint the whole picture for you. If you can do that, then you actually understand something. If you can think about which limbs of the tree you're going to end up in a bad situation, it more easily highlights the proper limb and the proper, so talk about what could go wrong.
- 3.24** Again, it's that certainty equation. It's like, let's admit that these things can go wrong, and let's discuss them. It opens up the door to a more frank and open conversation throughout an organization to talk about the things that may not work. Nothing's easier than self-deceit. This is the ultimate obstacle of human beings and processing of information. Here's how deep this is. This is unbelievable to me. He's an evolutionary biologist. His name is Robert Trivers. And he wrote a paper, I think mid '70s, early '80s.
- 3.25** It's called "Self-Deceit." In this paper, he discovered that, so human beings are very good at perceiving when other people are lying. We pick up on little subtle signals in terms of the way you glance around when you're lying, and you won't look somebody straight in the face, and get jumpy and sweat. There are all these things that are going on biologically when you're lying. Other human beings are wired to perceive that. So here's how deep self-deceit is. In order to counteract the fact that other people can tell when we're lying, what we have evolved to do is be extraordinarily good at lying to ourselves.
- 3.26** Because if you can lie to yourself and believe it, then you don't actually realize you're lying. And the consequence of that is, you can hide the lie when you communicate with somebody else. So self-deceit is really, really deep. It's way down in our psychological profile. Because as we manipulate the world, and as we navigate the world, is a better word, we're actually doing this to ourselves to some degree all of the time. It's little bits and pieces of information. You can't have the perfect picture of every situation. So everything is truncated and curated a little bit.
- 3.27** All right, so where does this go wrong? The unexamined self, the people who don't understand how self-deceit can influence the way that you perceive information. Here's the sequence that occurs upstream of a really bad decision. The first is they use what are called heuristics or intuition and jump to a conclusion really quickly. Then once they decide that that's the answer, too quickly without taking the time to really carefully process things, and they land on that answer, then the ego comes in and says, "We need to be certain about this," because a lack of certainty implies that it might be a mistake.

- 3.28** It starts looking for information, confirmation bias-driven information, that will certify the certainty of that decision that has actually already been made subconsciously. There's a reason why we're wired to do this. We'll talk about it in a second. It's evolutionarily efficient. If you see a glimpse of a tiger in the woods and he's running at you, you don't want to think, was it an African tiger or an Asian tiger? And what's the weight of the tiger? You don't want to overanalyze this, you want to just run.
- 3.29** And so there's a reason why we make decisions quickly that's evolutionarily advantageous. You just want to be aware of the fact that you're doing it. And we'll go through how to be aware of that. So you make a decision quickly. Tiger, run. And then you start confirming why that preconceived decision was a good decision. And when that happens, your ego comes along and it starts screening information for you. So what it does is, it says, "This tells me that the decision I've already made is the right decision. And this over here tells me the decision I've already made is bad and wrong."
- 3.30** And so it goes, "Don't look at that. I don't even want to hear it. I need more of this. I need more of this information over here." And so what actually ends up happening is that your neural processes will pre-screen information that it doesn't like, and it won't allow you to carefully consider it because it's offensive to the ego decision that's already been made. We all do this. It's unbelievably deeply built into our construct. It hides information it doesn't want to see, and it looks for more information that supports the decision that was already made without being analytical. You can see the danger in that happening.
- 3.31** So, heuristics. What are heuristics? I'm just going to read this Wikipedia definition: "Heuristics are the strategies derived from previous experiences with similar problems. These strategies depend on using readily accessible, though loosely applicable, information to control problem solving." What does that mean? It means— there's actually a great book on this topic. It's probably not as good as *Being Wrong*, although it's pretty close. It's written by a Nobel laureate called Daniel Kahneman. It's called *Thinking, Fast and Slow*, and it goes through exactly the depth of understanding of the difference between the jump-to-conclusion part of the brain and the deliberate, long-term, careful consideration part of the brain. He calls them System 1 and System 2.
- 3.32** So, System 1 is the heuristics. It's the tiger, run. It's oversimplified, straight-to-the-answer decision-making. System 2 is deliberate, careful, calculating, analytical, conscious, requires effort. This is actually a problem because, I'll go into why it's a problem in a second. The basic gist of System 1 is that something looks more or less like some other problem that you've seen, close enough. Your brain goes, "This is pretty close to this problem that I've seen before." And it just goes straight to, well, "This solution worked for this problem," and it just applies that solution. It goes, "There's the answer."
- 3.33** In the complex reality of the real world, that intuitive initial response, frankly, just doesn't work most of the time. The over-application of the solution to the old problem

is not a good one. So what's good and bad about System 1? The brain is an extraordinarily high consumer of energy. So, actually, to think very carefully and take your time and go through System 2, which is the more deliberate process, requires a lot of glucose. We're wired to avoid overusing our brains. Let's just simplify this and go with a simple approach.

- 3.34** So our neurology is actually wired to do this. And if you think about it, it's pretty simple. You're sitting there in your chairs. You're getting something like a million neurological signals per second. So there are signals. If you think about it for a second, the balls of your feet touching the floor. If you think about the balls of your feet, you can feel them. But if I didn't point it out, you would not be thinking about the balls of your feet, right? Same thing with your butt in the chair. The perception of electrons hitting your thermoreceptors in your skin is happening. This is how we determine if we're hot or cold.
- 3.35** But it's literally thousands and thousands of little signals all over your skin going warm, which warm is too many electrons hitting thermoreceptors. Cold is too few. And so your brain is processing all of this information and screening out all of the things that it thinks you don't need consciously. Otherwise, you would go crazy. Imagine trying to process a million things a second. You wouldn't be able to do it. So your brain is actually really good at pre-screening what it hands off to your consciousness, to say, "I think this is important, and we need to focus our higher-order resources on this."
- 3.36** The problem is when it does that for you, and it shouldn't. That's what you want to try to avoid. So System 2 is just paying attention to information, processing data, looking at things far more carefully and analytically, using doubt and reason to second-guess yourself, logically. It's different perspectives. It's gathering all the information to really try to understand what are the nuances of the good decision in all of that. Rather than, here's the answer, everybody shut up, let's just do this.
- 3.37** Confirmation bias is the real trap door that will catch you in all of this, because once your ego decides that this is what it wants to be true, it decides that it's true. Then it will ignore information that's counter to that, and it will seek information that reinforces it. Look, I'll be honest. If you're going to do really complex things in life, if you're going to run a good-size business, if you're going to get into science, you have to understand probability. Because probability is a way of understanding what complex information is really telling us.
- 3.38** You really want to try to become an expert on the challenge of statistics and deeper, more complex things in statistics, because it will tell you the real answer if you learn how to pay attention to it. Confirmation bias is inversely proportional to belief. The more strongly you believe in something, the higher the likelihood that you're going to be more confirmation bias-driven in terms of the ultimate decision. The more you like an answer, the more you're going to hope that that answer is the right answer.

- 3.39** That's actually the real tricky part of the trap. The more you hope something is true—we're looking at a new drug, and it's something that I think could be really cool and actually make a transcendent difference in the world in terms of the way that it works. I'm going to be more inclined to get excited about it. All right? So once I get excited, I'm going to become exuberant. Well, think of all the diseases we could cure. Think of all the things that we could do with this. It influences my decision-making process in a way that's counterproductive because it makes me want it to work, rather than make the decision, "Does it actually work and what's telling us that this is actually going to work?"
- 3.40** So you want to be careful about—I think that you can't get rid of your belief system. You can't say, well, "I'm not going to have a belief about something or other." You can't make that go away, but you can understand that it's there. That's actually the trick. If you understand that it's there, it's the impetus to seek information from other people and data sources and whatnot to counter your own preconceived notion. There are all kinds of problems that are ultimately the consequence of this. We self-censor.
- 3.41** I would say the ultimate trick is to, when you're facing a big decision, the ultimate trick is to try to understand, what is your preconceived notion? What do I want to be true in this situation? If you can understand what you want to be true, then that's a path to actually figuring out what is true. Because you have to know that that thing that you want to be true is a ghost of your own neural system that can cause you to go down the wrong path. You want to really try to be able to understand your own bias and your own belief. If you can acknowledge that, it's the path to finding objective truth.
- 3.42** So, how to engage System 2. Take a step back. Reconsider all the information objectively. Don't have a preferred solution. Don't rank-order it based on what you hope and what you don't want to be true. Don't do any ranking at all. Just take a step back and say, "What is this information actually objectively telling me?" Ask questions, ask questions about your own perspective. Look at multiple different sources of data.
- 3.43** We're working on a drug, and it's for a gastrointestinal disease. We run a study, and it helps with abdominal pain. But it doesn't help with diarrhea, or gas, or bloating, or any of these other things. That's an indication that it's probably not working very well. You don't want to go, "Well, it helps with abdominal pain, so therefore I'm just going to focus on that." You want to get the whole picture of all of these different things that are telling you, "Hey, this is actually not working." And you need to be aware of that. You don't want to just focus on the one thing that's working and say, "Well, I'm going to pursue this because I have one signal of something that's working." You want to broaden your mind and broaden the information flow and honestly assess your own bias.
- 3.44** We're going to go for a fun example here because, I think, the more a particular thing, a particular belief, the more deeply held it is, the greater the opinion and the confirmation bias-driven assessment and the emotion involved in the situation. So I've

intentionally picked a lightning rod situation here to give you guys an example of how information affects the way your brain interprets things.

- 3.45** Climate change! It's going to be fun. I promise this is going to be fun. There are two different perspectives that are pretty divergent on climate change, right? One is, and I'm not taking a position here. I'm not espousing one point of view or another. I'm literally just laying out the information and the way that it gets processed by your brain and the way that it flows to you. You have one point of view, which is, the Earth is heating up, and it's a really big problem. And we need to do all we can to counter this by getting rid of carbon dioxide because excessive carbon dioxide is absorbing more radiation, black-body radiation, from the Earth. It's warming up. We have a really big problem.
- 3.46** The other point of view is, that's all wildly exaggerated nonsense. And the Earth isn't really warming that much, and this is a political issue, etc. It's happening, or it's not happening. Regardless of how you feel about that issue, here's what's actually happening behind the scenes. Most of the data on things is actually being manipulated and left out. Most of what you get to see about the world is screened out, either by Google or whatever it is, or social media, or your own brain in confirmation bias, censorship, looking for what it wants to be true.
- 3.47** Or both, actually. You have to understand that. You have to understand there is information I am not getting to see. And the higher you get in an organization, the more responsibility you have. This happens more. We've gone through a lot of this this morning. People do not want to tell you when something's wrong or when they've made a mistake. The amount of information that actually reaches you, the higher your level of responsibility, it gets foggy and screened out more and more. You want to be aware of that. So here's the underlying, the basic underlying message here is, try to understand what you wish is true versus what is true. The way to do that is to access information.
- 3.48** If you're really concerned about climate change and you get on Twitter, what you're going to see is a graph that looks like this. You're going to see a red graph, and you're going to see a dramatic increase in temperature, and you're going to see it over and over and over again, to the point that it's actually impacting your neurology. It's manipulating your emotional and neurological systems. And it happens a lot, particularly—you know this. You have a conversation with your wife and your iPhone is listening to you. And then you get on Instagram or something, and you start getting advertised the things that you were just talking about. You're like, "What is going on here?"
- 3.49** It's curating information based on what it's perceiving about you. And this is getting worse and worse as the algorithms get more and more sophisticated. So you're getting to see what it is the AI algorithms want you to see. Now, if you have the other worldview, you're like, "Well, I don't think this is happening at all. I don't think this is a

problem. Not a big issue. We're all going to be fine," this is what you get to see. This is a graph of temperature going up and down some, but overall, over the long term, not changing.

- 3.50** Now, how is it possible? How is it possible that one graph is going up by 2 degrees Celsius, and the other graph is not changing? How can they both be true? From here to here is what's happening to you in an organization. The information that you get to perceive is being curated. And so it substantially impacts what you have access to, in terms of how you make a decision. What is reality? How is it possible that one person pulls out a graph that says temperature is not going up, and another person pulls out a graph that says it's going up dramatically?
- 3.51** How is that possible? Here's how it's possible. Here's the reality of the world. This is 33 different graphs of historic temperature over the last thousand years. Here's what I do. If I want to convince you that climate change is real, I pull the color graph that shows temperature is going up dramatically. If I want to convince you that it isn't real, I pull the temperature graph, this climate change, that shows that the temperature is going down.
- 3.52** This mess, this spaghetti mess, is true of most complex things in the world. So what you have to do is look for the spaghetti. Because the spaghetti is this ultimate source of truth. Like, maybe, once you distill all of this complex information down, there is one really clear answer. And again, I'm very intentionally not trying to take a position on this because I want to illustrate the point of how things are being manipulated in terms of information that you get to process. Be a cynic. It's the key here.
- 3.53** What is it that people are doing, whether it's in your organization or somebody who's trying to—if we're looking at a new drug, and these guys want us to invest in their company, what's the likelihood they're going to tell me something that's bad? Zero. They are not going to come to me and say, "Here are all the reasons why our drug is probably going to fail." They're not going to give me any of that information. Now, a really deeply honest person will. That's what you're looking for. They'll say, "Here are the things that are the signals that tell me that this is working, but here are the things that we should pay attention to that say it may not." That's a really honest person. They're not marketing to you. They're not curating the information in the background.
- 3.54** But as a decision-maker, you need to understand that these things are going on, and you need to get good at digging in and figuring out, "What is the spaghetti map that's behind all of this, that's the reality that I'm not being told?" Just some more examples here of things where information gets curated for us and we get to see what it is that we want to see. I think the last bullet on this page is the super secret. How good are you at looking at, perceiving, integrating information that you don't like, that's completely contrary to what you want to be true? Because that's the thing that you need to pay the closest attention to.

- 3.55** If you can allow yourself to take in that information— it really upsets you. You don't like it at all. It's completely contrary to what you want. But if you can start to pay attention to it and value it, objectively, and again, it's like we talked about earlier. If you can take the decision and put it on the table and say, "This information says this. This information says that," and try to place the decision into an objective framework that is just rote processing of the different pieces of information, it'll be a transcendent mechanism for you to be able to navigate the world.
- 3.56** Reality is messy. There's spaghetti maps everywhere. Pay attention to the people who openly acknowledge the spaghetti maps. They tell you the good and the bad. Those are data-oriented people. They understand the world is messy and that statistics cannot tell you whether or not you're going to get a seven when you roll the dice. They can tell you the possibility of a seven. They can't tell you when you're going to roll a seven. People who admit that, "I don't know if we're going to roll a seven," they're the advisors you're looking for because they tell you the good and the bad.
- 3.57** People on the other side of the equation, the overconfident chest beaters, they'll do the opposite of that. They'll confidently state that they know what's true, and they'll discount every bit of information that counters that point. They'll just pooh-pooh it and pretend like it's not really meaningful. Set that aside. It doesn't mean anything. You're looking for those signals. The honest person will tell you both sides of the equation.
- 3.58** The "fake-spert" is the overconfident. Here's some signs of the pattern of arrogant overconfidence. One is they'll appeal to the consensus. "Lots of people agree with me." Lots of people can be wrong. Lots of people often are wrong. So that doesn't actually mean anything. You want to be very careful with the consensus. What's the truth? Tell me how you know that's true. Explain it to me like I'm five. Don't tell me you're the authority and you have this degree and you got educated here. I know all these things. I'm the world's expert on this, and all these people agree with me.
- 3.59** Well, why? Why do you think what you think? And if they can explain it to you, then they have real expertise. If you challenge, and here's another good sign. We've talked about a lot of these things—bedazzle with jargon, appeal to authority. But another good way to discern is that the person who doesn't actually know the underlying thing really well is they cannot tolerate nuanced technical confrontations. They don't like to be challenged. They try to avoid challenge, and they do it with appeals to authority, and they do it with all of other things.
- 3.60** But if you ask somebody a tough question and they go, "That's interesting. I hadn't thought about that before." Maybe they start exploring the question legitimately and they start challenging their own thinking, that's a sign of thinking. They'll allow information to come into their worldview that challenges their worldview. Here's the secret. When that happens, you know that person is on the path to trying to find the truth, because the truth is never what I want it to be. It's never what I think. I can get, hopefully, relatively, probabilistically closer to it. But if I'm not directly on the truth, if

I'm not making the best bet in Vegas, I want to know that. A better bet is a better decision.

- 3.61** And so the way to do that is to correct myself when I'm wrong. It's constantly course correct. So genuine confidence, I've pretty much covered this. It's people who welcome other information. They don't self-censor. They'll admit they may not know something. There's an expression of humility built into the way that they communicate. They can explain things simply. They can teach you something about their world. They understand that their worldview, their area of expertise, that you may not know it. They'll distill it down into something that they can communicate clearly to you. We're going to see if we can do this with statistics tomorrow.
- 3.62** Here's an overall, overriding message on this whole thing. If it hurts to hear it, there's some truth in it. If it makes you happy when you hear it, if it's comforting, there's more lie in it. If you can transcend this, if you're a CEO, if you're a senior person in an organization, people want to make you happy. How do you make somebody else happy? You tell them things they want to hear. How do you make somebody unhappy? You challenge their thoughts. You challenge them with things that they don't want to hear.
- 3.63** You want to find that person. You want to find the annoying person that bothers you a little bit, like Weng, when she argues with you like crazy on something. It's actually super helpful. One of the biggest obstacles here, and we're going to go through personalities more tomorrow morning and how different personality-driven worldviews affect the way all of this happens. I'll give you a really good example of something that, I'm just pulling out one example here. But more agreeable people seek consensus. It's actually the goal.
- 3.64** If you get agreeable people in a room around on a conference table, and you assign an agreeable person the responsibility for moderating a discussion on some challenging topic, they perceive consensus as the objective of the discussion. What's the problem with consensus? There's a really, really big problem with consensus. If you have a different perspective than me, and we're going to come to a consensus, I have to, I have to sacrifice my position a little bit. I've got to negotiate and compromise to something that you're satisfied with, and you have to do the same thing. You have to change your worldview a little bit in order to reach a middle ground that we're both happy with.
- 3.65** Now add 10 people to the room, and we're all doing that, and everybody is changing the perspective a little bit. What the moderator is actually doing at the end of the day is tailoring and driving the whole entire conversation to the point where everybody's happy. Well, everybody being happy is not how you make good decisions. Often, it's exactly the reverse. It is, the truth makes everybody unhappy, including me. I mean, I'll give you, again, I keep using the same example. This happened to us a year ago, roughly.

- 3.66** We had invested in this drug. We were advancing it through. We had a meeting with the FDA, and there were multiple things that the FDA told us that made me realize that the complexity involved with getting this drug approved just shot up astronomically. They added a whole bunch of things. Maybe you could disagree with the FDA. It's not about whether or not the FDA was right, or we were right. The truth is, we would have to do many of these things. So it completely changed the profile of what I was willing to do with this project. I ended up having to cancel it. I killed the drug.
- 3.67** So the entire team that was working on this, no one liked that idea. Why would they? They've been working on this for three years. We've made a ton of progress. It's exciting, and we could maybe do something really good for patients who actually really need this. The built-in confirmation bias to continue the project, the sunk cost, is overwhelming. It's literally, almost the entire team is going, "We need to continue to do this." They're very, very frustrated with me. Why? Because I'm the one guy in the room who says, "No. We're not going to."
- 3.68** It was the right decision. I stand by that confidently because I could tell from a data perspective what it was that the FDA was requiring. We were going to fail. It doesn't matter that everybody wants to do it. It doesn't matter that if we could be successful, it would change people's lives. What matters is that we're not going to be able to be successful. There's nothing that we could do to make this work. It has to do with some statistical complexity. I won't go into it, but the ultimate requirement from the FDA imposed on us a trial design that statistically we could not make the math work.
- 3.69** The honest truth is, this isn't going to work. We can't continue to spend money and time and energy on this. No one liked it. So the consensus would be, we need to keep working on this. The disagreeable, tough guy, real, honest, truthful information person says no. That didn't make me very popular. But it was the right decision. The ability to do that is to, and let me tell you, it's not easy because I want to work on things that are going to change the world and improve people's lives. That's what our mission as a company is.
- 3.70** But if we have that be our mission without being critically second-guessing in terms of whether or not something is likely going to work, I'll bankrupt the company. What if I just said, "Well, we're going to work on all these different things, regardless of how complex they are because we want to make something successful that's going to change the world," well, that's a path to fail. We're going to fail on every single one of them because we're going to spread ourselves too thin. We're going to do all of these different complex things that are impossible to do, multiple, very complex things at once. We're going to sacrifice the long-term success of the organization.
- 3.71** Understanding consensus is—the pathway to potential failure from consensus is a big thing. I spend a lot of time on consensus there. I clearly have an opinion on a consensus. So the way to counter this is, find disagreeable people who will tell you the

truth. Again, Weng is a good example. She'll just tell me when it's not a good idea, and then we fight. It's a lot of fun.

- 3.72** Look for people who can explain why they have a different opinion than you do. They can tell you the reason. They can tell you all of the individual pieces that lead them up to the decision that they've ultimately made or the advice that they're giving. They're not appealing to authority, all the things that we talked about earlier. That is the end of the how to conceptualize data and information. It doesn't look like we have anybody asleep, so maybe we accomplished the goal. Any questions?
- 3.73** Agreeable people encourage compromise, and it sounded as if you may disagree with that. I'm curious if you could color that a little bit because I am thinking that compromise is generally a good thing.
- 3.74** That's a good question. I think that compromise, it depends on the context. It really depends on the context. If you're negotiating a real estate deal or something, you definitely want to come to a consensus with the other party. It's going to require that you compromise some and they compromise some in order to negotiate a deal and get something done. So compromise has its place, and it's an important aspect of the way that human beings interact. Absolutely. I would not argue with that point.
- 3.75** But in terms of good decision-making, the process, within an organization, if you're making a decision, should we invest in this? Should we do that? Should we hire this person? Should we build a new manufacturing facility? Should we develop this new product? Whatever it is, what you want is to find the person, the single person, or maybe it can be a couple of people, with the highest level of true expertise in that area. Because they're going to have seen all of the mistakes that people make in the process of pursuing that particular thing. That's a very different thing. You're not looking for compromise or negotiation.
- 3.76** Because look, I don't mean this to be rude, but if we are trying to figure out a new marketing program, I'm not going to ask the scientists. They're terrible at marketing, right? So you don't want to ask for, we covered this earlier with the topic on allowing different worldviews. But you want to find the people that have true expertise. Let me see if I can differentiate that. It's the person who actually knows what they're talking about, and they can explain to you why they know what they're talking about, rather than, let's get five people that generally seem to be experts and see what the consensus is among those five people.
- 3.77** In World War II, the Allies were bombing Germany, and they would bring the planes back every night, and they would be shot up by the Germans. They would have holes in the wings and all kinds of different issues. They took all the planes, and they did a really complex statistical analysis, and they figured out where the holes in the planes were most commonly occurring, in different aspects of the wing and that kind of thing. They created a pattern of holes in the wings.

- 3.78** And all the statisticians got together and they said, "We need to create armor plating and cover the areas in the wings that are getting hit." And so they started working on that, and they wanted to cover wherever there were the most bullet holes. And one guy, it's just like the guy in NASA, one guy stood up and said, "No. What you want is exactly the opposite of that because the planes that didn't come back got shot in the areas where there are no bullet holes in these planes." A picture of this will help a little bit. We can go through it.
- 3.79** You want the person who actually really has figured out the honest-to-God underlying truth. Not, "Well, 100 statisticians are telling me this, and this one person over here says something completely different." It's actually true. It's not easy.
- 3.80** The one thing that you were mentioning before is that there's a graph that says one thing and then a graph that says another, and then there's the graph in the middle. It's like the lie, or there's a story, your story, my story, and then the truth. But who is behind even the graph in the middle that's saying, well, this, the spaghetti model, is the truth? If everything's being manipulated and there's lies everywhere, because to me, it's a lie. Whether it's in business, you're being shown information that's not truthful to make decisions for people that want to make more money—
- 3.81** I would say that if there are human beings, there's no human being on the face of the Earth who's not confirmation bias-driven. None. Zero. The reason I say that, there's no such thing as a robotic objective, completely objective human being who can absolutely process information and find the truth because, again, it's all being curated for you. So your personality perspective is influencing the way that information comes into you, and it's influencing it uniquely for you, in your worldview. And so that is true of all seven billion of us.
- 3.82** And so there's not one person out there who's absolutely got the certain underlying truth. And so the consequence of that, of those seven billion slightly different opinions, is that every single person has a desire to have a positive relationship with other people that is in part a consequence of not challenging the relationship too much. So that challenge to the relationship is, look, if you're doing something and I think it's wrong and it's a really bad idea, and I say, "Why would you do that? That's really stupid," you're not going to like me very much.
- 3.83** So I have to modify the way, I have to mollify the way that I approach the—so the consequence is, I'm not actually giving you my hard-core, really deep underlying opinion. I'm in part trying to maintain a relationship with you. So that influences the way that I convey information to you, which influences the way that it's conceived or perceived. So you just have to get, what you want to do here, I would say, is you want to understand that these things are happening, and you want to get better at perceiving that these processes are taking place, and the information is being curated for you. But there's no way to absolutely solve the problem and say, "Here's the truth. That's it." It's not that easy.

- 3.84** How do you remember all these things? You got to make a lot of decisions on a daily basis, and they seem very sensible, but it's when the situation comes to you, it's like, do you remember the lesson, in that moment?
- 3.85** Each of us has a different ability to perceive different types of information. And so the key is to find the thing that resonates with you as an individual person. Because then it's not challenging to remember things. Because if you end up working in an area where you're actually genuinely interested, it's like there's an exuberance to that. I love looking at new drugs and trying to figure out whether or not we can make something successful. A lot of people will be like, that sounds like the most boring, nerdy thing on the planet. But to me, it's really exciting.
- 3.86** And I spend a ton of time with the data and the statistics, and people are like, "Why are you doing this? Why are you digging in?" Because we're going to invest a lot of money in this. I want to understand if it's really going to work. I'm excited about it. And so I would say the key to remembering things is to do something that you want to remember. It's easy if you're doing something that resonates with your guitar string.

Lecture 4: Personality and Perception

At a glance — Personality and perception.

- 4.1 Today, we're going to go through ourselves and try to understand ourselves. Because to understand the self is to acknowledge the positives and negatives about how you perceive information and how you perceive others, and then how others perceive you. If you can become better at understanding those dynamics, it gives you the ability to absorb information more accurately from other people, and then communicate well in terms of how other people are going to perceive the way that you're communicating with them.
- 4.2 The best way to do it is to understand a basic construct that's been developed in psychology over time with millions of people. It's called the Big Five personality profile or Big Factor Five. Basically, through millions of different people, they've asked tons of questions without any preconceived notions in terms of trying to guess what it is that's going to drive individuals' behavior in the way that they see the world. They literally just ask tons and tons of questions to millions of different people. They found that there were patterns across different individuals that emerged from the data.
- 4.3 From that, they were able to refine it over time. This has existed for quite some time, to the point where the model got better and better and better. They were able to see more clear bubbles around certain types of questions that indicated some insight about the way that you view the world, and that there's variability in terms of different people and how they view the world. So the bubbles that emerged from all of these different questions over millions of people.
- 4.4 The fascinating thing about this model is it was developed cross-culturally. So everywhere in the world, these basic general personality parameters are really, really consistent. You don't see dramatic differences in terms of cultural—individuals that grow up in different cultures, each still fit within these bubbles of personality. And there are similar personality profiles, regardless of your cultural background.
- 4.5 The ultimate model that came from this is, the acronym is OCEAN, and it stands for openness, conscientiousness, extroversion, agreeableness, and neuroticism. So OCEAN is an easy way to remember it. And then each of those different basic essences of our personalities is broken down a little bit more in the model. We're going to go through this a little bit, and then we're going to have some fun talking about how different personalities—there's two things to really understand about your own personality.
- 4.6 One is how you perceive information in other people, and the other is how other people perceive you. If you can understand that, it's a wonderful tool for being able to navigate the world because it teaches you that—one of my favorite sayings is, "It doesn't matter what was said; what matters is what was heard." And if you can come to understand that the way that you say things and the way that you communicate is

going to be perceived completely differently by different people, it really helps you communicate better. You need to understand your audience at the end of the day. I'm going to learn lots about you guys here before too long, and maybe I'll completely change the way that I approach communicating with you. We'll see.

- 4.7 I've gone through some of this, but basically, it's just, it's word associations and general ability to differentiate personalities into these different constructs. It's existed for 50, 60 years, and it's become very, very popular. Most recently, Dr. Peterson has really emphasized the importance of this model in terms of understanding people. What we've tried to do within our company is familiarize ourselves with this construct so that we can assess the ways that different people within the company are going to perceive information and make sure that we're communicating accurately.
- 4.8 Let's start with, this is not in the OCEAN order, but let's start with conscientiousness. Conscientiousness is basically just getting stuff done, frankly. It's how much of an impulse you have to get up and attack your day. You know you're conscientious if—here's my ultimate sign of a highly conscientious person. They will make a checklist after they've done things in order to check the things off of the checklist, because they so enjoy checking things off of a checklist that they want to have the checklist even if they haven't made it before they do stuff.
- 4.9 So they'll literally go create a checklist after they've done a bunch of things and check them off. That's how you know you're extremely conscientious, is the post-doing stuff checklist checklist. So there's a spectrum. Each of these personalities, you're graded or assessed on a spectrum of zero to 100, in terms of where you fit as a percentage of the general population. If you're 10% in conscientiousness, 90% of the population is more conscientious than you are. If you're 90%, then you are 90% more, you're more conscientious than 90% of the population. That's the way to read the test. That same scale is used with each of the different personality factors.
- 4.10 Conscientiousness is broken down into two basic pieces, which is industriousness, which is, I think of industriousness as execution, moving towards goals, just basically getting tasks done throughout the day. And orderliness. Orderliness is more oriented towards cleanliness, basically. It's making your bed. When you use the dishes, you immediately wash them off and put them in the, you'd like for everything in your environment to be clean, neat, orderly all the time. You can easily recognize an orderly person because you walk into their office and everything on their desk is lined up, and there's no dust.
- 4.11 I particularly enjoy messing with orderly people by just walking into their office and just tilting something, just a little bit. And what you'll see is, subconsciously, without thinking about it, they'll reach out and straighten it. It's literally happening. It's like, that's out of alignment. They'll reach down and just straighten it without even realizing what's happening. That's how deeply built into our construct these types of things are. Just a list of general things that indicate high in conscientiousness. Pay attention to

details. You want to get it all just right. You make a really good editor. Your grammar is always perfect. You're always on time. Follow a very tight schedule.

- 4.12** Lower in conscientiousness. You're okay with a mess. You're not concerned if things aren't in perfect order. If you're late, you're more inclined to procrastinate. It's very tempting to take each of these personality traits and make judgments about them. Why would anybody want to be low in conscientiousness? It would be better if the world were neat and orderly, and everybody got things done on time and nobody's ever late. So why would you want to be low?
- 4.13** Well, you want to be low because if you spend all, we want to have people who are lower in conscientiousness in our environment, because if you spend all of your time straightening the world, it doesn't give you the opportunity to create art and music and read books. It constrains you to constantly being task-oriented. You know this because really conscientious people, they have a very hard time enjoying a vacation. They can't sit on the beach and just enjoy themselves. They're like, I got to go do something. I got to go to the gym. They're constantly, anxiously driven to get things done.
- 4.14** Agreeableness. This is a fun one. Agreeableness is, I like to think of it as how relationship-oriented you are. So more agreeable people are more compassionate. They have empathy towards others. They develop relationships far more easily. They extend trust very easily. They're basically polite, kind people in the way that they interact with the world. They will tolerate. If you make a mistake, they're more tolerant of your errors. There's a high relationship-oriented component of being high in agreeableness, which is very positive.
- 4.15** The negative of agreeableness is that you tend to—we talked about this a little bit yesterday—you tend to seek consensus over truth. So it can become more important to you to make sure everybody agrees about something than necessarily confronting someone with something that you know they're not going to want to hear. So it can be difficult. Necessary confrontation is difficult for agreeable people. They don't want to upset the apple cart, so to speak. They like for there to be consensus and happy environments where everybody's satisfied and having fun, which can be a bad thing in terms of—I'll give you a good example that I use with my kids sometimes.
- 4.16** It's that if you, you come out of college, and you're 25 years old, and you're working at a bank. And you're a teller, and you're doing a very good job. And you get promoted. Now you're a manager, and you have two tellers that are working for you. You have a very good relationship with these people because you were colleagues. You've got a friendship. Your friends go out on Wednesday night, and they go to the club, and they stay up till 3:00 in the morning, and they come in completely hungover on Thursday morning. And you're their boss.
- 4.17** A really agreeable person will say, don't worry about it. I got it. It's okay. I'll cover the counter for you while you sleep under the desk and hope nobody notices. A very

disagreeable person will say, if you ever do that again, you're fired. Now, that's a really big spectrum. Which is the right answer? It's hard, right? Because once you're somebody's boss, you have to be more disagreeable for them to respect you. Because if you are hyper-agreeable and you don't tune that a little bit, what's going to end up happening? Everybody in the room knows.

- 4.18** What happens with the tellers the next Wednesday? If you're just like, I got it. Don't worry about it. What happens? What do they do? They go partying again. Then it becomes Tuesday, Wednesday, Thursday. Who cares? My boss doesn't care. I can stay out all night. And so agreeableness tends to be too tolerant. I mean, it's a wonderful trait. You're loving and kind, and you help people. But if you let it go too far, you end up with tellers who go to the club every Tuesday night. You have to know this about yourself because it can be very frustrating if you are that person and you're that boss.
- 4.19** Because what can end up happening is, you ask—usually, what more agreeable people do is, rather than having a tough conversation I outlined a second ago, is they will ask nicely. Please don't do that again. It's really not very helpful to me if you stay out all night, but that doesn't work because it's not actually a threat. It's not powerful. It's not potent enough. It doesn't sink in.
- 4.20** Each of these things, again, they're tremendous positives and negatives of extroversion, basically two aspects of extroversion. One is enthusiasm, which is your exuberance, excitement about the world, just high energy. My enthusiasm is very high. You can tell I'm walking around up here. I'm hoping to get you guys excited about agreeableness and extroversion. And so your enthusiasm is your desire to explore, really, what you find exciting and fun. You get excited about different aspects of the world, and they tend to make you enthusiastic.
- 4.21** Assertiveness is your tendency to want to take control. If you're in a group of people and it's not clear who the leader is, you'll step up and say, Hey, I'll take this. I got it. Let's start here. Everybody take a piece of paper. We're going to approach this this way. You start answering questions, you ask questions. Assertiveness is your authoritarian tendency. We all know people who are high in extroversion. They're the ones who went out on Tuesday night. The tellers that went out on Tuesday night and stayed out to two o'clock in the morning. Introverts don't do that. They're not at the club till two o'clock.
- 4.22** Openness. There's two aspects to openness. Openness is the willingness to accept new information. And there are really basically two different types of information that dictate openness. One is information itself. So books, ideas, podcasts, discussions like this. That's the intellect side. And then the openness itself is the tendency to seek new experience in terms of art, music. It's more of the aesthetic side of openness. Intellect is books; aesthetic is art. It's basically the general construct, the way you can think about it.

- 4.23 Pursuit of ideas, constantly wanting to learn new things. Very, very curious. Open people are extremely curious. Why would you not want to be open? Well, what's the downside of being low in openness? Well, it's hard to be extremely high in openness and extremely high in conscientiousness. That's a real challenge. Because what ends up happening is you're constantly exposing yourself to new ideas, and then you want to go implement each of the new ideas and put them into practice.
- 4.24 Well, let me tell you, if you try to run a company, I have somewhat of that problem. If you try to run a company that way, you will drive everybody completely insane because you're changing your mind. You're mercurial. You're constantly trying to create new things and new drugs, and you'll just drive everybody crazy. And then you want everybody to follow up on it and get it all done. So what you'll do is, you'll create 100 different things and you will impose them on the organization, and nothing will get done.
- 4.25 There's a balance to each of these different aspects of who we really are. Neuroticism is basically sensitivity to negative emotions. So when something bad happens in your world, how much does it affect you emotionally? Is a way to conceptualize neuroticism. There are two aspects of it. One is volatility, which is, volatility is how much of a reaction you have. How quickly do you get angry or upset if something is relatively minor? If you go to your favorite coffee store in the morning and they don't have your flavor of coffee, does it ruin your day? Withdrawal is the tendency to go into your room and disappear from the world when something bad happens. So your reaction is to close yourself off from the world and be by yourself.
- 4.26 So let's have some fun. Different personality types perceive information differently. Let's start maybe with agreeableness. Who would like to, who's particularly high in agreeableness? We got an agreeable group over here. Who's particularly low? Is anybody particularly low? Oh, this is perfect. We'll have a fight [laughter]. We got the agreeable. We got the nice people over here and the mean people over here. It's wonderful.
- 4.27 All right, so, a good example would be, we're working on some new information that tells us something about the drug that we want to convey to doctors that is different than what they're used to seeing. So it's going to confront them and challenge their worldview. The question becomes, how do you effectively do that? If you're more disagreeable, what you're going to do is just tell the truth. Here's the truth. Now, the problem with that is, if I walk into a doctor's office and say, "That's wrong, here's the data, this is the right answer," what's going to happen?
- 4.28 You're a doctor. What are you going to think of me? Yeah, right. Okay, but I'm more on the, I'm a left side of the room guy. And so you have to learn this about yourself because you're going to interact with the world, foundationally, based on who you are. So your tendency is going to be to just tell what you think the truth really is. Here's the right data. That data is wrong. Here's what this says. It's clear. If you're agreeable,

you're going to be much more inclined to take a very soft, polite approach with that conversation and not confrontational. That's going to maintain the relationship much better.

- 4.29 These guys are going to screw up the relationship. They're going to make the doctor mad. The risk with being too agreeable is that the message doesn't come through. Because you soft-pedal it too much. Would anybody like to share a particular story where—a good example of what happens when you're too agreeable is that people will take advantage of you. They'll figure that out, and they'll do something to take advantage of the fact that you're too nice. Anybody got a good story that they would like to share?
- 4.30 When I first went to college, my first semester, I'd finished up classes and I could not find my car, where I parked. There was a gentleman who helped me to my car. He noticed I was young and naive and asked me for some money. I didn't have any cash, but he pressed me. I ended up giving him a ride to the ATM, gave him some money. He was able to leave me with his briefcase and said I'll pay you back tomorrow. He called me tomorrow. I met up with him, and he asked me for more money, and I gave him more money. I had this suspicion, like I was being hustled, but I didn't have the ability to object to it. And so went to the ATM the second day, gave him more money.
- 4.31 That's actually a perfect, that's an excellent example of exactly the problem. And the challenge that, the great part of that is that, look, you cannot navigate the world if you do not extend trust first, because if you distrust everyone, they will not trust you. This is the challenge that more cynical, more disagreeable people have, is they're less inclined to extend that trust initially. The trick that I think of when I have that, when I'm working hard to extend trust, is, I try to extend the trust and then watch carefully. If somebody does something that, and be honest with myself about whether or not I'm being taken advantage of, as honest as you possibly can be.
- 4.32 Again, when you're more agreeable, it will get extrapolated more. You have to learn to really be very honest with yourself about what's actually happening. It is a good example. How about on the disagreeable side? What's going to be more likely to happen is that you're in a classroom and you get an argument with one of your classmates or your professor or a girlfriend, and I say girlfriend because the biggest difference between men and women in terms of all of the Big Five personality profile is in agreeableness.
- 4.33 Women are about, they average about 60 on the scale, and men average about 40, which doesn't seem like that big of a difference. It's actually a huge difference. If you look at the 90th percentile of human beings, basically, almost all of them are women. If you look at the 10th percentile of agreeableness or disagreeableness, almost all of them are men. This is why 90% of the people in jail are men. It's pretty simple. They're more disagreeable. They're more willing to argue. Anybody got a good example of when being disagreeable got you in trouble and caused you problems in terms of

relationship because you were so focused on, so committed to your perspective and just stubborn? Being disagreeable, you can think of as stubborn.

- 4.34 I guess just building off of what he mentioned from college times. I had a course, American Presidency and History. By this point, I had my own business. I was doing well, so I wasn't always the most open-minded and just liked debate. Professor basically said how she thought Bill Clinton was the greatest president in US history, so I decided to do my year-end thesis on why Bill Clinton was the worst president in US history. She wasn't too happy about it. I ended up going back because I disagreed with her marks on it, but it definitely caused me some marks on that paper.
- 4.35 That's a great example on the other end of the spectrum, right? I think of it this way. When you're very disagreeable, you will sacrifice relationships and outcomes for what you believe to be the truth. I conceptualize it this way. My agreeableness score is low, but I conceptualize it as willingness to consider information that you don't agree with for the sake of a relationship. That's how I think of low agreeableness. The trick, in terms of navigating the world, you're going to have that proclivity no matter what you do. It's built into your personality. There's nothing you can do about your stubbornness to stick with what you believe to be true. But you can reasonably make judgments that writing a paper that is intentionally inflammatory for your professor is probably not a very good idea. It doesn't mean that you have to sacrifice the truth, or what you perceive as the truth, but it does mean that you can manage the way that you navigate the world and tone down your tendency to be inflammatory in terms of, well, I'm going to tell you what I really think.
- 4.36 Look, you do have to judge this. It's situational. It's context-dependent. Because if you are, well, we're dealing with the FDA, and I completely disagree with them on something, right? Well, how well is it going to go if I say, "You're a bunch of stupid idiots and you got this all wrong, and here's the right answer." Even if I'm completely right. I'm going to get a C minus on that paper, is what's going to happen, and we're not going to get a drug approved. You do want to work on the—being more agreeable, you want to work on the skill of protecting yourself and setting boundaries. You have to set boundaries. It's a good idea to set a boundary and define it.
- 4.37 If you help this guy out, and he comes back the next day, your boundary should be, I helped him out, but that's it. And he comes back the next day. Nope. And just practice that skill. Set a boundary and then stick with it. This is the thing that agreeable people struggle with, is to let the boundary move. They get talked into changing the boundary. On the disagreeableness side, you've set too firm boundaries. That you insist that's where the line is no matter what, even if it's not in your best interest to do so.
- 4.38 That's where you get in trouble. So you have to, again, these are—at the end of the day, what this is really all about is when—people on the left side of the room talk to people on the right side of the room—the opportunity for misunderstanding. This group is going to be more direct, and I don't mean to over. Are there any agreeable

people over here that want to move over? I don't want to be over here anymore. You're going to be more direct, you're going to be more straight to the point, and it's going to ruffle feathers.

- 4.39** Over here, you've got to set boundaries. You've got to try to focus on what you think is really the truth. You have to, I think, acknowledge the tendency to allow something to affect you for the sake of a relationship, even if you don't agree with it. That's going to be your tendency. So, neuroticism. Anybody particularly high in neuroticism? This is a fun one. All right. Anybody want to share crazy stories?
- 4.40** Whenever I feel like I'm impacted negatively by something, I just, alone, I just stay to myself. I think that it's very hard for me to express those things to other people.
- 4.41** That makes a lot of sense. That makes a lot of sense because, are you relatively high in agreeableness as well?
- 4.42** I'm 54.
- 4.43** Yeah. So that's still high enough. So the combination, somebody in your world does something that upsets you. Really, what you need to do is prioritize the confrontation because you have to correct it. If you defer that correction, if you go to your room and defer that correction, it'll ultimately cause you a really big problem because it'll compound. Because it does tend to make you more and more upset the longer you delay the necessary confrontation. You have to practice the confrontation, which gets hard, right?
- 4.44** Because if you are a little higher in neuroticism, then that confrontation itself will be really tough to manage, because it's going to upset you. If it's a spouse, set rules for the way that you're going to negotiate through a confrontation. One good rule, Dr. Peterson talks about this, I think it's a great rule, is, I'm going to tell you the way that your behavior is impacting me. If you would please tell me what you think I said, back to me, until I agree with your assessment. Then I'll pay the same favor to you. You tell me how you're perceiving this situation, and I'll rephrase it to you until we're in agreement.
- 4.45** Then, at least, you're talking on the same page. It doesn't matter whether this is a spouse or a friend or somebody, a colleague in a work environment. It's a good tool for being able to make sure that you're clear on the communication front. How about anybody that found out something about themselves on the Big Five personality profile that was a surprise, that you think actually isn't true?
- 4.46** Though I agree that I'm very nice and I do believe I'm able to set the boundaries, and maybe I just do it with more love than an aggressive way, or maybe that's not even the right word. So I was particularly, like, I'm not very agreeable. I don't agree with that. I feel like I'm the person—
- 4.47** You don't agree with your agreeableness score?

- 4.48 No. And I feel like I'm known. A lot of my friends and my family know that I'm the person that's going to challenge what you're saying. So for me, it was very surprising. Not the part, like I said, of being very trusting because I do believe in trusting until you're proven not to be trusted, and being loving and kind and compassionate. But I also know, and I don't know if it's maybe something that I've learned or the balance that I create through my own practices, but boundaries—
- 4.49 Is this something that you have thought about before and intentionally set boundaries so that it is more manageable? You do this naturally.
- 4.50 No. I feel like I've always been the person that challenges. Maybe what I have learned has been to be loving and kind and compassionate because that wasn't something that came very natural to me. So maybe I was more disagreeable at some point in my life.
- 4.51 Why do you think that happened?
- 4.52 I think because the way that I delivered my message was always very harsh and hurtful. Through my own personal development, I had to develop a way to communicate with others in a loving way where they didn't feel diminished or less than me, or I was the only one that was right, or even in the whole leadership so that they were able to feel comfortable enough to come to me in openness and make the mistakes.
- 4.53 That's a really interesting, thoughtful perspective, because I think what it appears that you've done, here's the point of this whole exercise, is that if you can understand yourself— you can't completely change who you are. You can't change your fundamental reactions. It's like how we were talking about information and jumping to conclusions with system one versus system two. You can understand your proclivity to react a certain way and, with intentionality, tweak that to some degree.
- 4.54 I think that's the positive that comes out of an exercise like this. You can't change who you are, but you can understand who you are and understand how you're going to react. At least maybe, if you're fundamentally more disagreeable, you take your time to consider how you're going to convey the information to others because you realize that your initial reaction is maybe too confrontational. That's a good thing. I don't know if that's an accurate description, but it's a good thing.
- 4.55 Yeah, it makes sense. It feels like it's the practices that I have set for myself to be able to be compassionate and loving and understand people. Any other surprises?
- 4.56 I was pretty high in extroversion and really low in neuroticism. I expected both to be dialed in the opposite direction. But it's not very surprising to the extent that I had made an observation because I was taking a stimulant medication for personality disorder, ADHD. So I stopped taking the medication because I no longer had a need for it. The last time I had taken a test like this, I had seen exactly that, lower in extroversion and higher in neuroticism. On the flip side, now that I've stopped taking it, I see that

both of those metrics have changed. That was an observation I had made myself. And then I was pleasantly surprised to see that the score reflected that.

4.57 You've got three different tests, before, during, and after?

4.58 There were two personality tests. One before stopping, one after.

4.59 I don't know enough about the literature on how various CNS or central nervous system-acting agents can affect scores, but it would be a fascinating analysis to go through because I suspect you're bringing up a point that is fundamentally true. The only literature I know obliquely is that they did some work with psilocybin mushrooms, I believe, and found that they actually can have a big impact on openness.

4.60 They can make people far more open. I'm not recommending that. I'm not saying everybody go do ayahuasca or anything. But it's a fascinating question that's probably fundamentally true. I mean, you think about alcohol, caffeine, all of these things are probably impacting, well, definitely they're impacting our nervous systems, and in all likelihood, hyperbolizing individual aspects of our personality. Any broader questions or any other particular dynamics?

4.61 I have a question about the methodology and then just in your experience or in your utility. Have you found that there is a correlation or a discrepancy between self-reported scores and external representations? Just in terms of the utility of this kind of a metric, I wonder if there's a high relationship, because if you don't know yourself, maybe your perceptions can be out of sync.

4.62 That's a really good question. That's a really good question. We actually use a little bit more sophisticated version of this type of test within our company. It's called the Predictive Index. It does draw exactly the distinction that you're highlighting there. The distinction that it draws is between self-concept and basically external concept. It's fascinating. It's the point we were making a minute ago, about how you've maybe adjusted for your disagreeableness, is that you do have two different sides of the self.

4.63 One is the deep, real way that you conceive of who you are in the world and who you are in your most natural state and most natural setting. Then basically, the other side of it is the persona that you present to the world. The most clear distinction is going to be in a work environment because part of what happens in a work environment is that you realize that you need, or people at least believe, they need to present a certain aspect of themselves to their boss or their colleagues or whomever it is, that may not be exactly who they really are.

4.64 It's a great question. We do absolutely make these modifications. But it does, I think, the challenge—and it's good to look up the Predictive Index when you when you have time, sometime, because the challenge is that, if you are trying to do something in the world that is vastly different than your basic personality, and the way that you can tell

this is when there are very big differences between the way you perceive yourself and then the persona that you present to the world.

- 4.65 If there are really big differences there and you're not being your true self, you're not doing the right thing. You're doing something that's actually generating a lot of deep angst because you're trying to constantly present yourself to the world in a way that it's just fundamentally not who you really are. Very good question. Yeah, that's a good one.
- 4.66 All right, so just a wrap-up. Agreeable people form relationships easily. They seek consensus, even if it's wrong. That's an important thing to know. Even if it's not the truth, consensus can be prioritized over the truth. You want to be aware that you can fall into that trap. Disagreeable people tend to do exactly the opposite. They will seek what they believe to be the truth, even if it damages relationships.
- 4.67 More neurotic can be absolutely fantastic in compliance, highly focused on rule-following and making sure that everything's done really, really well, avoiding getting in trouble. But they can also avoid risks. Sometimes we need to take risks in life in order to grow a business. You've got to be entrepreneurial. You've got to do things that are risk-taking. An example— I think the overriding thing here is to try to understand that, if you're more neurotic, and you think that your boss is doing something that's risky, you want to take a step back and try to understand whether or not, in the grand scheme, your perspective is not necessarily the most accurate one.
- 4.68 You want to try to understand that you're perceiving something as really highly risky, and why is he or she doing that, when it's actually maybe a good decision for the business, even though you don't necessarily agree with it. Try to step outside of your personality. In order to do that, you have to understand who you really are. Extroverted people are salespeople. They get out and engage with the world. They're excited about things, but they can be overconfident, really easily. They can take charge and take an authoritative stance on things over-exuberantly. What I mean by that is, to make a decision because you want to be the person in charge, even if there's not necessarily a really good formation or reason for that particular decision.
- 4.69 Open people are creative, but can be very inconsistent. They can jump from idea to idea and change their minds and drive everybody crazy. Conscientious people can also drive everybody crazy by being really demanding. If you have a very conscientious boss, they will set unrealistic deadlines, and they will expect everybody in the organization to be as focused on the task at hand as they are. And they can be tyrannical, right? Because they want to get everything on the checklist done as fast as they possibly can, and not everybody is wired that way. And so they can struggle with relationships.
- 4.70 On the other hand, if you have really conscientious people in your organization, you can get a lot of stuff done. It's extraordinary. They will execute. So, positives and negatives of each. And I think that is—any other questions, anything you want to discuss? One more. WWe don't have a manual for life. These are ways of us to learn

on our own how to better communicate, how to better connect to others. But then we have to depend on others to also do the same things, in order for it to actually work. If we don't have a manual for life and we just go to school to learn certain topics or about laws, pretty much is what we're educated on, we're basing our experiences and our behavior on what we've experienced in life.

- 4.71 Everyone has a different perspective and a different life. We all have that primitive tribal need to belong in a tribe. That's where a lot of disagreeableness comes in because you don't want to rock the boat or else the tribe might ostracize you. Is there a way, in your opinion, that we can create a system? There are laws for rules and things like that, that we can create a basic system where everyone can—
- 4.72 Negotiate this?
- 4.73 Where everyone can take a test when they're young and as they grow, to identify their basic personality type, and then refine it to where it can interact correctly with another personality type on the same basis and rules. That would then keep the society in a flowing way where there's no crime, where there's no authority figures that are abusing their power with their underlings, where there's the person that's insecure and traumatized that can't belong. I'm trying to think about— we're being taught what it is, but can we be taught actually how to do it as a society? I know that's a Brave New World-type of thing, but not like that it's exercised as a law, but more of, in school we're going to learn this or in society, and we're all going to cooperate with this system.
- 4.74 Can we create a utopia? Look, I think the closest you can come, the fundamental problem, there's a challenge, there's a tension in life that transcends the spectrum of different personalities, that I don't think you can oversimplify to one basic construct other than free speech. I think free speech is the mechanism that ultimately can resolve the tension. I think it's the only mechanism that will resolve it. Because if we don't have the primacy of free speech, you end up in a situation where, if—
- 4.75 Society creates the rules. What if the rules are wrong? Because we do get it wrong. Look at Nazi Germany. Look at Solzhenitsyn did a very good job of describing communist Russia. So we get it very, very wrong. And so when society gets the rules wrong, the only way to resolve that is to have very disagreeable people who otherwise aren't going to, who are, by nature, inclined to challenge the rules. And you need those people. They can be a pain because they're constantly challenging things and arguing, and they make great lawyers. They love to argue.
- 4.76 But you have to have that because if you don't have that independent fundamental— we talked briefly about this with the airplane and World War II and the one engineer. The 50 engineers who say we need to cover these bullet holes and the one engineer who's actually right, says no. And you have to be disagreeable to be that person. It's the

only way to do it because otherwise you're not going to stand up and say, you're all wrong. Here's the right answer.

- 4.77 No, I'm not implying that everyone should be the same, like vanilla, like a controlled society. No. What I'm saying is, let's say you are very disagreeable and confrontational, and you like the truth. You don't like to sugarcoat it because of someone's bruised ego and trauma and problems and agreeableness. But what I'm saying is, how about we all learn, very clearly, about everyone's fundamental self, so that there's no more being offended and vengeance and I'm being discriminated or this person doesn't like me because I'm a woman. It's to understand really where this person is coming from and to not allow the abuse of power. I'm not saying that the person can't speak up.
- 4.78 So here's the fundamental challenge with your question. The reason it's so difficult is because part of what's built into your worldview and the construct—and I don't mean you, I mean the general you, our worldviews—is that different types of personalities fundamentally, absolutely struggle to understand the worldview of a person who's completely different. Why don't we just get people to understand each other? It's like, well, that would be great. We've been working on it for roughly four million years, and it's not easy.
- 4.79 I think the extreme of the "everybody get along and follow the rules" is ultimately tyranny. Because if the rules tend to tilt towards corruption in larger and larger organizations, just "everybody do what I say." Somebody at the top ultimately says, everybody do what I say. I think on the other side of the equation, if you get too disagreeable and too individualistic, then you tend to go to war. I'm right; you're all wrong. We've been balancing between tyranny and war forever.
- 4.80 I think if you took somebody who was extremely disagreeable and put them in a room with somebody who's extremely agreeable, there's almost no way for the two of them to understand each other's worldview because the agreeable person is going, why are you such a stubborn idiot? Why can't you just change your worldview a little bit and come along and modify it? Just for the sake of consensus and group cohesion and good social, why are you breaking the rules? And the really disagreeable person goes, why are you trying to tell me what to do?
- 4.81 I don't know how you solve those, honestly. If you have an idea, I would love to hear it. But I don't know how you reconcile that, other than to let the group of all of us have a social dynamic which encourages us to talk about it. And that's why I go back to free speech. It's the only mechanism. One over here we got.
- 4.82 I would say that one of the ways that, as you mentioned yesterday, is humility.
- 4.83 Yeah, that's a good idea. Yeah, that's a very good point. I think that helps a lot.
- 4.84 Which is more of a religious type of view in a way, too.

- 4.85 Humility and forgiveness. I don't mean to be too Christian in the point, but if you're just going to analyze Christianity from a secular perspective, it does prioritize the right things in terms of answering your question, which is humility, which is, you're not so great and you might be wrong, and forgiveness. I'm not so great, and I'll make mistakes, and you should forgive me when I make mistakes, right? If you can do those two things, it actually tilts us towards a functional society.
- 4.86 I think humility is a good answer because if you can have that construct within your worldview, and if you can continue to reinforce it, it does set a mechanism for you to be able to more carefully consider another person's perspective. And I think, particularly if you're disagreeable. So, it's good advice for me. I work on humility because I am more disagreeable. And so I realize that my proclivity, and I'm assertive, and I'm the boss. And so, it's a struggle for people to challenge me. And I know my stuff. I don't make decisions without thoroughly understanding the underlying data.
- 4.87 So if you're going to challenge me, you better be prepared. You better know what you're talking about. That could be really intimidating. It could be extraordinarily intimidating. If I'm going to create an open environment where people are willing to express their worldview, I have to calm that rabid dog down a lot. But I know that, and I know I have to do it. Humility is the fundamental basis of that decision to intentionally not be confrontational. Good point. It just wanted to touch on, I feel like a lot of humans get caught up on, like, the world should be perfect and there should be no war. We wouldn't evolve if there was no problems. We wouldn't evolve if we wouldn't find disagreements, and we wouldn't look for creation or innovation if we didn't find problems. I think what's great about this is just it gives you the insight to know yourself and, just like how I told you about my disagreeable part and how I brought some of my own compassion and understanding people. Because before for me, it was like, take responsibility for your life and change your life. It was very like, now I can show more love and affection and all that.
- 4.88 But I feel like this gives us an insight to work on ourselves, to be able to, like you said, we've been working on ourselves as a human race, just to be able to connect with others. That's the work. It wasn't easy for me to start being more agreeable and be more nice and seek compassion and understand why you went through pain through your life and you have trauma. That was hard for me. I had to learn that. And setting boundaries, too. It's not easy to say no. It's not easy to set boundaries. You have to learn that, and that's the work. And I feel like a lot of people are just looking for an easy way or like, oh, take my five-day course. Everything's going to be, no. That's a journey for life.
- 4.89 I think you're making a really good underlying point. I'm going to rephrase it and tell me if I got it right. We're going to practice the friendship mechanism. So I do, fundamentally, I would hypothesize that the reason for this distribution of different personalities is evolutionary. Richard Dawkins would completely disagree with this

point, but so be it. I think that if you conceptualize human history, human evolutionary history, as the ability to be successful from a fitness and reproduction perspective, in the world, and competing with other animals, etc., then a distribution of different personality profiles is actually really advantageous.

- 4.90 Because if you have a very disagreeable person, they're going to go hunt and they're going to kill things, and it's not going to bother them from an empathy perspective, to kill. Well, we need to get food, right? And so you want that. If you have a person who's very high in neuroticism, they're going to be more likely to see the attack from a tiger that's coming at the group than anybody else. They're going to be more sensitive to potential threat. You want that. You want that alarm bell going off. And so if you're more agreeable, you do a much better job of taking care of children, way more empathetic with, and children are extremely demanding.
- 4.91 They're basically like, take care of my every need for the first two years, more or less. I don't care what you're doing. I need this right now. And so, that requires an extraordinary amount of agreeableness. And so we really need this distribution. It's advantageous for us in terms of navigating the world as a social group, as a pack, so to speak, as a tribe. You used the word earlier. And so that's a good thing. YYeah, it was perfect. I think even in business, like, how you were saying, something that I do with my sister in business, we're complete opposites, and she brings a perspective that I never even see when I bring up my perspective.
- 4.92 Absolutely, the introverted, highly conscientious scientists who are in the lab and they're willing to pipette and test multiple different antibodies until they find the right answer over and over and over again. Well, if I take a really extroverted, highly relationship-oriented salesperson and say, I need you to sit in the lab for two weeks and pipette for 10 hours a day, how long will they last? Yeah, exactly three minutes. You take the scientist and put them in the field and say, I want you to get up every single morning and go out into the field, and I want you to go to these conferences, and I want you to talk to 50 different doctors when you go to the conference. It's not going to work.
- 4.93 It's the same dynamic in today's world. You need all of these different people to run a company well, frankly. Because look, an introverted rule-following accountant is a really good thing to have in the organization, but they're not going to be a good salesperson. They know that, and you know that. And we do tend to navigate the world. And for the most part, I think we do a pretty good job of ending up where we should be, even without understanding the Big Five personality profile. But I think this is an advantage, the standpoint that it makes it a little bit more explicit and a little bit more clear. Yeah, good question. Good point.

Lecture 5: Fundamentals of Statistics I

At a glance — Fundamentals of statistics I.

- 5.1** The next lecture is going to be on the most exciting thing that any of us have talked about in quite some time, which is data and probability and statistics. And I think I described earlier that the defining characteristic to look for in someone who's a true subject matter expert is whether or not they can take something that's complex and distill it down and de-jargon it and explain it in a way that's more readily accessible to people who aren't experts in that area. And so I've challenged myself with exactly that task. And so we're going to practice a couple of things.
- 5.2** One, I'm going to practice doing this and explaining some pretty complex things. And as we go through this, it's going to get more and more and more complex to the point where we're probably all going to be exhausted. And so what I would like for you guys to do is exercise the advice that I gave yesterday on asking. I don't want to get too distracted on questions. But if you would just raise your hand if there's something that's just completely, sounds like Greek, and I'll take a step back and see if I can, help me get better at my job, which is to try to explain these very complex things. You have to tell me if I'm doing a bad job because there's no other way for me to know it. You have to communicate with me.
- 5.3** If you're going to run a business, if you're going to run any complex business, these three things are pretty much essential: accounting, statistics, Microsoft Excel. And what I mean by these, like, why accounting? Why do you need to know accounting? Because accounting is the language of business. If you don't understand the accounting, you do not know what is going on in your organization. You cannot run it well if you do not know where the money is going and where the money is coming from. Because those are the leading indicators of the decisions that you need to make in order to run the business well. You have to have a basic fundamental understanding of accounting, because otherwise, the money will disappear. Because you're not keeping track of it. You don't know how to understand what's happening.
- 5.4** Statistics. This is just the world that we're in. Everything is getting more and more complex and complicated, and we're exposed to more and more data from a marketing perspective, from a social media perspective, etc. So, you have to understand how to manage information using data in order to make good decisions. Now, different businesses are more or less complex. In the pharmaceutical world, it is the top end of the spectrum in terms of being able to understand statistics well.
- 5.5** But I would say, look, if you want to manage your business and you want to understand what's going on, you should have a basic fundamental concept—I want to stress the word concept— rather than the ability to actually calculate a standard deviation. You want to know what it means and be able to interpret it, not necessarily have to do the

math. We're not going to go through how to do the math exactly. We're going to go through what these things mean and what they inform you of, in terms of a decision-making process.

- 5.6** Microsoft Excel is a great tool for being able to analyze both these things. I would learn this stuff. Data is foggy. This is why we need to understand it. It's a probabilistic distribution of uncertainties. It's like, what does this mean? You roll the dice, you get a seven. What are the odds you're going to get a seven? What does it really mean? You need to understand how these things impact your world, because if you don't understand how to properly assess the information that is coming to you from a data perspective, you will make bad decisions. You will be too easily misled into the wrong direction. Because if you over-rely on the data, you'll make mistakes. So you need to understand what it's really telling you.
- 5.7** Here's the fundamental mistake that almost all of us make with data. They think, we've got a whole bunch of fancy words, and we've got sigmas, and t-distributions, and blah, blah, blah. This guy must know what he's talking about, and he gives me the answer to all of these complex numbers on the page. And, therefore, I can have a higher level of confidence that what I'm being advised about is true because it's all complex and there's a whole bunch of numbers and math. That's a mistake.
- 5.8** You do not—the over-analyzing things and cutting and slicing the data many, many, many different ways does not increase the accuracy of what you find out. In fact, it's the opposite. The more you look at something, the higher the likelihood a small signal is going to mislead you. So we're going to go through how that happens and how to be aware of that trap.
- 5.9** So I just touched on this, but you do something many, many, many times. Data, probability, statistics can give you a distribution of what the most likely results are going to be. But likely is the key word in this sentence. Likely means more probable. It means, that's the thing that's most likely going to happen. That does not mean that's the thing that is going to happen. Data is not about certainty, and this is apparently my favorite word. You want to be cautious about certainty because you're not looking for the answer. You're looking for the best answer and distribution of possible answers.
- 5.10** But you need to know. The reason this is so important is because, once you decide to seek certainty, you stop accepting new information. That's the key catastrophe that can occur. And you want to continue to be able to accept new information because probability says, well, I've made this decision, but I might be wrong. So I should revisit somewhat regularly when new information comes along that tells me that I was wrong. We start running a clinical trial, we get halfway through, we get enough data to take a look at how it's going, and the drug is not working at all. There's no signal whatsoever. You don't want to go, well, I still think it might work. You want to intelligently second-guess yourself.

- 5.11** So when you roll a dice, here is a normal distribution of the outcomes. These are two dice. The reason Vegas wins when you roll a seven is because it's the best odds. It's the way to think about this. Normal distribution, standard deviation, blah, blah, blah. Here's the fundamental math. It's actually not as complex as they make it. Okay?
- 5.12** The math is, you roll two dice, right? Okay, so roll one dice. You roll one dice. You have six different possibilities. One, two, three, four, five, six. You roll two dice, it's six times six. You have 36 different possibilities. You can get a two, a three, four, five, all the way up to 36. The likelihood that each number is going to occur is this math. It's the total number of times that that number can appear. Let's take something like a two, snake eyes. There's only one possibility of getting a two. It's a one and a one. So you've got one possibility out of 36 possibilities. It's one divided by 36, which is 2.78%.
- 5.13** All right, let's take a three. Where you can get a one and a two, or a two and a one. So there are two possibilities of a three. So it's two divided by 36. Does this make sense? Raise and let me know if it doesn't. Can we study all of this stuff and plug it all into a computer and have it tell us an answer? Well, people have been trying to do this with the stock market for as long as the stock market has existed. Here's a distribution of 10-year returns over the last 80 years: 4.7% over a 10-year period, 4.3, 17.7.
- 5.14** What's the outcome if you invest in the stock market right now? What's the return going to be for the next 10 years? Is it going to be 11.7 or 17 or 9.5 or 4.7? This is the problem. Somebody who comes and tells you, "We've got this fancy model and we've done all this work. It's going to be this," they're wrong. Because there's no way that you can take this variance and distill it down into a clear answer. It's impossible.
- 5.15** The key to all of this is to be comfortable with the uncertainty. You can model it out and say, well, I think it's going to be somewhere in this range. Somebody who says, I think it's going to be somewhere in this range, and they introduce the uncertainty to you, they're trying to tell you the truth. They're honest about what they think the outcome is going to be. It's just a good example of the world is very difficult to predict because it's complex. You've got interest rates going up and down. You've got potential war in Russia. You've got all of these things that have a huge impact on the stock market that you cannot predict. That's why this happens. So the fundamental thing to consider about data conceptually, it's almost always misinterpreted and either intentionally or unintentionally manipulated. We saw this the other day with a spaghetti graph. You're getting to see the things that whoever is doing the basic analysis wants you to see. Now, that may not be nefarious. It may not be ill intent. A salesperson, a marketing person, is going to come and tell you the things that are positive about their marketing program, of course.
- 5.16** They're not going to come and say, we did this great program. It cost a million dollars, and it didn't work, and it was a complete catastrophe. That's never going to happen, especially from a marketing person. It's not going to happen. So you're always going to see curated data to some degree or another, in almost every respect. A scientist who

runs an experiment that they have been working on for their entire life, who gets an outcome that is not the answer that they want to get, what are they going to do?

- 5.17** Are they going to just completely stop and throw out 10 years of work? No, they're going to do the experiment again. And they're going to do the experiment again, and they're going to tweak it, and they're going to do it again, until they get the answer that they want. This is not evil. It's not lying. It is human nature to seek the goal that we want. If your goal is to demonstrate that this particular drug works, it's completely natural to try to work on something until you find an answer that is what you want. This is just fundamental human nature.
- 5.18** But the key, you can't get rid of that aspect of who we are. But what you can do is try to understand it and try to acknowledge when it's happening. The basic concept here is to be somewhat cynical. We're going to go through a real-world example of what we call the activation rate within our company. This is data that we have access to internally. Activation rate is the number of prescriptions. We have a drug that we sell in the market. And the salespeople go out, and they try to find new patients and get new prescriptions. And the new prescriptions come in, and they don't always get processed.
- 5.19** There are all kinds of things that can happen. The insurance company denies it, or the patient ends up deciding they don't really want it, or the doctor doesn't follow through with all of the prior authorizations. You know this world. You know all of the complexities that can occur with trying to get a prescription filled. Two years ago, roughly, we changed what's called a specialty pharmacy. Kind of think of it as like an Amazon. It's not CVS. It's a central warehouse. The details don't matter, but we changed.
- 5.20** Our average time from when a prescription came in to when the shipment goes out to the patient doubled, from 25 days to 50 days. Our sales and marketing, our commercial team, appropriately freaked out. Our pharmacy, they get paid on prescriptions. So ultimately, the commission of salespeople tells you where they focus, as it should. The consequence of this is, this stressed everybody out tremendously.
- 5.21** Now, here's the fundamental problem, is, you have to look at something like this and go, is this true? It's data. It's thousands of different data points all over the country. Is it real? You have to seriously second-guess things. Because, for example, just a reasonable question to ask is, what is the likelihood that the average time between when a prescription comes in and when it ships out just doubles? That's a huge difference. Is this really true? I guess the first lesson would be, don't just take this thing on the spreadsheet and believe it's true, just because there's a bunch of numbers and fancy analysis. You want to go to your intuition and second-guess it and go, does this really make sense?
- 5.22** So don't abandon your intuition. It can tell you things that all the magic numbers don't. Is this really true? No, it wasn't. It's not true. In fact, what happened is the new patient

shipments were actually up 30% during this period. New patients are going up, but it's taking twice as long to process each patient. How is that possible? The fundamental answer to that question is, it's not. It's not physically possible to be doing a worse job of processing each patient and have more of them, with the same system, with essentially the same number of people processing things, etc. It can't happen. What actually was going on is that the way the new specialty pharmacy processed the information was completely different. So when a fax came in from a doctor's office that was a new prescription, they immediately recorded it as a prescription. But the old pharmacy did not record it until after they had gotten all of the information they needed in order to be able to process it, prior authorization or medical necessity or whatever it is.. So they were recording the start date of when the clock starts differently.

- 5.23** So it wasn't anything to do with any actual change. It was just accounting for things differently. It took us a while to figure this out, but it's a good example of, this is telling me something that doesn't actually make sense. I need to appropriately second-guess it and see if I can find out the underlying real truth. So, ask the obvious dumb question. The data seems weird, second-guess it.
- 5.24** A true analytical thinker will do this. It's the person who will raise their eyebrow and go. They'll focus on the priority. They'll focus on the thing that really matters. These are just signals of people who are really trying to get to the fundamental core, underlying answer. They don't get distracted by a whole bunch of different variables. That's key. Someone who, once something seems weird, they go lateral and they start bringing in more and more and more information from different areas that aren't relevant. It just makes it more complex and more confusing.
- 5.25** What you want to do is focus. The true analytical thinker will focus on the thing that is most clearly and explicitly the priority. They can explain it to you clearly. They can tell you why they think it's the priority. Again, it's that simplification process, and they can explain the sequence of how they got to the decision to prioritize that particular thing. This is key, too. They're mercurial. Mercurial means willingness to change their mind. They will change direction when presented with new information. These are the things that you're looking for.
- 5.26** All right, so this is right here in the middle of this probability lecture, some life advice. This is the way that I make decisions, which has worked out pretty well so far, that is much more reliant on intuitive, basic, high-level concepts than a whole bunch of analytical research into things. And that is this. I'll take a drug as an example because it's more familiar to me. If you're in the pharmaceutical business and you're presented with a new idea by a smaller company and they want you to invest in it, if it does not work, if that drug isn't actually doing what they think it's supposed to do, it'll be obvious really quickly. The data won't separate. It won't be really, really clear.
- 5.27** For example, if the placebo response is 30%, and the response in the drug is 38%, that's not good enough. It's not like, blow you away, unbelievable. So if it doesn't work, it's

not obvious. If it works, it's obvious. So it's going to be like the—it's going to be really, really clear. You don't even need a spreadsheet. Response in the patient population with the drug is 70% and placebo is 30%. I don't need to do any fancy math. It's really clear. If it's in the middle, it's 42%. It's a little bit better than the 38%, but here's the thing. If it's in the middle and it's mediocre and it's a maybe, possibly, might, could be, if we do enough work and pencil-sharpening and digging, it doesn't work.

- 5.28** If you have to work hard to convince yourself that something is going to be a good investment and crunch numbers over and over again, stop. You don't need to do all that. Warren Buffett says it this way. He says, if I can't figure it out on the back of an envelope with two or three calculations, I don't do it. Don't bring me 500 spreadsheets of data. If you need 500 spreadsheets of data, you've already told yourself the answer. At a high level, you want to look for things in life that are really obvious and just blow you away.
- 5.29** A practical analysis of some statistics here. We'll have some fun. Fun with statistics. Are men on average taller than women? I've picked this particular topic on purpose because intuition tells us the answer. Who in the room doesn't know the answer to that? It's obvious. But how do we prove this mathematically? The reason I'm doing it this way is because, if you don't know something, if I ask you, is this average symptom improvement with this drug, with a standard deviation of this, different than what this placebo, blah, blah, blah, average standard?
- 5.30** I don't know what you're talking about. I don't know the answer to that. But in this case, you do know the answer. You actually know the answer. Now, here's the danger with both of those things, is that a lot of times you're looking at things in life where it's not intuitive. It's not obvious. You don't actually know that there's a difference in height between one group and another group. But you can make the same data mistake if you over-rely on the data. I'm intentionally picking an example here where we all know the answer, but we're going to go through the mathematical way of differentiating these two.
- 5.31** There are two important points that we need to decide whether or not there's a difference in height between men and women. One is the average. The average is the midpoint of a group. It's the single number that best represents the entire group. It's the average of all the different heights of women. The standard deviation, the de-jargoned version of the standard deviation, is it's the "spread-out-ed-ness" of the data. So, think of it as the average distance from the average. I'm going to go through an example here.
- 5.32** Look at the bullet points down towards the bottom. So a tight distribution with a small standard deviation. Let's say we want to know the average between one woman who's 5 foot 5, one is 5 foot 7. We're going to probably all do that math, right? The average is 5 foot 6. The standard deviation is 1. So one woman is 5 foot 5, and the average is 5

foot 6, so she's one inch away from the average. The other one is 5 foot 7 and the average is 5 foot 6, so she's 1 inch away. So the average of 1 and 1 is 1.

- 5.33** Now, the reason this is important, and the reason you need to understand what deviations really mean, is because they tell you something very fundamental about the data that the average does not tell you. If you take one woman who's 5 feet tall and another one who's 6 feet tall, they average 5 foot 6. Is there a difference between 5 foot 5 and 5 foot 7 that averages 5 foot 6 and 5 feet and 6 feet that averages 5 foot 6? Obviously, really big difference. The standard deviation tells you the difference between those two groups.
- 5.34** A woman who's 5 feet tall, and the average is 5 foot 6, she's 6 inches away from the average. Same thing for the woman who's 6. On average, those two are 6 inches away from the average. We good? Raise hands. No? Let me see if I can rephrase this. Another way to think about it is how different, if you take a group of people, how different are they from each other? If you take the whole group, you want to get an idea of how much difference there is in the whole entire group. So the more difference there is in the group, the bigger the what's called variance. So variance is a measure of different-ness.
- 5.35** Why is this useful? Because most data are normally distributed, which is a "jargony" word. I think that one of the things to understand about normal distribution, rather than going into central limit theorems and why this is the case, I think the simple gist is, it's like gravity. Almost everything in life is normally distributed.
- 5.36** If you want to be a data analytics person, or at least conceptually understand these things, you need to fall in love with the normal distribution and understand what it means, because everything that you look at, especially in biology, is going to be normally distributed. There's going to be an average of blood pressure, and a standard deviation. I don't know what the average systolic blood pressure might be, 120 or something, maybe something like that. Then there's going to be a standard deviation of 20. Most people are going to be within 100 and 140. The average is 120. The standard deviation is 20 away from 120, so that's from 100 to 140.
- 5.37** You see it everywhere. It's unbelievably prolific. It's a more certain law than gravity, actually. But if you roll two dice, a normal distribution is a bell curve. We've all heard about the bell curve, right? So you can see this looks just like a bell curve. There's a peak in the middle. The thing that's in the middle is the thing that is most likely to happen. The things that are out on the edge are the things that are least likely to happen.
- 5.38** So, a seven—there are six different possibilities for rolling a seven if you roll two dice. There's one possibility of a two, which is snake eyes. So this is a way of comparing the possibilities of different things happening. Again, with the highest possibility right in the middle. When the distribution is normal, the standard deviation can tell you some

fundamental things about the underlying distribution. Because of the way that the math works, it actually tells you how, the standard deviation tells you how packed together the data is or how spread out it is.

- 5.39** I'm going to go to a picture here because a picture will help even more than the math in terms of understanding this. So the average is this point right here in the middle. The standard deviation is the width of the bell curve. So let's use women one more time. If every single woman were between 5 foot 4 and 5 foot 6, and the average is 5 foot 5, and everybody is between 5 foot 4 and 5 foot 6, then the standard deviation is 1, and the average is 5 foot 5. Now, in reality, it's more spread out than that. We know some women who are 5 feet tall and some, a fair number, who are 5 foot 6, 5 foot 7.
- 5.40** So what that means is there's a broader standard deviation. There's a broader distribution of people who are farther away from the average. Their height is not in the middle. And the standard deviation can tell you the percentage that are close to the middle. In the real world, in the US, women average 5 foot 4.5 with a standard deviation of 2.5. What that means is that 68% are between, you want to take the middle, which is the average, which is 4.5, and subtract 2.5, so that's 2 on the low end, and add 2.5, which is 7 on the high end.
- 5.41** All right, here's the cool thing about this tool. What it does is it allows you to predict the possibilities of different things happening. Once you know enough about a population, I can tell you the likelihood of different occurrences. I can tell you how likely it is to get snake eyes. It's 1 in 36. I can tell you how likely it is to find a woman who's 6 feet tall. It's 3 standard deviations away from the average. It's a fancy way of saying there's a 1 in 300 chance of being 6 feet tall if you're a woman. It's really rare.
- 5.42** We're in Miami. Maybe they're more 6-foot-tall women in Miami, but it's still unusual. You can actually predict things about the world the more you know about the statistical parameters of the way that these things behave. Am I doing okay? We're good? All right. Who cares? Why does it matter? It's a bunch of bell curve words. It matters because if you can understand this, it's actually a really powerful tool for navigating data flows that come to you.
- 5.43** It could tell you the way that different things are related, because the pattern of behavior of normally distributed data is consistent. It doesn't tend to behave completely erratically. It gives you a really good mechanism for being able to predict likely outcomes if you understand the basic underlying data. You want to learn this stuff because it can help you navigate the world from a data perspective.
- 5.44** What does this tell us? What it does is it gives, the standard deviation and the average give us the ability to assess two different bell curves. So one bell curve would be men. Another bell curve would be women. If we put them both on a graph, they look like this. Men are in orange, women are in blue. So, is there actually a difference between

the average height of men and the average height of women? And again, we all know the answer intuitively, but here's the fundamental problem.

- 5.45** This area here, there's huge overlap between 5 foot 5 and 5 foot 9. There are a whole bunch of men and a whole bunch of women. That's actually a problem because it makes it harder to mathematically prove that there's a difference. The bigger the overlap in terms of the behavior of two different groups, the higher the likelihood that you will get an answer when you go through the math that is wrong.
- 5.46** We're going to go through how to prove this mathematically. What we're going to do is we're going to perform a mean comparison with samples from each population. What the heck does that mean? What's a mean comparison? I'm going to try to de-jargon this a little bit. It's the average, if we randomly pick people from one group and we randomly pick some people from the other group, and we compare those two, can we tell if there's a difference? The answer is, it really depends.
- 5.47** We're going to talk about the null hypothesis. This is a really important scientific term, but I think it's actually very important for all of life. The null hypothesis, I'm going to read this super jargony definition: "In scientific research, the null hypothesis is the claim that the effect being studied does not exist." What the heck does that mean? It means that the thing that you're studying, that you want to be true, is not. If you're going to study whether or not a drug works, the way that you have to do the statistics is to presume that it doesn't. The null hypothesis is that this drug doesn't work.
- 5.48** The null hypothesis about whether or not there's a difference between men and women is that there is not. Basically, it's a way of forcing you to consider that the thing that you're trying to prove, actually there is no difference. Is there a difference in height between men and women? The null hypothesis is no. There's not a difference. We got to come up with some math to prove whether or not this is actually true.
- 5.49** And so another way of thinking about this is, there's actually a likelihood that rare things happen that's pretty high. If you roll two dice and you get snake eyes, it's like, well, that's a 1 in 36 chance, which is what? 2.78%, I think? That's not going to happen. You're not going to get snake eyes. But you will. If you roll the dice enough, you're going to get snake eyes. On average, 1 time out of every 36 times you roll the dice. What we're going to try to do here is we're going to try to prove that, when we do the math, it's not just an unlikely random event. It's actually a true difference.
- 5.50** There's some ways that we can do that. How are we doing? What we're going to do, I had the computer do this. We're going to randomly pull 4 samples of 3. These are some men. Each sample is either men or women, but we don't know what they are. We pull 1 sample of 3 people, and they average 66 inches. We pull another sample, they average 66.7. We pull another sample, 66.7, and another sample, 67.3. They're between 66 and 67. Which group is men and which group is women?

- 5.51** The answer is, you don't actually know. Because statistically, these aren't different enough to be able to make any determination. The fundamental thing that you want to understand is that, unless things are really, really distinct, you cannot draw conclusions. The answer is, the first 2 are women and the second 2 are men. Probably could have guessed that, but math doesn't tell you that.
- 5.52** One more concept here. The average is the midpoint. The standard deviation is how spread out it is, and the sample size is the number of samples that you take. So if you take 3 samples, you have an average and a standard deviation. If you take 10 samples, you have an average and a standard deviation. The fundamental point here is that the more samples that you take, the more certain you can be. Because these are two distributions that are almost the same. The average is right here in the middle. The standard deviation is the breadth of the blue line. The average of the orange is right here in the middle. There's a whole lot of overlap in this space.
- 5.53** What you're looking for, from a picture perspective, and what is, how much overlap is there? If there's a whole lot of overlap in the data sets, it's very hard to distinguish between the two. If there's not a lot, it's a lot easier. This is actually intuitive. If you take 10 people and they're all 6 feet tall, and you take 10 other people and they're all 5 foot 2, it's actually a whole lot easier to make, versus the data I showed you a second ago. Everybody was, the average was 67, 66. It's close. But if you have groups that are much more distinct, their pictures are going to look more like this.
- 5.54** The more distinct the pictures are, you don't even have to work through the math. The picture tells you that there's a difference. The fundamental thing to look at is, how much overlap is there here? That's the thing that tells you how closely related the two groups are. The bigger the overlap, the more samples you need, to draw a distinction. If we do 10 samples of 3, here's one way to think about this. These are 10 different random samples of a distribution of women in Excel, and the average is, well, here are each of the averages of the groups of three, 61 to 67.
- 5.55** The total variance in this group is, we take different averages. I think this is the point. One group of 3 women was 62 inches, and another group of 3 women was almost 68, completely randomly. When you randomly sample things, you're going to get vastly different answers. I think that's the fundamental point that you want to understand here, is that a sample of something only tells you a little bit, unless you make the sample bigger. If we do 10 samples of 10 women each, the variance goes down. It gives us more certainty. You can actually see this.
- 5.56** Now I think, in the data a second ago, it was 61 inches to 67 inches. We got different averages. Now it's tighter. The difference between the samples of 3 was 61.7 to 67.7, which is a variance of 6. When we draw 10 samples, it's much more tightly grouped. Why? Why does more samples make things more tightly grouped? I'm going to go through the math. It'll be pretty obvious, I think, if we go through the math. Let's say

we get one group of 3. We've got 64 inches, 64 inches, and 69 inches. Now, if they were all three 64, the average would be 64.

5.57 If you throw in one 69, and the average moves up to 65.7. So that one person in a group of three moved the average up by 1.7 inches. If we take a group of six, and five of them are 64 and one is 69, that one 69 only changes the average by 0.8. So the more samples you have, the less impact each individual sample has on your overall sample set. We doing okay? We having fun? We're going to keep, tell me when to stop.

5.58 The z-distribution. The jargon is "the spectrum of probabilities for different values of X, like heights, given a known average and standard deviation." A z-score is equivalent to 1 standard deviation. The z-score is, when you take 1 sample, let's take somebody in this room who's 5 foot 2, and the average is 5 foot 4.5. That 1 sample, that one person, is 2.5 inches from the average. That's actually 1 sigma or 1 standard deviation, which is one z away from the average.

5.59 A z-score is a way of translating each individual sample— if we take 10 different samples—how far away are they from the average of the samples? It's a simple metric to translate it to a universal value. Because if we're dealing with height, we're measuring things in inches. If we're dealing with weight, we're measuring things in pounds. A z-score translates both of those things to a universal metric that you can analyze, regardless of what the underlying measuring tool is.

5.60 Here's a way of thinking about this. If you take a sample, and you get the z-score, which is the number of standard deviations that that sample is away from the average, what that will tell you is the likelihood that that thing, or that sample of a person that height, will occur. If we take a woman who's 6 feet tall, which is three z-scores away from the average, the likelihood that we're going to find a 6-foot-tall woman is 1 in 300. The z-score tells us the probability that different things are going to happen.

5.61 If you tell me the z-score of something, in all cases, without exception, I can tell you, you just tell me the z-score, I can tell you the probability of that event. Whether it's height or weight or dice or coin flips, if you tell me the distribution of that data set, I can tell you, and the z-score of one particular thing, what is the likelihood that you're going to flip, that you're going to get 10 heads in a row if you flip a coin?

5.62 You can do the math and you can calculate it. I won't go through the math. It's roughly 1 in a 1,000. It's actually a pretty unusual event. It's 2 to the 10th. If you tell me the distribution of something, I can tell you the likelihood of different things occurring. If I could tell you the likelihood of different things occurring in men, what's the likelihood that I'm going to have, if I randomly sample a group of men, what's the likelihood that I would get a man who's 5 foot 6? I can tell you the likelihood, because I know the z-score of 5 foot 6.

5.63 If I randomly sample a woman, what is the likelihood that I will get a woman who's 5 foot 6? If I know both of those things, I can tell you the likelihood of whether or not one

is a man or a woman. Now, that doesn't mean, it goes right back to the dice. That doesn't mean that I know. That means that I have an informed guess. What's the likelihood that a woman is 5 foot 9, which is the average height of men? It's roughly 1 in 250, something like that. So that's really unusual. That's not going to happen very often.

5.64 What's the likelihood that a man is 5 foot 9? It's 50%. It's like a coin flip. Half of men are going to be 5'9 or taller. If you said, Derick, here's somebody who's 6 feet tall. Are they a man or a woman? Like, 1 in 300 chance, or 1 in 2? It gives you a way of comparing these two things. When you understand that, you can go, well, I'm going to bet on the 1 in 2 because that's way better than 1 in 300. We're doing okay? We're going to keep escalating here.

5.65 The p-value is a way of comparing what is expected to occur with what actually happened. So it's a way of assessing the likelihood of an outcome. If I draw a random sample of women, and this person is 6 feet tall, I can tell you the likelihood of a woman being 6 feet tall is 1 in 300. I can use this same tool in other areas. If I want to understand whether or not a drug works, the placebo has a response of 30%. So I give a group of 100 people a placebo. It's a sugar pill. A third of them, 30 of the 100 get better.

5.66 Now, I take another group of 100 people, and I give them a placebo, and 40 of them get better. And I take another group, and 25 get better. And I take another group, and 35 get better. Then I take another group and 27. This is what's going to happen in the real world, right? The average is 30%, but you're going to randomly get different values that are going to be 20, 25, 30, etc. You want to understand that random possibility, and you want to compare that possibility to your drug, to try to understand whether or not the outcome that you got with the drug is because it's real, and people actually got better, or is it because you got lucky? Is it just a random circumstance? And so, you can do this mathematically. If I take 100 people and I give them placebo, and 30 get better, I take another group and I give them placebo, and 40 get better, then I take a group and I give them the active drug, and 42 get better—did the drug work? It's not clear, right?

5.67 This tool gives you a way of mathematically assessing that with more accuracy. Anybody want to nap? Anybody want to ask a question? Yes, please. Let's get a microphone.

5.68 How often do you rely on data visualization to make a decision as CEO?

5.69 That's a good question. I would say visualizing is actually really helpful because it gives you a picture, which in some ways your brain can process more easily. Because if I tell you the average is 5 foot 4 and the standard deviation is 2.5, it's like, what does that really mean? I don't know. I don't have any way of really picturing that in the world. I think the graph helps make it more accessible.

- 5.70** I can tell, if I just look at this, there's a really big difference between these two groups. It's obvious. I know that, if I get an answer that says that there's a difference, and I work through all the math, and it says there's a difference, and I look at this, I'm like, that's probably real, right? Because these things are so clearly distinct. I do both. I like to look at the picture of the distribution, get an idea for what it's telling you, and then do the math.
- 5.71** In society nowadays, one thing that you notice is that oftentimes on the news and in public discourse, you hear very statistically trained people say things like, "This proves." And one of the questions that I have as well is, when you look at this distribution, you can very clearly amend it by redefining what a woman is. And so I'm curious in your experience, what kind of patterns or what you've observed in terms of how certain choices are being made in the utilization of statistics?
- 5.72** Yeah, that's a really good question. I'm going to answer it in the next section, but I'll give you a prelude. The fundamental thing that I would say is that the problem with statistics is that it is very, very easy to prove things that aren't true. Almost way too easy. If you give me any data set, in any parameter, anywhere, height, weight, whatever, I just demonstrated, I can cut and slice the data different ways and take different samples and blah, blah, blah. I can do all kinds of things, and I can give you the answer that you want to hear, period. Without exception.
- 5.73** Here's basically the reason why. If I flip a coin 10 times in a row, I think it's 1 in a 1,000 that I'll get heads all 10 times. That's really, really unusual. What if I flip a coin a billion times? Right. It's going to happen a whole lot. It's going to happen 100,000 times or something like that. Right. It's going to happen. So this is the mistake that people make with statistics, is they take 5 million people, and they look at 300 different genes. And they say, we found this gene causes heart disease. It's the same thing as flipping the dice or flipping the coin. They've done so many calculations.
- 5.74** If you just keep calculating and calculating and calculating, statistically, some unusual thing is going to happen in that data. And so, all you have to do is run lots of math to find a relationship. And I think you really want to understand that the likelihood that something unusual will happen increases with the number of times that you do calculations, the same way that flipping a coin more and more and more, eventually, you're going to get 10 heads in a row, just statistically. Does that make sense? Yeah, we're going to get into that in the next section.
- 5.75** So, do you find there's a Pareto distribution in the utility of data from big business to small business? So for small operators who might be working with smaller data sets, less availability to that?
- 5.76** Yeah. I mean, I think that—you run a gym, right? So you've got the only 3 sample problem. And so you want to be—it's a great question. You want to be careful about, over-ascribing value to things when you don't have a very big sample size. Now, on the

big business side of the equation, it's the opposite end. They get to flip the coin so many times, they can find a relationship. And they tend to just get so bogged down in overanalyzing the data that it can be confusing on the other end of the spectrum. So, yeah, good question.

- 5.77** Michael Berry famously predicted the stock market crash or the housing market crash in 2007, 2008. And you said earlier that it's finding these really nuanced details is not so productive. But in his case, he would look through bonds and find these really, really garbage loans. And because of his due diligence in that regard, he had a very successful trade, or I'm not familiar with the jargon. But my question is, yeah, a great movie.
- 5.78** The Big Short, right? But are there exceptions to that rule you said earlier? If you're doing too many calculations, too nit-picky, not worth it?
- 5.79** So it's challenging, right? Because there's a problem with The Big Short. I think he actually figured out something and made a very good bet. But there's a lot of hindsight bias in it. What I mean by hindsight bias is, all of the people, you don't see the data of all of the people that made bets that didn't go very well. So, Michael Berry makes a really big bet and does extraordinarily well because of the subprime market, but it was that could have been random.
- 5.80** It's hard to discern whether that's random good luck, or was it actually a very good analytical decision? It's really hard to differentiate those two things mathematically, because if you have—here's why. It's the same thing as the coin flip. If you have 100,000 different mutual fund managers, and the market collapsed, and five of them make a billion dollars, it's like, those five are geniuses. Well, no. Five people in any situation are going to do really, really well in an unusual circumstance. It's hard to tease out those two things.
- 5.81** I'm not saying that wasn't because he figured out something very important in the data. That's not my point. I'm saying that it's difficult to discern whether or not that's the reason. Warren Buffett describes this well. He says when the underlying market becomes disconnected from reality, it's really hard to predict what's going to reconnect it. You don't know. When it gets tulip bulb, hyperbole, you don't know what's going to happen that's going to cause it to go, I'm going to go back to reality now. But Michael Berry figured it out in that circumstance. I don't know if he'll figure it out again, but I do think my guess would be, he figured out a good underlying analytical reason. That's a guess.

Lecture 6: Fundamentals of Statistics II

At a glance — Fundamentals of statistics II.

- 6.1 We're going to go through about 10 or 15 more minutes of some heavy data concepts, and then we'll move on to some more conceptual things around how to differentiate between people who are trying to advise you to make good decisions versus trying to convince you that they're right and they have the answer. A little bit more data stuff here. I think the overall high-level point you want to make sure you take away from this is that a good statistician can mislead themselves by crunching numbers over and over again.
- 6.2 They can cut and slice things until they get an answer that is going to make you happy. That's actually, subconsciously, what is occurring. Because no one likes to come and say to the boss, I got this answer that you really don't like and you don't want to hear. It's no fun. And so, again, it's not a mistake that people try to mislead the world. It's a natural occurrence of the human proclivity to try to enhance social interactions by providing people with what they're looking for.
- 6.3 This is a well-known phenomenon. It's called p-hacking or p-hunting, which is—another way of thinking about this, is you just keep flipping the coin until you get 10 heads in a row. If you have long enough and you have enough data to cut and slice, you will get there. You will get the answer that you're looking for. I'm going to give you a real-world example of this. This is a drug we were looking at a couple of years ago. They ran a clinical trial. It didn't work. The p-value is 0.28. Basically, so that means there's a 28% chance that this drug was better than the placebo, randomly.
- 6.4 We had one group that was on placebo, and we had another group that was on placebo. This one's better than this one. There's a 28% chance that it would be that much better. What you do with the math, when you run a clinical trial, is you are looking for a random event that has less than a 5% likelihood. If you run 20 clinical trials, there's 1 time out of 20, your drug is going to randomly be better than the placebo, even though it isn't. Another way of thinking about this would be, if you ran 20 clinical trials of placebo versus placebo, 1 time out of every 20, this one is going to be so much better than this one that it convinces you that it's working, even though it's just random.
- 6.5 What happened here is that the company that was looking for us to invest, they went p-hunting. And so they took the clinical trial and said, well, this failed. It wasn't actually different than the placebo. The drug didn't work. But if we had only given it to these people, instead of this whole group of people, if we'd only given it to these few people, well, look, it really worked. If we had known beforehand that it was only going to work in this group, then we would have only given it to those people, and they got better.

- 6.6 They got a whole lot better. So therefore, let's run another clinical trial and only enroll the people who got better in the first trial that we did because they were just way better than the placebo. Here's the problem with that. It's that it's the error of omission. What an error of omission is, is all of the things that were calculated that you aren't being told. Here's what actually happened.
- 6.7 They take the group of people and they say, well, let's look at the people who had upper GI disease. Well, how about people who had middle GI disease? Well, what about people who had more severe disease? What about people who had less severe disease? They just keep looking at different groups. Here it is. It worked here. Then they come and tell you about that. What they don't tell you is, it didn't work here. It didn't work here. It didn't work here. It didn't work here. You don't get the whole spaghetti picture that we talked about yesterday.
- 6.8 The only thing that's presented to you is the thing that is indicative of what the presenter wants you to hear. This happens in data all the time. The most important thing here is not necessarily to be aware of how to do a p-value calculation and calculate this sophisticated stuff. The thing that you want to take away from this fancy math is that this happens very often. You want to be a little cynical and a little suspicious when somebody says, I did all this fancy math, and here's the answer.
- 6.9 No. What about all the math that you did that you're not telling me about? Why aren't you telling me about the math that you did that you didn't get the answer that you want to tell me? That's what's missing in the communication.
- 6.10 Here we go. Genome-wide association studies. What does all that fancy stuff mean? What it means is that, if you take five million people and you analyze 300 different genes, you're going to get a p-value, period. That's what 1.5 billion different variables that you're calculating. So that's flipping the coin over and over and over again. You're going to get 10 heads in a row, period. It's going to happen. So a Bonferroni correction is a way of raising the bar to correct for the number of calculations that you run.
- 6.11 That's the important thing to understand there, is that if you do more calculations, the likelihood that you're going to show something positive goes up. You want to increase the calculation of the null hypothesis. You want to make the null hypothesis more unusual in terms of disproving it. What's another way to say this? Here's a simple way of thinking about it. If the goal is to get a p-value of 0.05 or better, you divide the p-value by the number of calculations that you run. That's a simple way of thinking about it.
- 6.12 If you do two different calculations, you don't want to believe that it's true unless you have a p-value of 0.025. If you do three calculations, 0.0125. You basically want to make the goal harder and harder and harder, the more times you flip the coin. This is just a quote to make this point: "With four parameters, I can fit an elephant, and with five, I can make him wiggle his trunk." The point here is that the more sophisticated you

are with statistics and the more data you have and the more different ways that you cut and slice and look at things, you can absolutely find a relationship in all circumstances.

- 6.13** How do we calculate a p-value? What's the likelihood that we would get a random sample of three women with an average height of 6 foot 7, and a standard deviation of 3.1, when the true average for women for a whole population is 6 foot 4.5? First, we're going to calculate the t-score. Here's a way of conceptualizing this. A t-score is a way of assessing the, think of it as, like, the standard deviation of the standard deviation that is a function of the number of samples.
- 6.14** The guy who figured this out, he actually worked for the Guinness factory. He worked in the beer factory, and he was testing the beer in the vat, to make sure it had the right amount of malt or whatever. He realized that he would do 3 samples and he would test it. It would somewhat routinely be a bad batch, and he'd have to throw out the batch because it didn't meet the standard or whatever it is.
- 6.15** What he figured out is that, if you only test something a little bit, the likelihood that you're going to—again, it's the random group of taller women. It's actually common enough that, if you are testing your Guinness batches every day, over and over again, it's going to happen relatively routinely. So he came up with a way of adjusting for this. And the way that he adjusted for it is he took the standard deviation and he created a new distribution based on the number of samples that he took.
- 6.16** Here's a way of thinking about this. If I take two women, one is 5 foot 6 and one is 5 foot 4, and their average is 5 foot 5, the likelihood that that's a true average and a true standard deviation is pretty low. It's only two. If I take 100, I can have a lot more confidence that that's a true average and a true standard deviation. But the average and the standard deviation themselves don't tell you anything about the number of samples that you took. They don't tell you how confident you can be in that average and the standard deviation.
- 6.17** This is what a t-distribution does. It takes into account the average, the standard deviation, and the number of samples. So it modifies the bell curve. A way of thinking about this is, it changes the shape of the bell curve, and it spreads it out or centers it, based on the number of samples that you take. The more samples that you take, the higher the likelihood that it's a true answer, that it reflects what's really going on. The fewer the samples you take, the lower the likelihood.
- 6.18** The z-score, the normal distribution, if I take one sample, what's the likelihood that I'm going to find a woman who's 6 feet tall? It's 1 in 300. If I take 3 samples, what's the likelihood? It's lower. Because average women are going to pull down that one really tall women. They're going to pull down the average. If I take 10 samples, it's lower. This is a way of assessing the likelihood of an event, given the number of samples that

you've taken. The more samples you've taken, the lower the likelihood that something really unusual is going to happen. That's the way to think about a t-statistic.

- 6.19** I'll just go through this math really, really quickly. But here's the actual math. What you do is, you take the average of one group, the average of the men, you take the average of the other group, the average of the women, and you divide it by the standard deviation, divided by the number of samples. If you take some time and you go through that math, what you'll see is that, as the sample size goes up, because n is in the denominator of s , as the sample size increases, the s relative to n gets smaller and smaller. The consequence of that is the t-statistic goes up.
- 6.20** The greater the t-statistic, the higher the level of confidence you can have that you're actually measuring something that's real. The more samples you have, the greater your confidence. This is a way of mathematically assessing your confidence based on the number of samples. There are really three parameters we're ultimately looking at here when we're doing these comparisons. How far apart are the means? Things that are way far apart are more likely to be different. Things that are closer together are less likely to be different. It's obvious.
- 6.21** Then we take into account the standard deviation. Things that are more far apart and tightly packed are more likely to be different. If they're way far apart, but they have a big standard deviation, they're spread out, the bell curve is really spread out, there's a higher likelihood that they're going to overlap in the middle. So, far apart, more likely different; less spread out, more likely different.
- 6.22** Sample size is the third parameter there. Sample size says the likelihood that this level of spread-out-ed-ness and far-apart-ness is actually real goes up the more samples you take. It's a way of just calculating that mathematically. Because if I say, well, I've got 3 samples and I found this. And I took 10 samples and I found this, well, what's the likelihood that each of those is actually real? What's the likelihood that they're meaningful? It's hard to just picture that in your head. You can't do it.
- 6.23** What if I took 100 samples, how much better? What is the amount that 100 samples is better than 50 samples is better than 10 samples? This is a way of actually calculating that. You can actually measure the amount of how much better it is, the more samples you take. I won't go through all the math, but basically, the probability that a sample of 3 random women would have an average height of 6 foot 7.7, when the true average is 6 foot 4.5, so it's 3.2 inches greater than the real average, is 0.22.
- 6.24** There's a 22% chance that if you randomly sample a group of women, they will have an average height of six, seven and a half, whatever that is. So 22% chance, so that's low, but it's really not that low. If you flip a coin twice, the chance of getting heads twice is 25%. You have four different possibilities. You can get tails, tails; heads, tails; tails, heads; and heads, heads. If I did that right. One time out of four, 25%, you're going to,

22% here, you're going to randomly sample 3 women, and you'll get an answer that tells you that the average height is 5 foot 7.7, when in reality, it's actually 5 foot 4.

- 6.25** All right, so just to summarize the data, objective data often is not. It's these repeat analysis, doing it over and over again can subconsciously mislead us. So you need to be aware that that's happening with the people that you work with. You need to be aware that it's happening on Google. It's happening everywhere. These things are, you want to, well, I think at a high level, conceptually, you want to second-guess this stuff. And then if you really want to know your stuff and you really want to make these more complex decisions, you want to dig into this data and you want to understand what a p-value really means.
- 6.26** Because it gives you a level of confidence in what you are being told. And it gives you a measuring stick. What statistics really is, it's a measuring stick of confidence. It gives you a way of saying—it gives you a way of increasing your level of certainty in the underlying data that you're being given. We'll get through this.
- 6.27** All right, polysyllabic nonsense. What's the difference between somebody who, if you're not really a math person, you're not a statistics person, a whole bunch of the stuff that I just went through sounds jargony. I tried to de-jargon it right, but maybe I didn't do a perfect job. What is the difference between somebody who does that and lays out these complex things and says, "I'm an expert, so if I say this drug works, you should invest in it," versus someone who owns shares in the stock, and is just telling me the data that tells me the drug works, and is leaving aside all the other stuff, and is using all the jargon? "Here's the p-value. This thing works," and blah, blah, blah here.
- 6.28** How do you tell the difference? It's not easy. It's one of the most challenging things in life is to figure out who actually knows what they're talking about. I make that mistake regularly. I tell you, it happens all the time. I think the first step to understanding who is the real expert is to understand that it's not easy and that it is difficult to differentiate and that you should be aware of the difficulty in differentiating. We talked about "fake-spertise," but let's talk about communication patterns because this is a concept. We don't even need to get into—if you don't know enough about statistics, you don't have a measuring stick for determining whether or not I know enough about statistics.
- 6.29** I just sound like I'm just throwing around big words, like, do I actually know? And so I think that here's a couple of different ways to conceptualize this. A "fake-spert," the trap that we can fall into is that human beings ascribe knowledge based on the way that it is communicated. If I stand up here and have a loud voice, and I'm assertive, and I seem to be able to navigate through these things pretty well, it seems like I know what I'm talking about. Hopefully, I hope I've given that impression.
- 6.30** But that's the same impression that somebody could give you who doesn't actually know. If they're good at it, if they're practiced in all of these fancy concepts, they can actually communicate things in a way that sounds almost exactly the way that I just

went through it. So, who's really telling the truth? Here's a couple of examples of how to, well, here's an example of how common this problem is. This blew me away when I read this study.

- 6.31 They actually did a study on fake expertise. It's pretty recent. It was from Cambridge. Here's what they did. They took a whole bunch of different publications, on all kinds of different topics—science, psychology, sociology, biology, all kinds of stuff. And they took a really fancy-sounding, jargony, statistical, mathematical sentence from a completely different, totally unrelated paper that made no sense whatsoever and did not relate to any of the calculations that were done, and if you truly understand the underlying subject matter was completely nonsensical and absurd, and they stuck it into the abstract of papers. So they created fake papers.
- 6.32 Here's what happened when they sent them in for reviews. The more the nonsense, but really fancy-sounding there was in the paper, the higher it got ranked. These are reviewers, for scientific publications, and they are using the same heuristic that we all mistakenly use, which is, this sounds really fancy and really complex, so this person must know what they're talking about. That's the trap that you want to avoid. It's unbelievable. In all of science, this happens.
- 6.33 And so, you need to understand how prevalent this is. It's like, this doesn't just happen to people who aren't—the fundamental point here is, this does not just happen to people who aren't experts. It happens to experts, who are, like, well, I don't really understand that particular realm of science. The way this is conveyed, it really deeply sounds like this person knows what they're talking about. But, actually, the more random complexity that you throw in there, that's super jargony, the higher the likelihood of misinterpreting it, over-interpreting it, over-ascribing value to something where there is none.
- 6.34 What you're actually looking for is the pattern of, I call it "fake-spertise." One of the most obvious things is there's just way too much data. They have multiple different tables, and you look at the, they come in and give you charts, and there's chart after chart after chart after chart. It's a fire hose of too much. A fire hose of too much is a sign that somebody doesn't actually know what they're talking about, and they're just trying to impress you. So someone who actually knows, they will take the one thing that's really valuable and super potent, and they will describe that one thing in clear language without bedazzling you with all kinds of fancy words.
- 6.35 And so, again, the overconfidence, big words, lots of data. It creates an impression of certainty. We did this, we cut this multiple different ways. We did this analysis, this analysis, this analysis. And we did this Bonferroni. I threw that in there on purpose. We did this Bonferroni, right? And we, it was true here, and it was true here. They just bowl you over with way too much that's communicated in a bravado manner. Too much data, no priority. Big words, lots of brash statements. They talk about their authority. I

know what I'm talking about. I went to this school. I did this. I achieved this. I talked to these people. I know these people.

- 6.36 They make appeals to their relationships and their authority, rather than clearly communicating from a data perspective why they know what they're telling you they know in more simple terms. So even if I didn't communicate it perfectly clearly, all that fancy stuff we went through, if I hit the goal of trying, if I give you the impression that I'm doing my best to simplify some really complex things, that's the pattern that you want to look for. You want to look for somebody who's saying, well, I know this is complex, but if you think about it in this more simple reference frame, it's the direction of the communication pattern that you're looking for.
- 6.37 A true expert, as the corollary to the fake-spert, is they understand there's a gap between their knowledge and yours, and they do their best to try to close that gap. They don't want to bedazzle you. They want to educate you. That's the good way of thinking about it. They're not trying to impress you. They're trying to communicate the best they possibly can with something that is pretty technical. As we just saw in the last hour, it's not easy, right? It's not easy to talk about something that's really, really complex and distill it down into an easily interpretable framework.
- 6.38 You just want to look for somebody who's trying, who's trying to simplify things. I think a couple of days ago or yesterday, we went through the way to do this, is to ask the—hey, tell me what this means like I'm five. Just ask that question. What does this mean in really simple terms? If they can translate it to pretty simple terms, they know what they're talking about. What you're really looking for is somebody who's trying, maybe 10 rather than five. I ask that question all the time.
- 6.39 If they can simplify, clearly explain things, metaphors, examples. Examples are always really good. When you actually look at the bell curve, it's more clear why these two things are different than if you, it's a picture that helps you understand the underlying meaning. You're looking for different ways people communicate. A whole bunch of fancy jargon, again, is a sign of "fake-spertise." Here's a key point that I think is really super valuable, is that most people who are bravado-driven "fake-sperts" is they don't know that they're just rotely going through the calculations without a depth of understanding of how they got there.
- 6.40 They're just methodically doing the math, and they don't realize that that's not actually informative information that can influence a decision. They're not thinking about whether or not the information should be construed in a way that will influence a decision. So they're hiding their own bravado from themselves. They don't realize that they're not necessarily interpreting things correctly, etc.
- 6.41 A couple of questions you can ask to tease this out. Please tell me the single most important aspect of your analysis. What's the one thing that you take away from what you did that you think is really valuable, that I should be listening to? What's the one

graph that really explains this well? If you could summarize this in a simple fashion and explain what you think this underlying message should be? What should I be relying on to make this decision? If you can explain it to me, then I would appreciate it. Because I'm going to become more knowledgeable, and maybe it'll help me make a better decision.

- 6.42** People who are overexaggerating their competence, and they can't do this. Because what ends up happening is, here's what they'll do when you ask that question. Tell me the one thing that you think is really important in all this that I should really be paying attention to. They'll immediately give you lots of stuff. What that tells you is, they don't know the priority. Because basically, think of it like this. I ask you a question on a test. Any random question on a test. I can't think of a random test question, but I ask you a question on a test, and there's one specific answer, or there's an opportunity to provide 10 different answers, one of which may be right. And either way, you get the question right.
- 6.43** What fruit is red? The answer is apples. But I can guess apples and blueberries and strawberries. There's another one. And I can guess 10 different things. Well, then, what that tells you is that I don't actually know the answer. So I'm guessing as many times as I possibly can, in order to hope that the right answer is in that big group of answers somewhere. So a person who really understands the underlying data, they will focus in on the one most important thing, and they will communicate that to you clearly. That tells you they know what they're talking about.
- 6.44** Any questions on Bonferroni corrections? It's got to be on Bonferroni. No, I'm kidding.
- 6.45** I wanted to know if there was an app that can be used for a layman person. Is that the way to say it? To actually apply these things to anything in life, to find the average, whether it's, what's the likelihood of this to happen if I do this or if I move to this city? What is the statistics on the people of my age and my race that make this kind of money?
- 6.46** Statistica, I think, is a pretty good site online that's got a lot of statistics on a broad range of different topics. And then if you want to get into a little bit more complex calculations without going crazy, there's, I think it's called GraphPad. I think it's the name of the site. I'll have to check that.
- 6.47** Then the other part was, if the more you test, the more accurate you're going to find that answer that you're looking for—
- 6.48** The more you sample.
- 6.49** The more you sample.
- 6.50** The more you test, the higher the likelihood that you're going to be wrong. You're going to find the wrong answer. So the more you sample, the greater the likelihood that you're right. The more you test—is that distinction clear?

- 6.51** The thing is, is that, for a person to be biased and they want to present an information to you, I thought that the more they did something, you're going to eventually find the answer you want. What I want to know is, what is the magic number to get the truth? Like that spaghetti. Is this really happening? I don't want answers of people's biases and egos.
- 6.52** There's no truth in statistics, unfortunately.
- 6.53** Well, the average would be the most likely.
- 6.54** You have to dig into the math to know the—I think the answer to that question, if you're trying to solve some difficult statistical equation, is to find a statistician who is driven by the truth, who's really trying to give you the true answer, even if it's not what you want to hear, and let them do the math. Because it's going to be hard to, because the answer to your question is that it depends. And it's not because I'm dodging the question. It's because, what is the likelihood of flipping a coin and getting heads five times in a row? It's completely different than the likelihood of getting heads 10 times in a row.
- 6.55** The judgment of the likelihood is going to depend on the way that you've gone about testing and the thing that you're trying to test. It varies, unfortunately. The overall idea here with the statistical concepts is to engage in a testing program that gives you the confidence that what you have done is probably the right answer, given the question that you—so it gives you a way of judging the likelihood that you've found a true answer. Another way to think about this is it's closer to truth. Probability gives us the ability to assess how close something is to true.
- 6.56** My question is, when you receive data and just visuals in general, how important is it for your data analyst or data scientist or whoever works in data for you, how important is it for the storytelling and when they tell you the story about the data, how important is it to you? And do they sometimes persuade you to change your answer?
- 6.57** Well, I do a lot of the number-crunching myself. I feel like I have to understand this because we're making very big bets on things. And so I want to really understand whether or not it's a good bet. And I don't want to rely on the possibility that this kind of thing may happen. But I would say that, if you're not a data scientist yourself or not a statistician, you want to—there are a couple of things that you can do that would be helpful. One is to say to your data folks, go back and re-cut this data 10 or 15 or 20 different ways, and tell me what all those different answers are and tell me how you've done it.
- 6.58** And particularly, I want to focus on the thing that tells me—like, if you're trying to assess, I don't know, where to put a new store, and you're trying to assess the demographics of this particular city and where's the optimum location and traffic flows, you're probably looking at all of these things. They come and say, I-75 at this exit, or whatever it is. They tell you the answer. Well, don't just give me the answer. Tell me

the ways that you've considered all of the different possibilities, and why this answer that you've given me, you think is the best answer. So they— make them show their work, would be the high school equivalent, I guess. Did that answer your question? Okay. Yeah.

- 6.59** It's a two-part question. I'll try and keep it pretty concise. You reference the Cambridge study in 2023, the nonsense math effect. You may have mentioned, I may have missed this, but do you recall how rampant it was in academia? Do you have any, maybe intuition is not the right word, but any sense of how rampant it is in industry?
- 6.60** It's completely rampant everywhere. I don't remember the exact name of this publication, but this effect has been assessed multiple different times in science, and it is a fundamental deep flaw in science. Basically what they did, here's the reason there's a deep flaw statistically in science. It's because it's the error of omission problem. If you are looking for a p-value of 0.05, what does that mean? That means there's a 5% likelihood that this thing that you discovered, this new relationship between this biological mechanism and that one, or whatever it is that you're analyzing—this drug works, that one doesn't— the likelihood that that's going to occur randomly is 5%.
- 6.61** So what does that mean? It means one out of every 20 times that you assess this, you're going to say there's a relationship when there is not. Now, here's what happens. Here's the problem. What gets published? The things that get published are the relationships. No one wants to publish something that says, I studied this and found out there is no relationship, and that's it. It's boring. It's not interesting to discover that you didn't discover anything. Nobody publishes, I did this study and I didn't find anything meaningful.
- 6.62** The only papers that get submitted to the journals are those that have a statistically significant p-value. What about all the others that didn't? They don't get submitted. You don't send in something that doesn't show a relationship. Here's the problem, is that if there are a whole bunch of scientists out there in the world, and it's the same thing as flipping the coin. They're all doing a whole bunch of different experiments. Well, one out of every 20, it's going to be statistically significant, randomly.
- 6.63** And so here's what these guys did. They did a spot check, and they repeated the experiments. What they found is that, across the board, in all of science, I don't remember the exact number, but I think it was something like 90%. Don't quote me on that because I'm not sure that that number is right. Some huge percentage of studies, the effect could not be demonstrated when they re-ran the experiment. I'm talking about the New England Journal of Medicine. They redo the experiment and they find there is no relationship.
- 6.64** So this is a really, really deep problem. It's part statistics, part a lack of broad understanding of what p-hacking really is and what it means. I think what you're looking for, if you're in the field of science, and a true scientist, what you're looking for

is somebody who really aggressively second-guesses themselves. And they re-run the experiment, and they do it again, and they check it different ways. And they run it different ways, and they truly absorb the null hypothesis and say, the thing that I want to be true is probably not. And they really believe it. That's what you're looking for. Did that make sense? There was a second part of your question.

- 6.65** Does that mean that something like a meta-analysis could be an aggregate of potentially misleading data in some sense?
- 6.66** Absolutely. Meta-analysis, yes. Absolutely. The problem is the p-hacking effect. Here's the problem with the meta-analysis: what studies were thrown out? All the studies that didn't support the meta-analysis is what's thrown out. I can tell you that. That's what happens.
- 6.67** I think that, with statistics, you're trying to—one of the underlying assumptions here is you're providing an answer to a question. Does climate change exist? The answer is portrayed in this through these different statistical graphs. My question is, in terms of another overarching question is, which questions can be asked is another statistical topic. For example, you're not allowed in society to really do any more research into intelligence because there are societal implications. And as such, there isn't really any appetite for those questions to be asked, even though there may be convincing answers that are very strongly supported by statistics. In your experience, how do you describe that phenomenon of un-askable questions?
- 6.68** I'm not sure I should answer that question [laughter]. So climate change and IQ, that's what you want me to answer. One of the things that I think you can do is to try to determine whether or not somebody is driven emotionally or by confirmation bias, is you can ask them to play the devil's advocate. Tell me why what you believe could probably be wrong, statistically. A true, honest scientist will be able to give you all the reasons why.
- 6.69** Somebody who's just trying to manipulate you, they'll just start listing their credentials and they'll tell you, they'll appeal to authority, and all these people agree with me, and you're an idiot for asking the question. They'll just dismiss you. That's somebody who's lying to you. Somebody who—look, a true, an honest person is open to challenge. And there's a really, really good reason for that. And we've been going through it the last two days. The reason is that an honest person wants the truth. If I'm trying to decide whether to invest in a drug, and the truth is it doesn't work, it's much better for me and our entire company if I know that.
- 6.70** So even though I may not like the answer, it's way better if somebody gives me that honest truth. And so, you want people who are open, very open, to appropriate challenge, and they will reconsider their own thinking when you challenge them. That's what you're looking for. For somebody who just censors you and shuts it down, you

almost can't have a conversation with them because they're not, they're not willing to reconsider things.

Lecture 7: Creating Resilient Organizations

At a glance — Creating resilient organizations.

- 7.1 Most importantly, for the concept we discussed the last couple of days is how to put those tools and knowledge into our decision-making. We will share some of the stories. Knowing what we know, it doesn't guarantee successful outcome. Sometimes we have to make a decision, knowing there is a possibility of failing. But nevertheless, with this mindset, this is a perspective. This is an anticipated outcome. I think we will share some of the real-life experience with you. Hopefully, that you can learn something from our experience.
- 7.2 This is about five years ago, knowing I have no background in sales. I was a scientist in training, and I was managing technical aspect of the team, for most part, but Derick decided for me to take over the sales team. The very first interaction from our sales leader is to ask me for approval to fire a sales rep. Itself is surprising to me just because the salespeople are so nice. They are always very kind, very pleasant, and nice to be with. I was really shocked that he asked me to fire one of the team members.
- 7.3 Before I even got over that shock, I asked, I said, would you please provide the justification? What is the base for you to want to fire this individual? She's like, very easy. I will give you this, and we need to do this immediately. There are basically five different areas. The first one was that this person is delinquent and did not register calls. That's a requirement. She didn't do it. That's one reason. The second one is that she did not perform up to expectation. Because we have an expectation, this person is under-performing.
- 7.4 The third one is that, one of the objectives is for the salesperson to develop new offices. So this person did not develop new offices. Instead, she was just milking the existing offices. So that's unacceptable. Another one had to do with the expense report. Because the expense report, company policy, is that you can provide a training program Monday through Friday. This person provided it on Saturday. That is a violation. Then another one is, this person had called on behalf of the offices to the pharmacy, and in disguise, and did not disclose who she really was. That's a misrepresenting of the caller. It is a violation of compliance.
- 7.5 All these five make this person not really suitable to be a QOL employee. We should dismiss her immediately. I was like, Derick, what did you do to me? This is my first day on the job.
- 7.6 That was only a little bit. It gets worse. I'm going to do one quick tip. If you look in the margin of Management for Dummies, there's a tip over on the side. The tip in this circumstance is that, we talked about this yesterday, it is the sign of someone who doesn't have a very good reason for something. The sign is lots of reasons. Here's the long list. Somebody who really has a good reason for something, there's usually one or

two very good, really clear, very explicit reasons. So, the long list is a telltale sign to pay attention to. So go ahead, Weng.

- 7.7** So with this information, I said, where do I start? I have no experience. I have no idea what she's talking about. I don't even know the vocabulary, what logging calls means, all of this. But there's one thing that I always rely on, and I was good at, is look at the data. Now I have the head of sales telling me this. I went back to data behind the scenes, to pull all the information and to see if this is validated. To my surprise, that even the first one says, delinquent in logging calls, and this individual logged the second highest calls at the QOL team. We covered the entire country, 50 states, and this person had the second highest calls. Then I was like, that is interesting. The first one, I make a mark. I have a question.
- 7.8** Then the second one is that she did not perform up to the expectation. I pulled the three years of data to look at where her ranking was. She was consistently top five of QOL's sales history. It's just make me really surprised. I was like, wait a minute. There's something, it didn't add up, right? Now, the third one, she did not develop new offices. She's actually milking the existing offices. I look at the data shows the new offices. She was the number one of developing new offices for QOL of the year. I was just completely shocked. I couldn't even comprehend this. I was like, what's going on here?
- 7.9** Then look at the program. It has to be Monday through Friday. But I look at that, although it's true, most offices only work Monday through Friday, but this office works on Saturday. Also that day, they have a huge program. They have all the regional offices from everywhere. If you want to do a program for education, it is actually a good thing to have the program on Saturday and not during the week. And that there's all documentation throughout the correspondence to justify that.
- 7.10** I was like, why do they try to punish an employee who's willing to work on the Saturday on behalf of a QOL? That should be rewarded, right? Then the last one, the call on behalf of the office, that wasn't a crime. Because I look at the entire QOL database, that plenty of people call on behalf of office, and there's no evidence to show she had misrepresented herself. She never did that. There's never evidence of that. So, I have this one request for firing this person, and those are justifications. And then my research completely dispute any of this. With this, I went to Derick. I think we have a problem.
- 7.11** We did. And so the question, we've been talking about people and data over the last few days. Now we have a situation where what you're being told and what the data is saying are completely different. The most important fundamental point here is to capture both of those perspectives. Be careful about relying on just what you're being told because everyone has a story. They have a narrative, a goal, an objective, and the data that you're going to be presented with is going to be that which most strongly supports their objective, the list of all the reasons that were wrong.

- 7.12 And so then you have to ask the question of, what's the people issue here? Why would somebody want to terminate one of the top-performing sales reps? How would that happen? Why would a boss want to do that? It doesn't seem to make sense. It's illogical. And I think that, when you get into the understanding of people and personalities, what you can take away from this is that more agreeable, so this particular boss is more agreeable, and a more agreeable person does not like to be challenged. They view it as an offensive affront to their worldview.
- 7.13 They don't like confrontation. And so this is a particularly strong-willed, confrontational sales rep. She was good at her job. She still is. And she would challenge things. Her boss didn't like that very much. So that was basically an authoritarian play on, just do what I say; do, and don't fight me. Well, if you see that in an organization, you want to be very wary, because it's inevitably going to propagate and cause really big problems. Because if every other sales rep sees that we terminate one of the best sales reps, and they know, right? They know.
- 7.14 Because that person is challenging the hierarchical structure and not just, not just going along with what the boss says no matter what, but is willing to convey their real opinion on things— that openness, you want to have that in an organization to an extent. They see that this happens. Well, what's going to happen with the next sales rep, when they see something that maybe we're not doing properly in the organization? Are they going to raise it? It's not going to come up. Then, what happens is, the problems grow.
- 7.15 And this is what I meant on the first day, when I talked about how being receptive to information that you don't like, which is at this level of the sales management position — someone who tells you something you don't like, you actually want to be receptive to that. Because, look, if we're having a problem with a particular payer challenge or information flow through the pharmacy or whatever it is, if we don't know about it, how are we going to fix it? But if you allow the authoritarian control to propagate in an organization, what ends up happening is that the information flow stops, because people won't tell you about things that are wrong, if they're scared.
- 7.16 As a next step, remember, this is just day one, right? As the next step, I asked Derick if I could visit the top five sales reps. They're all throughout the country, to really get what's going on in the trenches. Derick was very supportive. I went just to visit, to spend a day or two, with our top sales reps, to get what it's like for them to have a day with offices and also get their feedback about what's their impression of our sales organization and leadership. And that was very, very rewarding and eye-opening for me.
- 7.17 So by the time I came back, Derick and I, we realized we have some work to do. Because, just like Derick said, the organization was running by a very nice person, but by fear, which is completely a surprise to me. Because I couldn't even comprehend, how can someone so nice instill so much fear in our team? Because of that, we decided

that we need to bring an experienced, seasoned, more competent sales leader, and that required some reorganization of the company.

- 7.18** I think that the most significant takeaway there is that, at a very fundamental deep level, and in any organization, what actually happened there is that those people that Weng spoke with trusted her. Because despite their fear, they were willing to tell her what was really going on. They were willing to tell her the truth. I think that you have to work hard to instill that openness in an organization. And it's an eternal battle. Because as you build a hierarchy and you have more and more people in structures, it is inevitable that you will have people in the organization who rule by too much tyrannical, authoritarian control.
- 7.19** But at the highest levels, if you can continue to communicate in a way that people believe, you believe, in the value of open communication and trust and the pursuit of truth, then they'll do what they did with Weng. Actually, they do it better with her than with me. I'm a little bit more intimidating, I think. They will very openly express their true opinion with Weng, which is great to have. What a valuable thing to have in any organization.
- 7.20** Well, I'll toot my own horn. It's hard to realize that she's better at it than I am. That's a challenge because it's like, it's frustrating. But at the same time, when you know that's true, to allow that to happen and that communication, it enhances the way that the entire organization functions, even though it's not necessarily, I'm not necessarily as good at it as she is.
- 7.21** At this point, we know we have a problem, and we know what will be the solution, and we actually went to work and identified a national-level experienced sales leader to replace our current sales leader. This is where Derick and I had our own gray shoe/pink shoe moment. The current leader, who used to be a trainer and who is excellent at training the sales teams, really good with the personality profile, really good in providing lectures, just phenomenal. Although he's not a really good sales natural leader, but we do need a trainer.
- 7.22** When we talk about what to do with him after we bring in another leader, my suggestion is that we will reassign his role. We're not going to change his title at the lower level. We're not going to reduce his salary at the lower level. We just move him to a role that is more suited for his talent. Derick said that's not going to work. I don't think that's going to work. But I could feel in my bones, that's going to work because I am so for second chance.
- 7.23** I said he's going to be very grateful. He has family with children. I said please, please keep him, and keep him in a different role. He's going to be so supportive of the new leader, and he's going to really shine at this new role. Derick said, not sure that's going to work. Anyway, we just agreed to disagree. But ultimately, I persuaded Derick to

allow me this opportunity, knowing what we know, just to see if it's going to work out or not. We don't know. What did you think at that time?

- 7.24** You expressed it well. That's not going to work.
- 7.25** We followed my suggestion. We introduced a new leader, and we reassigned the current leader to a different role and gave him a big piece because a sales meeting coming up. We need to train the sales team. There will be an opportunity for him to shine, and I was just so excited to give this task to him. It's a pretty big task. The date to the sales meeting become closer and closer and closer. There wasn't any deliverable.
- 7.26** He didn't do anything. It turned out that he turned really bitter and really resentful, became destructive. At the last minute that Derick and I have to struggle, try to pull this education material together, because the new leader just got here, didn't really have the background. It was extremely, extremely stressful, humiliating for me, and it's embarrassing in a way, but did I learn a lesson?
- 7.27** I did that on purpose. I figured what would probably happen. We can make guesses. It's very rare in any organization where, if there's any perspective that authority is being taken away, or you're being semi-demoted or anything like that, it's a crushing blow to people that is just not easily overcome. I don't know that I've ever, maybe in one circumstance I've seen it overcome. I figured what would probably happen. But at the same time, there's some experiences in life that are just much better learned yourself, and just go through it yourself.
- 7.28** I sat back and watched this happen to poor Weng, but on purpose. It helped us both grow through the whole process in terms of, it was a real-world, right in front of us, very genuine experience that we were going through at the same time with preconceived notions about what the likely outcome would be, and talking about it pretty regularly. Then, I was right on that one. I'm not right on all of them, but I was right on that one. I've been wrong plenty of times. Don't worry.
- 7.29** Yeah, but still, I'm very grateful that Derick allowed me to have this opportunity to learn and basically crash at the end. But I think an organization with this culture, as you can see, that allowing employees to make mistakes, as far as we learn from these mistakes and not to make the same mistake again. So this is one lesson I would like to share.
- 7.30** I have three daughters, and I say to my daughters, when somebody tells you who they really are, listen to them. You want to be forgiving. Because if you're not a forgiving person, you'll exclude everyone from your life. But you don't want to forgive and continue to have a close relationship where you have to trust someone who has not extended genuine, real contrition in terms of their understanding.
- 7.31** What happened here was, this person didn't say thank you for the opportunity that Weng opened up. It was exactly the opposite. It was, at every chance along the way,

I'm going to get you. I'm going to take every chance I can to undermine you and get you. Well, you have to protect yourself from that. At the end of the day, you want to be cognizant of the reality of what's really happening and not talk yourself out of how clear that reality is. Ready for another story? Quickly?

- 7.32** Any questions on that one? We got lots of questions.
- 7.33** Do you believe in giving second chances?
- 7.34** Absolutely. Yeah, absolutely. So the key to second chances is when someone understands the mistake. And people make mistakes. We've been talking about this the whole entire time. People make mistakes. If you're not forgiving of mistakes, you can't do anything in the world because they're going to happen a lot. So the key is to try to understand whether someone understands the mistake and has learned from it. And that's the underlying key, what you want to make sure the organization understands.
- 7.35** The mistakes are fine. What's not fine is covering them up, pretending like they didn't happen, and hoping nobody notices, and hiding under your desk. And then they just get worse and worse and worse. That's not okay. But making a mistake is absolutely okay and should be forgiven.
- 7.36** I think yesterday you made a comment like, you can't take your accountant and put them on the front lines to be a trainer. It's very different. So I'm just wondering—
- 7.37** So, why did I put Weng in charge of sales team? [laughter] So I think when I made Weng COO, which was, do you remember when that happened?
- 7.38** 2018.
- 7.39** 2018. So I recognized that her skill set was not pipetting and doing science, although it very much is. Her skill set was really managing people and processes and creating good operational flows within an organization. That's really more foundational than just the more technical aspects of what she was good at. So I realized that her skill set was much broader than just the technical. So that's the reason why I made that decision.
- 7.40** So we were talking about the groups yesterday in tribal. Everyone has a role that can be beneficial to survival. And I think that, when I read about the five personalities, mostly people that are in top positions, no offense, tend to be more conscientious and less agreeable. And that's what gives them that dominance. However, unfortunately, sometimes they can lack that empathy and that agreeableness or that understanding of other people's needs, or even if an employee challenges them, the ego.
- 7.41** So it's very difficult for companies to, let's say, work openly and honestly because there's no one to monitor that. Like I was saying yesterday, who can monitor, who can watch how that can play out? That's why people just don't confront it. Now, at the same time, I understand that Dr. Weng was being overly compassionate and giving this person another chance. But I also think we need people like her to open up those opportunities.

- 7.42 I think you answered your question.
- 7.43 Well, in a way, but what I was saying was, I think that people that have predatory intentions are very obvious. And if he was a person that was already firing someone because of the challenges and not really because of her abilities, then it's clear that he's on the narcissistic scale and probably predatory, things like that. My question is, how can bosses be confronted on this issue without getting fired? Because the thing is, they can find a way to fire you, if it's not for that. They can use your weaknesses against you, instead of being compassionate to help you grow, so they can exploit that. Not everyone's like Dr. Weng.
- 7.44 Do you want to answer the question? Let's both answer it. Let's see what we say.
- 7.45 Well, I think that there is a place to be nice. There's a place to be direct and transparent. I think depending on the situation. I think, by the end of the day, if the company has certain objectives, if that objective is being, I would say, interfered, it doesn't really matter how compassionate, how nice you are. But in this case, it's the opposite, right? So we have someone who is a high-level performer consistently, but with a more prickly personality, and the boss didn't like, and tried everything to remove that person. I think that could be a toxic culture.
- 7.46 This is what Derick mentioned earlier, is that we want to have a culture that people feel comfortable to come forward, and we recognize that. If anyone in the organization in fear of being honest and truthful, that organization is toxic, and that organization ultimately will not be successful. To change that culture, that's what we try to do, is to really remove the fear, that I know there are problems on a daily basis, and I get calls all the time. They feel okay to relate to me the problem. In turn, I will call Derick when we have problem, and I just know he's not going to fire me because I relate the problem.
- 7.47 So I agree with you. If, in a very individual basis, that fear exists, the problem is on the top. We need to fix it from the organization perspective, change the culture. We will remove the fear, and people feel comfortable to be honest and provide the honest feedback.
- 7.48 One is, you want to be encouraging and open, and you want to create an open, honest organization. But there's a limit to that because if everybody just expressed their opinion on everything all the time, I wouldn't be able to get through my emails in one day. So you can't have just, I think this. I think that. There's a constant competition of ideas that would be a huge problem.
- 7.49 So the way to manage the need, the goal of an open organization where people feel like they can trust their boss, is to make sure their boss understands this. That's actually the key. Because the boss needs to do the same thing that we're doing, which is encourage people to express their opinion and be responsive to it in an open way, but

also control the hierarchy is really what you want to do to make sure things are functioning well. So it's both pieces.

- 7.50** Derick, you mentioned that you didn't have a ton of success moving people laterally within your organizations, especially when it's interpreted as a demotion. So how important does understanding temperament when hiring, to avoid that, to make sure you're getting people in the right seats?
- 7.51** Well, I think understanding temperament when hiring is important in terms of making sure that you're aligning the job with the personality of the person. So it is a factor. I wouldn't say it's like, you can't just put things on a spreadsheet and score them and say, here's the ideal. You should take it into account. But it's not the end-all, be-all, by any means. I don't know that there's a personality profile that would be okay with a demotion. So I'm not sure that exists. Yeah, that's a real challenge. Do you want to keep going or do you want me to tell a story?
- 7.52** Yeah, you're going to tell a story.
- 7.53** You want me to tell a story. This one was fairly early on, and it had to do, when I first became involved with the company, it was very small. We had four employees. This was late 2010, early 2011. The company had a history of a reasonable amount of success. Things were going pretty well, despite its small size. We had a couple of consultants in place that had been working with the product and working with the FDA for quite a while. Again, everything was going well.
- 7.54** In 2013, 2014, we had to change manufacturers. The company that was making the product for us decided they couldn't accommodate it anymore. When you change manufacturers with the FDA, they come in and re-review everything and make sure that the new manufacturer is doing things in a compliant fashion that they put in place with the old manufacturer. Well, they came in and re-reviewed everything. Part of the challenge in the world of pharmaceuticals is that the standards are always evolving. The standards that exist at one point in time may not be exactly the same at a future point in time. They tend to grow, as with all bureaucracies.
- 7.55** The new manufacturer was not in compliance with the many of the evolved standards that the FDA had gotten to at that point. I was over my head. I didn't have the expertise to really understand what was going on. I also didn't understand that the people that were working with us at the time, who had been working with the product for quite a while, didn't understand the new challenges. From their worldview, the FDA was being overly onerous and restrictive and causing problems for no good reason.
- 7.56** I should layer in on top of all this. We didn't have any quality problems with the product. We didn't have any safety issues with the patients or anything like that. It was just a matter of new regulatory standards that the manufacturer needed to be following. The advice that I was getting, a couple of different things occurred simultaneously. One, we started getting letters from the FDA that were—a strongly

worded letter from the FDA, as a pharmaceutical company, telling you that you are not doing things as you should be doing them is really scary. It's not a lot of fun.

- 7.57** We were getting these "you guys are screwing up" letters. At the same time, the people who were responsible for fixing this were telling me everything's fine, and we're going to get through this, and the FDA's being unreasonable. I'm getting mixed messages. I'm reading the letters, and at this point, there's a fair amount that I don't totally understand. I don't have the experience to really understand why the FDA was so concerned with where we were versus where we needed to be. This all culminated in a meeting with FDA in January of 2015.
- 7.58** We had our manufacturing and compliance people there and me, and there were 25 FDA people in the room. And they were all yelling at me. They weren't even paying attention to the people who were explaining any of the technical stuff. They were just yelling at me. Basically, it was a lot of fun. Let me tell you, it was one of the most enjoyable experiences I've ever had. Now, I want to make one point here. They were actually right.
- 7.59** The most important thing to understand when you're in a situation like this is to take a step back and say, well, what if I'm interpreting this? Because I was more or less on the side of the manufacturing guys. We've always been doing it this way. Everything's fine. It's all working. What exactly is the problem here? I remember one of the quotes that I'm getting yelled at. In the middle of this, I'm, like, please help me understand here because I don't know what the real concern is. I laid out some of the reasons why there shouldn't be a big concern. No safety issues, no product quality issues.
- 7.60** They said, we have unknown concerns. How am I going to solve an unknown concern? I didn't have the concept of why they said that. It didn't make sense to me. Fortunately, I'd gotten to know Weng pretty well, maybe for about a year at this point. I knew she had a lot of knowledge. I wanted another opinion because the manufacturing folks at the time were telling me that the FDA is crazy. And as a matter of fact, at one point said, we should sue them. I'm like, don't ever say that again. Literally, that was my reaction. I think I threw in a couple of other words, maybe, that I probably shouldn't say in the Peterson Academy lecture.
- 7.61** And so I called Weng. I said, help me understand what's going on here because I can't figure this out. I don't know why we're being challenged so vociferously here by the FDA. She said, Derick, what they want to see from you is that you understand what they mean when they use the word quality system. They want to know that you're using the same definition that they're using. It's not a matter of defining the product itself as being high quality. It is a matter of understanding that the whole entire manufacturing system is designed in a way that a high-quality product will emerge from that, no matter what happens.

- 7.62 Because if something goes wrong through the whole manufacturing process, you're going to catch it, and you're going to understand it because you've got checks and balances all along the way. So you need to convey to the FDA that you understand that whole construct. That's what they're looking for. And so the key here is to, when you're confronted—and by the way, in the middle of this, the FDA said, you're going into drug shortage. Which is the same thing as saying, your company is going bankrupt. That's what I heard in the middle of this meeting. I don't know if we're going to survive this.
- 7.63 Nine years ago, we were this close to not making it. I had to solve this. The key to solving it was to understand that the worldview that I had was wrong, so I had to change. And I had to figure out what I needed to do in order to, and so Weng helped coach along the way, and we ended up, we had quite an adventure from there. We went through one more contract manufacturing, and we ended up building our own facility at the end of the day because it was the only way we could figure out how to do what we needed to do, in a completely satisfactory manner. Would you add anything to that?
- 7.64 I think it's a pretty good summary, but just a little bit, I received the call. I remember the date. It was January 21st. This is 2015. Derick asked me, I want you to take over this project. It was the call of my life. I just had no experience.
- 7.65 It's the call of my life.
- 7.66 I know. I actually know very little about manufacturing or building manufacturing facility. I know those, but I just love a good challenge. Remember, only just a year or two ago that I had the full responsibility, I was dismissed. That is really a very energizing experience.
- 7.67 But by the company she used to work for, not by me.
- 7.68 From there, we worked very closely. I'm not exaggerating, day and night. Because I probably reached out to a number of different contractors to see the time frame. They always say at least 18 to 24 months. That's the time we don't have. So I called Derick. I said, you know what? We need to do ourselves. And we did it in nine months. It was unbelievable what we did. And it's a testament to what human beings are capable of when we decide on a goal and resolutely focus on it at the expense of everything else. But when you decide, this is something that I'm going to do, you can move mountains. We literally moved a mountain. We signed a lease for a brand new facility. It was an empty box, basically looked like this. In July of 2017, I believe, I think we signed the lease in July of 2017. We had a fully built-out manufacturing team, quality system, completed manufacturing facility, and approved in September of 2018. We went from an empty box, a lease on an empty—that's never happened in the pharmaceutical world ever.
- 7.69 And the whole entire team. So all really powered by the knowledge we learned about this, how to identify people, all of this, because company at the time was small, and we

didn't have really the whole team. So four manufacturing team, the QZ, QC, and regulatory, everything. So that's amazing.

- 7.70** To credit the FDA, they allowed us to continue to sell the product through that whole process from the old manufacturer. They gave us just enough of a lifeline that we were able to survive through that extraordinary challenge. In a way, they really worked with us. They weren't always nice, but I think they understood that we had the right goal and we were doing what we were supposed to—we were aiming towards doing what we were supposed to be doing from their perspective. They worked with us, which was great. Any questions on correcting a disaster?
- 7.71** You mentioned that the, people call it unknown problem. But, Weng, how did you get to the conclusion that the unknown problem was that they wanted to make sure that you have a way to prove that everything was—if there was a mistake in the process of manufacturing that you were going to catch it. How did you get to that conclusion?
- 7.72** Well, so what happened was that the product was approved 25 years ago. FDA has a very different standard at the time. But when we received FDA letter, this is decades later, FDA had increased the requirement from the previous standard to pharmaceutical standard. That is now you just go through the action of making product. You have to have a quality system in place which is documented in every step of the way of your manufacturing process. That is the mindset the previous manufacturing team didn't have. I actually just come back from, remember, I was manufacturing intraocular implant. Our requirement by the FDA is the highest here. Previous, the manufacturing is here, so now the requirement is here. So I'm very familiar with the quality aspect of the manufacturing process.
- 7.73** Weng understood the lexicon of what they were saying to us and said, this is what we need to do. That was how we answered it. She just had to convince me.
- 7.74** Two questions. One is about timeline. So I'm just curious between the first story where it sounded like Dr. Weng, you were in charge of sales, and in the second story you were in charge of operations, how those kind of order? And then for Derick, my question for you is, it seems very obvious that Dr. Weng is a very useful, is able to take on many types of challenges or many different types of projects. And as a leader, I guess, how did you identify that in the pool of people whom you were selecting at the point when you hired her? And also, I guess, how did you, as a leader, facilitate that flexibility, or what choices did you make to allow that situation to emerge?
- 7.75** Yeah, good question. So I would say it's a couple of things. One, you're looking for, so in terms of the sequence, the first question. You were COO in 2018. The first story that we told about the sales team was roughly 2021, 2020?
- 7.76** I think it's 2018. No, 2019?
- 7.77** No. No, it was sales management. When did you take over the sales team?

- 7.78 Oh, 2019, I think.
- 7.79 It was 2019. So that was the sequence. In terms of how to identify a really good operational person, I would say, it's a couple of different things. It's a pattern of getting things done and getting things done well. So, what you want to do, when you're growing an organization, is you want to extend trust and delegate carefully. So a little bit of responsibility, see what happens. A little more responsibility, see what happens. And so I basically did that with Weng.
- 7.80 First of all, she got a manufacturing facility built in a year, so that was pretty clear that she could get things done. Do people get things done and do really well? That's just a really clear sign. You can actually rely on the outcome in that circumstance. It's actually hard data that tells you, well, this is exactly what you'd hoped would be the result. It was done in an extraordinary fashion. It's not actually that hard to tell when that happens.
- 7.81 Then I would say, the second piece is when you watch somebody's decision-making process, and there's a process of extending trust and relationship with someone who's reporting to you that I would describe like this. It starts with the need for over-communicating because you want to understand that person's perspective and worldview and vice versa. She wants to know how I think and vice versa. We talk about probably more things than we necessarily should. Then, after a while, I realized that she's making decisions the same way that I would make them. I know I can trust her to do things really well and to come to the same conclusion.
- 7.82 Once you have that trust, you just can extend more and more responsibility because it's like, well, why wouldn't you? Somebody's doing a really good job, and they're thinking about things from your perspective that you think are right. On top of all that, she's also sometimes in the boundary of inappropriately challenging.
- 7.83 She can be confrontational when she feels like it's needed. And that's actually a super valuable thing as a leader because it's unusual. Most people just won't do it. And so you want to encourage it when it happens because, again, it's hard to find advisors that will tell you what you need to hear, rather than what they think you want to hear. Do you want to go through the investment we made and ultimately sold, how we got to know each other?
- 7.84 From my perspective?
- 7.85 Yeah. Why don't you kick that one off?
- 7.86 You're probably curious after I was dismissed from my role, from my own company, what I did next. The next step, I sent an email to inform everyone that I have now a new email address. Before it even hit, it's literally like 30 seconds. Within 30 seconds, I got a call from this guy that I worked on almost collaboratively from a large pharma

company before that he left his job, and he has now with, I think, several opportunities with the Mount Sinai Medical School.

- 7.87** He called me and he forwarded me two opportunities. One is a central nervous system, one is a cardiology project. I take a look. I said, Not interested. Do you have something else? Yeah, we have something else, that's already taken, it's not available. I said, Send it to me anyway. He sent it to me and took a look. That's it. That's the one I really want. He said, Well, it's not available. So I said, Then don't call me anymore. When it's available, you call. So a few weeks later, he called. He said this project is available. Would you come to New York City? So I went to New York City, talked to the investigator, and look at this project.
- 7.88** It's something completely interesting, and within my expertise. I could do it myself. He said the goal is that we're going to develop this into a product. We're going to sell it within 18 to 24 months, whatever. I took over, started to work on this, and a few months later, got a call that said, you need to come to New York to present your data to our investors. I pulled my presentation together, took a train to New York City, and presenting, the investor was Derick. That's how we met.
- 7.89** The longer-term outcome as that story grew was that we learned that the way that that product was being managed, there was a person between Weng and me. He was over his head and was having some struggles. Eventually, I could tell that Weng was trying to do things the right way and trying to make sure that all of the boxes were checked properly. The facility was built in a year, metaphorically. The guy who was senior was not driven by optimization, I guess, not driven by doing things optimally, more like, let's just piecemeal it together and we can sell it to a big company.
- 7.90** Ultimately, understanding that incompatibility took a little while because I didn't totally recognize that. I could see signals, but I didn't totally recognize it until I experienced some of the challenges myself. So Weng was experiencing more challenges. This was early. We didn't know each other that well. She was experiencing more of the tyrannical-boss challenges that we described in the story earlier and telling me about them. Keep in mind, I didn't know her very well. I didn't know that I should be trusting her worldview.
- 7.91** So it's hard when someone is telling you something that's inconsistent with what you see, in terms of the way that you're being treated and the way the person is interacting with you. It's incompatible. And so, it's hard to, well, what's the real truth here? And it's also hard because, as human beings, we want to ascribe motivations to people. Why would Weng be coming and telling me this overt truth about what's really going on behind the scenes, without some—does she want to be in charge? Does she want to be CEO of this subsidiary? What's going on here? You can't help but wonder, what's her true motivation?

- 7.92** As I got to know her better, I learned later that her motivation is just truth in doing things well. It's not for herself. Now, ultimately, if you pursue truth and you do things well, you'll benefit tremendously. I know it was frustrating, but we got to know each other much better in that process.
- 7.93** Sometimes I said that it took me such a difficult journey to get to know Derick. Even this opportunity was introduced by this person who was just really, really difficult. The thing is that he had two faces. He treated me really poorly, but treated Derick nicely. It's hard for him initially to recognize that. But I think his true colors ultimately revealed himself. The story is that the project was successful, we're able to exit. And he actually was out of the picture, and Derick and I started to work together. He called me many times, please come work with me. I said no. Once he said, but I'm the one to introduce you to Derick. I said, yes, I am very grateful.
- 7.94** Yeah, so I think the key to that is, once you do have that trust established with someone in your organization, you want to understand that you can rely on that. You can rely on their opinion because what you're really looking for at the end of the day in running a business well is people who have the same objective that you do, which is to run the business well, right? Not flatter the CEO, which is what happens in most companies. You want to run the business well. And that is that non-selfish motivation that is pretty rare in people. If you find someone who has that, try to work with them because you'll ultimately just benefit tremendously.
- 7.95** I do want to share one story, just to complete this. Just tell you how strange the situation. This person had no technical expertise, but didn't know that. He was, when negotiated with a contract with some of the contractors, and made a huge mistake. He decided it was my fault. He called me up. He said, you need to make sure I know everything. If I make a mistake, it's your fault. And to the extent that he actually reached out to Derick to fire me because I was responsible for it.
- 7.96** That's happened multiple times, by the way. I've had lots of people try to get rid of you all along the way.
- 7.97** I know. Isn't that amazing? I asked Derick. This is fascinating. Why I should be responsible for his ignorance? That's a lot of responsibility because he doesn't know. How do I know what he doesn't know? To take over that responsibility, what a job description is for one individual responsible for another individual's all the mistakes?
- 7.98** That's the opposite of good leadership. It's the exact opposite. What if I said to the organization, you're responsible for making sure that I don't make any mistakes, and if I make them, it's your fault? How long would you work for me? You want to do the opposite. The mistakes are my fault, the successes are your fault. That's someone people will follow. The \$800,000 column? Yes. It was a filtering system that cost \$800,000.

7.99 Right. He thought it was \$30,000, and he said it's my fault. I was like, who come up with the \$30,000? He did. I said but why would that be my fault?

Lecture 8: Leading with Integrity

At a glance — Leading with integrity.

- 8.1 So I'll attempt to sum all of this up and distill it down into some basics that integrate everything that we've gone through the last couple of days. I think the most important thing at top of the list is to try to understand it, to listen to what people say and why they're saying it, and how they say it, rather than who they are. Because I think who they are, if you rely on who they are, you too easily fall prey to the appeal to authority, which is a big trap.
- 8.2 So again, you want to focus on, well, this is a good example, right? Have we done a reasonably good job, the last couple of days, of trying to convey to you guys why we are saying the things that we are saying, and the depth of it, and things like humility and openness to information? You've got to be willing to receive new information without having a reaction to it. That's actually a significant key to—the fundamental of good decision-making is to allow new information to come in without having it cause you to respond. I'm going to start this last section with a particularly provocative example of exactly that.
- 8.3 We talked about climate change a couple of days ago and how it is a more emotional, more political issue for people that can instigate a particular emotional response in individuals. On the next slide, I'm going to present information that completely counters everything you probably believe. I'm not actually going to do that. I'm not going to do that. What I'm going to do is point out that that statement causes a certain emotional reaction in people. If you think I'm going to say something that's contrary to what you fundamentally believe, you are already preparing yourself to challenge the information.
- 8.4 It's happening already. It's happening in the cortisol in your body. If you think I'm going to say something that supports your fundamental belief, you're excited. You have a positive, enthusiastic emotional response. Both of those are wrong. What you want to do is fundamentally do what Aristotle told us to do 2,800 years ago, which is the mark of a great mind, is the ability to consider information without accepting it. You want to think about it objectively. You want to realize that the exercise that we just went through is what happens to human beings all the time.
- 8.5 We decide what we want to be true before we have the information that tells us whether or not it is true. All you need to do to overcome that barrier to good decision-making is be aware of the fact that it exists, so that you can competently second-guess yourself, knowing that it's going to happen. Ideally, the best way to do it is to try to understand your own bias. If you can clearly explain your bias to yourself, if you know, I deeply believe this, and I want this particular pathway to be true, then you have a reference frame for being able to consider objective information.

- 8.6** So that's the overall message. I don't have any graphs or charts. There are no more statistics. We're going to talk conceptually here for a little bit. Another key point, I think what Weng and I tried to make this morning is, surround yourself with people strong enough to change your mind. Those are people who can confront you with information that you don't like.
- 8.7** I think there's actually a scale here. The more you don't like the information, the greater your emotional response, the angrier it makes you, the higher the likelihood that some intuitive aspect of your deep personality is telling you, this is something that you actually really need to listen to. This is something that you need to hear. So use the emotional response as an indicator of the value of the information. It's a great signal flare that tells you whether or not something is valuable.
- 8.8** The opposite side of the equation is also true, which is, if the information is, you really like it, and it's reinforcing an opinion that you already have, what it's actually doing, if you're wrong, is it's reinforcing your biased perspective. It's creating more bias. It's adding to it because it's very easy to take that into account and say, I knew I was right. And so you want to be aware of both of these. Information you don't like, signal to try to accept it. Information you do like, signal to try to second-guess it.
- 8.9** Just one last point on that. I think the people, you do want to try to find people who are willing to challenge you. It's key to a well-run organization. I want to talk about focus here a little bit. This is Steve Jobs: "People think focus means saying yes to the thing you've got to focus on. That's not what it means. It means saying no to the hundred other good ideas." This is excellent advice. If you're going to be successful, you have to set a goal. There is no mechanism of success that does not start with a goal. No exceptions. That's an iron-clad rule.
- 8.10** You can't just make yourself better. You want to make yourself better towards a certain goal. Because what it does is, it concretizes, it gives you a mechanism for deciding what it is that you're going to do every day. Is this in alignment with my goal or is it not? You can't be successful sailing to a port that you haven't defined. Now, the hard part about goals is that goals are scary. And what goals do is they set the boundary for success. But the other side of that boundary is failure.
- 8.11** So once you set a goal for yourself, it's scary. You've got the possibility of not making it. And that causes a reaction to be wary of goal-setting as a practice. Again, for good reason, it's scary. But this is something you've got to transcend if you're going to move towards success, ultimately. When you do set a goal, another way to think about a goal, is to re-conceptualize it as a mechanism for screening out all of the things that you're not going to do. It's a way of deciding which activities fall into which category. Is this helping me get towards where I want to be, or is this an unnecessary side street that's going to distract me and cause me not to get there?

- 8.12** Now, I can tell you from personal experience that particularly open, creative people, if you are high on the intellect scale, in particular on the openness score of the Big Five personality profile, this is challenging because you're excited about ideas and new opportunities. You're enthusiastic about doing new things, which is a wonderful way to distract yourself. So it's an absolutely wonderful way of distracting yourself.
- 8.13** So here's my trick, if you have this challenge, for managing this. What I do is I try to channel that creativity within the bounds of what it is that I'm trying to achieve, rather than allow it to distract me with other things. Well, is this exciting new shiny thing on the path to the goal, or is it something that's just exciting and new and shiny, and should be moved aside? I often describe myself not as CEO, but as Chief "No" Officer. It makes people really like you in an organization when you're constantly having to say no. But that is the key to success, is to say no to all of the good ideas because they're a massive source of distraction in life.
- 8.14** My rule is a rule of three. I think there's magic to the number three. There's even more magic to the number one. But you want to limit the big things that you really want to focus on to one to three. If you are in the business of opening new stores, how many new stores am I going to open this year? What other goals? None. If you can just focus on that, then you dramatically increase the likelihood of success.
- 8.15** Why? You have 168 hours in a week. You have to sleep some. So whatever is leftover, let's say 8 hours a day, so one hundred hours, if you're really, really focused on something, you've got 80 to 100 hours in order to achieve it. If you have two goals, you have 50 on your primary goal. If you have three goals, you have 33 on your primary goal.
- 8.16** So the more you can focus your time and energy on the thing that's the most important, the higher the likelihood of success. Mathematically, we just went through it. It's pretty easy. Because every minute you're spending on something that is not targeted to the number one goal is waste.
- 8.17** What's the leader's job? Set good goals. Listen to people, understand data, analyze these things yourself, try to understand that different perspectives and worldviews are key. You have green shoe people and pink shoe people, and you want to listen most closely to the pink shoe people if you're a green shoe person. Because by definition, if you see a pink shoe and I see a green shoe, I cannot see the pink shoe. By definition, it is a blind spot for me. And so the only way to reconcile that, my inability to see the pink shoe, is to talk about it.
- 8.18** You've got to work through these things because the huge advantage of differing worldviews, if you can, if you can allow for differing worldviews to come into your perspective on life, you have more eyes. You have a broader perspective. You want to listen to the pink shoe people if you're a green shoe person and vice versa.

- 8.19** Try to identify the true advisors, not the fake-sperts. We went through some tips and tricks on how to do that, how to understand who's genuinely trying to give you good advice that's driven towards honesty and the truth and doing good things for you and for the company. Then once you have gone through the process of understanding the complexity, whether or not we're going to invest in this new drug, carefully consider how we're going to design the statistical analysis plan for a clinical trial.
- 8.20** Once you've gone through all that, make a damn decision and stick to it. Unless, here's the hard part, information comes along that tells you you're wrong. Because the absolute key is to be open and mercurial, which is the ability to reconsider. And this is why I have stressed over and over again, the word certainty is a cuss word. It is something you want to get out of your vocabulary. Because here's what certainty does. Certainty causes you to be less open to considering new information. And you need it.
- 8.21** You want to shoot for about 80% certainty. If you're over 80%, reconsider. If I'm so absolutely, unless it's just extraordinary, the 70% response right here and 30% response right here, you can be 95% certain in that circumstance. But most of the time, in most of life, if you are anywhere near 100%, you're not second-guessing yourself enough. It's a sign that you need to be reconsidering other information. You got to reconsider, and the only way to reconsider is to open the window of uncertainty a little bit. If you don't, you won't look at the information. You'll be confirmation bias-driven.
- 8.22** I'm going to go through a model here called capital allocation. This is something that is taught in business schools, and it's a framework for how to decide on where it is you should be spending your time and energy. It mostly is written for business, but the gist, the basic construct, applies to everything in life. What you want to do, the capital allocation construct says, what you want to do is create a hierarchy, a priority, of the things that you want to do in the world, a hierarchy of goals or a hierarchy of possibilities. Let's start with possibilities.
- 8.23** Let's say you haven't figured out what it is that is the optimum goal for you. You're 25 years old. You don't know exactly what it is that you should become in life. You don't know what your career is going to be, right? Well, you could set a goal that is career-possibility-enhancing, that's not the final solution. And that's, again, the open door to let other things in. So you just want to pick a pathway, and you want to stay on that pathway until the world is telling you that that's not optimum, and maybe you should tilt.
- 8.24** How do you generally align the pathways? Which door should I go through? What you want to do is take the realm of possibilities and rank-order them the best you possibly can. So this is, I think, the best. This is the worst. And in general, you want to take your top one, the rule of one to three, your top one to three possibilities, and focus all of your time and energy on number one until that bucket is full. So fill up bucket number one. If it's the most important thing that you should be doing, don't do anything else other than the first bucket.

- 8.25** So I'll give an example in pharmaceuticals. We have three different drugs that we're developing to try to get approval. Take the one that has the highest likelihood of approval. Take your best people and your capital resources and focus on that to the exclusion of everything else until you can't focus any more resources. You've exhausted your ability to put everything into the thing that's the most important.
- 8.26** And then it's not part on bucket one, part on bucket two. You go to bucket two after bucket one is full. So every bit of energy that you can possibly put into the most important thing, do that. If you do have some time and energy and resources left over, then focus on the next thing. You can get to bucket three. I would say, if you have more than three buckets, you need to throw a bucket away. Three is about the limit for any competent person to be able to navigate at all intelligently. Because, again, of the rule of 100%, 50%, 33%, you just get distracted. Three is borderline too many.
- 8.27** This is a good one. Shane Parrish: "One mark of a smart person is the ability to learn from people they don't like." It's particularly tough for the right side of the room, for the agreeable. It's harder. Now, if you're more disagreeable, you're more confrontational, so you'll immediately attack the information. If you're more agreeable, you may accept it, but then you reconsider it in light of the directness and maybe rudeness that a more direct person will communicate it. Try to set aside the personality profile.
- 8.28** Again, put the objective information out on the table and consider it independent of the personalities that have presented it. You're trying to just consider the information all by itself, essentially. All right, so the communication profile, how people are communicating with you first. Does this person really understand what they're saying? Are they able to explain it to me like I'm 10? Are they able to give me good metaphors? When I challenge them and ask them questions, do they just dismiss it and put it aside, or do they carefully consider what I'm saying and explain the background?
- 8.29** What you want is somebody who can paint the whole picture for you. They can explain all of the reasons why they got to the particular perspective that causes them to end up where they are, and they can communicate that clearly. Or are they telling you what they think you want to hear, or what they want to say to you because they want to influence you to come to their worldview? Are they trying to get you to believe what they believe? Are they trying to give you a better objective view of reality?
- 8.30** When you're in a more senior position, what they think you want to hear is one of the biggest problems. So in any context, when you're CEO, it gets tough. Let me tell you, it is not easy. When you walk into the room and you sit down at the table and there are eight people around the table, in most circumstances, what they're trying to do is figure out what you want to do as CEO, and then tell you that. Which, again, in most circumstances, and most people want to have their own worldview reinforced. So they like hearing information that tells them what they want to hear. So it causes the

organization to target that as the goal. People learn to tell the CEO what he or she wants to hear.

- 8.31** And so you want to be particularly—there are two ways to mitigate that. One is to be penurious or selfish with expressing your opinion openly, so people don't know. That'll drive the obsequious types really crazy. They come into the room and they're like, what is it that he wants to hear? I'm not telling you. I want to hear what you think. Not what you think I think. That doesn't help me. Just beware of the fact that when you're in a more senior position, that starts to happen. People start to try to develop relationships with you, based on what they think you want to know.
- 8.32** The signs of a data diviner, somebody who can find the water, so to speak, find the truth. We've touched on this, but clearly state the priority. A proliferation of too much information. If they're just giving you the kitchen sink, all the reasons on multiple pages with lots of graphs—"The answer is probably in here somewhere. I'm not sure exactly where it is"—versus a clear communicator. It'll be one clear priority. Here's the answer to the question that you're trying to investigate.
- 8.33** What you'll see, a sign of somebody who is distracted by too much information, is, if you say, I want you to go look over here, there are these 10 possibilities that we're considering as an organization, and I want you to come back and tell me what the top one to three really are. Where should we be focusing our time and attention?
- 8.34** When someone cannot discern the priority, which can happen for lots of reasons, maybe they don't have the skills or experience, whatever it is. They will come back with way too many answers. A sign of a lack of understanding of what the priority is, is too much. People who can discern do not give you too much because they understand the capital allocation rule I highlighted a minute ago. Let's focus on the one thing.
- 8.35** Maybe redundant, but this point is so important. Real dialogue is where two or more people become willing to suspend their certainty in each other's presence. That is opening the door to the possibility that you might be wrong, which is the key to being right. I might be wrong. If I am, I want to know; how else am I going to get to right? If I have a "damn the torpedoes" attitude, it's a balance. You do want to be stubborn and strong-willed and focused on the goal, but open to new information along the way.
- 8.36** We touched on this one some yesterday. It's a quick story. The picture helps enhance the point. It's a quick story about World War II airplanes. They were flying over Germany. They got shot at a lot. There were all these bullet holes in the planes. Bunch of engineers analyzed the bullet holes. So we need to put extra armor over the areas of the wings where there are bullet holes because that's where the plane is getting hit. One guy comes along and says, no. We need to put the armor over the areas that did not have bullet holes because those are the planes that didn't come back.
- 8.37** It's hard because 100 people are telling you to put the armor over the red dots, and one person comes along and tells you to put the armor over the areas where there

aren't any red dots. It's a challenge because most of the time in life, when things like this happen, it is that one person that you need to listen to. And so you have to be open to listening to that one likely disagreeable, cantankerous, rude, direct, more autistic, straightforward. This is the truth. Because otherwise, you're going to put the armor in the wrong place.

- 8.38** How do you maintain the one to three rule within an organization? You want to beware the "Hydra-headed idea monster." If you are too—you want to open the door to ideas. You want people to be able to express their opinion. But if you do that to the exclusion of drawing boundaries, it's a real challenge, right? I want to hear all the different ideas. We have 100 people. We got a sales meeting next week. I want to hear all the ideas. Well, I'm going to get 80 ideas.
- 8.39** What are the top three? I have to decide, ultimately. Now, if I open the door and say, everybody give me their ideas, and I get 80 ideas, and I pick three, which is my job—I'm the Chief "No" Officer—I'm going to say no to 77 ideas. What happens when the CEO says, no, I don't like your idea. I'm going to pick somebody else's. It's pretty offensive. People don't like—you think about it from the perspective of an individual. This is an opportunity to present something they've been thinking about deeply and working on and carefully considering to the ultimate big boss. I've been thinking about this. Here it is. They're giving you a gift. It's a really valuable thing.
- 8.40** But if you're not very, very careful with managing that situation, when you have to say no, because you do, you're going to actually damage the relationship. Because what you've done is, you've taken something from somebody that they've been working on and thinking about, and then said no. And you've cut it off and you've said, no. That's actually a bad idea. No, we're not going to do it. I don't say it that way. I say, that's fantastic, but we don't have time to do that right now. We've got to focus on these other things.
- 8.41** But regardless, you have to be careful with managing information flow because if you're not somewhat aloof, if you're too open in a more senior position, you will invite exactly this problem. What I don't want to have happen is 77 people upset with me because I've said no. The simple way to do that is to have it filter up through the organization. We want to hear your ideas, would absolutely love to. I am not saying that disingenuously. I mean it. If there's something wrong out there that we're not doing well, I want to know about it.
- 8.42** But there's a proper channel for communicating it. Let your regional manager know. They can distill the different ideas down. They can let the sales manager know. That process, that hierarchical process, ultimately, the top two or three ideas will take care of themselves without me, in a more senior position, offending anybody. You don't want to jump around the hierarchy. You want to be careful about the level of relationships that you have with different people throughout the hierarchy and keep it

on a strictly friendly relationship-oriented level, but not on a too much of a business level.

- 8.43** Because it can be confusing to people to believe that they have a close relationship with the CEO, which seems like an exciting, valuable thing from their perspective, but it's a dangerous thing to do. It'll disrupt the hierarchy really badly. So be a little bit aloof. When you ultimately communicate to people, you want to be pretty clear and direct. So you go through this whole process of analyzing all these things and data and p-values and fancy stuff, and then once you've done that, you want to simplify that message for the organization.
- 8.44** Because if you explain too much about the whole sequence, when you explain the entirety of the background and the reason and the why for your decision-making, what you're actually doing is you're inviting the ideas. You're inviting challenges. You're saying, here's the whole thing. What do you think? It's like, well, what's wrong with that? Somebody has a good idea. Why wouldn't I want to hear it? If we're going to invest in a drug and somebody thinks it's going to go wrong for whatever reason, I would want to know that.
- 8.45** The reason you want to be clear and direct and simple in your communication is because if you broaden it, you invite the challenge. And once you invite the challenge, you can't turn off the fire hose. Because if I describe in intimate detail all of the mechanics that have gone into why we're going to target our sales and marketing efforts at these particular doctors—here's all the work we've done. Here are the analytics and different products we've looked at and diagnostic tools they used and tons of analysis and data.
- 8.46** Somebody goes, why don't you do it this way? Why don't you think about this? If I get 200 of those, we're stuck. You can't be too open. You can't be too receptive to too broad a level of information. It's simple, here's what we're going to do, which is essentially a goal. Here's the goal. We're going to find this many new patients this year. State a time frame. This year, 5,600. This is our new patient goal for this year. Any questions? Super simple.
- 8.47** Then, what you want to do is divide that up and distribute it throughout the organization so that people know what they are supposed to be doing. Each individual salesperson has some piece of that goal. Here's your goal. Here's your goal. Goal-setting as a leader is actually comforting to an organization because people want to know what it is that sets the boundary for them being able to declare success. They want to know what they have to do in order to be able to have confidence that they're doing the right thing and contributing to the organization.
- 8.48** So you want to break up the pieces. Manufacturing team, I need you to make this many batches this year. Sales team, I need you to find this many new patients. So that everybody understands. They want to understand the high-level goal. These are the

new patients we're going to find. And then you break it up into the pieces and distribute it through the organization. Each individual understands what it is that they're supposed to be doing.

- 8.49** How to fail. The ego jumped to a conclusion too quickly. I think in some ways that is believed to convey intelligence, like a smart person will find the answer really quick, and they'll jump to the answer. That's a danger zone. You want to be careful with that. You don't want to jump to an answer too quickly. Once that single perspective, that worldview, has been decided upon axiomatically, then it causes people to incorporate certainty into that worldview. And so then they don't want it challenged. It becomes a "sacred cow" that can't be challenged.
- 8.50** And then what happens is, they look only at the confirmatory data that tells them that that worldview is correct. They will accept the "fake-spert" opinions that reinforce their worldview. So a consultant comes in. I want to invest in this drug. It's actually a bad idea. It's going to fail, but I think it's great. If I'm not careful, I invite an expert in, and most consultants will do this. They're people, right? They figure out, well, I really want to do this. Well, they're going to tell me that it's a great idea.
- 8.51** That's going to happen automatically because they want to be invited back. You have to be really careful with, I want an objective opinion, genuinely. You start looking for information and data and and "fake-sperts" that support your worldview. Then you ignore, challenge, set aside anything that challenges it. This is the path to failure. The reason I know this is because I've seen it over and over and over again. I can't tell you how many opportunities we have had to invest in new products that have occurred because of this. It's unbelievable. It's everywhere.
- 8.52** Then what happens is they wrap their ego into that decision. So they associate the decision to invest in that particular drug with themselves, so that anybody that challenges that decision is perceived as challenging you as an individual. That's really scary for people in the organization because it just shuts down challenges. They won't be open and honest with you because you defend the perspective like it's a threat to you as a person. Your persona can get really, really defensive of prior decisions.
- 8.53** And then, because of this whole sequence of bad ideas and gathering information from the wrong perspectives and tying your ego up with the idea, it fails. And you go, it's not my fault. I did everything right. I worked really hard on this. I've got expert opinions. I tried really hard to make this work. I know it's going to work. We need to run another trial. So, not only do you have one failure, but you allow yourself, because of the robust commitment to the prior belief, to continue. You have a sequence of failures.
- 8.54** The key to all of this is fundamentally the recognition of the reality of a failure. If you could do that in any aspect of life, it's not just investing in drugs. If you can do that in any aspect of life and acknowledge it, then you will stop the sequence of bad decision-making. Because it's, something in the world is telling you, hey, this wasn't done right,

and you need to listen to it. Failure is a pretty good indicator. It's a pretty good smoke signal if something wasn't done right.

- 8.55** And so, if you can do this, if you follow this page and follow this pattern of belief, of trying to avoid continuous failure, what it actually does is it implants in your brain a pattern of behavior that will lead to a failure. So what you start to do is you start to see the things earlier that, hey, this isn't a very good idea. You course-correct earlier. You realize that you're doing the wrong thing earlier once you realize that the thing that you're doing is more likely to lead to a bad consequence.
- 8.56** We're going to give a quick example. We're going to pick on this poor company. This is Wheels Up. It's a public company, but they own a private aircraft. I'll go to this slide real quick first. This is their stock chart. It's not inverted. That's actually what happened. So this is \$100 a share, I don't know, multi-billion dollar company, \$5-billion company or something. Now it's a dollar a share, right? The private plane business is very challenging. Planes are really expensive. Fuel is expensive. It's a whole lot of overhead, and it is not an easy business to manage. This company hasn't done very well.
- 8.57** The reason they didn't do very well is they got too excited about the growth opportunity. So they spent all of their money on sales and marketing, getting new planes and growing as fast as they possibly can, without someone in the process along the way going, are we really making money here? We go public. We get a bunch of money from the public market, and spend it all on planes and marketing, grow like crazy, and lose money. That was, in essence, the business plan that ultimately occurred.
- 8.58** Now, here's the problem with all of that, is that May 11, 2023, which is down towards the bottom of that green chart I showed you a second ago, "We are not considering bankruptcy. While we realize our recently announced changes may create some short-term pressure"—that's a cute word, short-term pressure—"we have conviction that the long-term outcome is strong for our business." Now, how self-deceptive and deluded do you have to be to make that statement when you're at the end of this chart? What is the objective reality telling you? You're losing \$150 million a quarter. Your company is going bankrupt. You have no money. Your stock price has collapsed. And you say, this is just a short-term problem. Everything is going to be fine. That is a seriously deep level of self-deception.
- 8.59** And this is the point that I'm trying to make. This type of thought process will lead this person to say, if we can just get, if we can just get a little bit more money, we'll turn it all around and we'll make it successful. If we can just do this, somehow it's magically going to improve. The reality is, this is one of the best disasters I could find. It's a complete and total disaster. And by the way, I think people have placed something like a billion dollars of deposits for future flights with this company that's now all gone. So your customers aren't going to be very happy when their indirect investments—it's just to accept the reality of what reality is telling you.

- 8.60** So ultimately, the sequence of good decision-making is to define and control a process. We gather information, try to understand the people that are giving you information. If you're going to make complex decisions, you have to understand statistics. You want to be a blind spot and a bias bloodhound. I should have said that out loud before I wrote it —a blind spot and a bias bloodhound. You want to understand the biases and blind spots.
- 8.61** When you're a green shoe person, you need to consider the pink shoe perspective are prevalent in every single human being. Understand capital allocation and how to prioritize those three things, and focus on number one most. Make the decision, revisit and revise when you need to, and then communicate at a high level the goal throughout the organization with limits.
- 8.62** Characteristics to look for if you're trying to decide whether to go to work for a particular company. What are the signs of a leader who has these things that we've been talking about? I think they're willing to mentor. They want to help people rise up through the organization. They exhibit humility. They're open to new ideas. They readily accept new information. They're decisive and clear, but willing to change their minds. They distribute credit for success, and they accept responsibility for failures. Most people in the real world do the reverse.
- 8.63** They're annoyingly insightful. People who can get straight to the point, like, how did they do that? It's because they're thinking at a deep level, and you want to notice it. They can be disagreeable and argue effectively. They do not attack people's personalities, which is an ad hominem attack. They address data and facts. They're stoic, low in neuroticism, typically. Can be slightly aloof and unfriendly. That's actually a good trait. It gets back to the point of prioritizing well and not accepting too much information too easily.
- 8.64** In all cases, the most important thing at the top of the list is a deep level of integrity. Integrity is when someone's words and their thoughts and their actions are all aligned. You can't see somebody's thoughts. Maybe when Neuralink comes out, we can walk around with our thoughts over our heads. That'll be fun. But you can't see their words and their actions. When somebody says, yeah, I'll do that. I'll get it done. Then there's no activity and no follow-up, and they don't actually follow through. Well, that's not somebody you can rely on because if you have to ask, again and again.
- 8.65** What you want is somebody who you say, we need to get this done, and they go get it done. I often tease the thing that I love about Weng is I say, we need to do this. I drop it, and by the time it hits the table, it's done. That's what you're looking for. My favorite word, certain. Not to be absolutely certain is, I think, one of the essential things in rationality. This is Bertrand Russell.
- 8.66** Just some final thoughts here. Practice saying, I don't know. Get good at it. Be comfortable with the willingness to be the stupid fool in front of other people. You

don't need to convey an exuberance of confidence unnecessarily. That's not actually how people—people will respect you deeply if you say, I don't know. Because they know it makes you human. It makes you real-world, right? And they know they can give you information. So it opens the door to a discussion, is what it does.

- 8.67** It's okay to make mistakes, and you want to absolutely forgive mistakes, starting with yourself. That's really what all of this is about, is that if you make mistakes, you need to forgive yourself because that's the way of correcting it. And you want to propagate that same attitude throughout the organization. But it's not okay to hide mistakes. When you hide them and you run from them, that's where things go badly wrong. They compound, they get much worse, and failure iterates to more and more failure. So you want to recognize the mistake early.
- 8.68** Certainty, again. It's the smoke signal of looming failure. It's the thing that you want to be most cautious of at the end of the day, is absolute confidence because it doesn't allow you to reconsider. With that, I'm going to invite Weng up to see if there are any last questions.
- 8.69** What would you advise when setting goals so you don't overlook things because you're so focused on that goal? I guess I'll give an example. In the 1960s or the 1970s, I think it was Ford or something. They set a goal of building a car that was under 2,000 pounds for under \$2,000 or something. In a certain time frame, they did reach their goal. But they overlooked something in the gas tank or something where the gas tank started exploding and all of this stuff.
- 8.70** Yeah, the Pinto, I think. I think the Pinto, yeah.
- 8.71** Then they became the most recalled cars in that time. I guess that is my question.
- 8.72** It's hard.
- 8.73** Do you think it's part of the process?
- 8.74** I think that stubborn focus on the certainty that this is the goal no matter what without allowing for—I'm sure there was probably some engineer along the way who said, this is potentially going to be a problem. You should pay attention to this. And whoever was in charge didn't listen to it. So it's somewhere in that 80-90% rule. You want to be focused and driven towards the goal, but take a breath and reconsider, pretty consistently, even get in the habit of it. Am I still on the right path here? What have I seen here recently that's going to potentially screw this up?
- 8.75** When you look back at your life and you think about what the potential was or who you would possibly become, did you have a clear image of what you wanted to do, or was it blurry? And if it was blurry, what did you use as reassurance to keep going?
- 8.76** That is an excellent question. I think for all of us, there are a few important decisions we need to make. Then we need to look back and reevaluate if we made the right decision or I'm on the right path. To me, I started as a medical doctor. I thought that

will be the only thing I will do for my entire career. Because at the time, I thought becoming a medical doctor, I could make the most impact, and it's the most fulfilling profession. Until I became one, and I realized if there's no treatment available, when the patient comes in, the only thing you could say is, I'm sorry, there's nothing I can do.

- 8.77** Almost overnight, that changed my idea of what is my goal or center of purpose. Because it's absolutely not acceptable to tell patients, I'm sorry, there's nothing I could do. That's why I decided to take the next level of education, try to develop tools for the doctors, make medicine, rather than just employ what's available on the shelf. So this is just one example to show you that, over the years, how this goal can change and what I feel like my life calling can also change because of this.
- 8.78** I would say, for me, I'm a more creative, open person. I like ideas and exploring different things. At a young age, I was more inclined to consider a very wide funnel of possibilities. And I think that's a danger for any creative person. We tend to struggle to concretize a particular identity. The thing that I would say that is most important is, don't try to pick the perfect path when you're at that point. Just pick a path and start going down it. It doesn't have to be perfect.
- 8.79** And that almost any path of learning and understanding people and all of these types of things we've been talking about will help you navigate the world better and better, and it will expose you to things that ultimately may be a better fit. So pick a path is more important than perfect path, especially at a young age.
- 8.80** There's a contrast to an earlier anecdote that you described, where there was what you described as, and this is for Dr. Weng, she described as an easy leader who was secretly a tyrant or the sales leader from the story who had a good reputation but had this secret identity. And so I have found that that style of leader is quite available, unfortunately, in today's society. And I don't really know how to distinguish between that style of leadership and an actual good leader because that person may be very effective at saying the right things. That's a good question.
- 8.81** How do you determine that?
- 8.82** How would you differentiate?
- 8.83** I think we'll be looking at what people say and what they do. It's fascinating because, for people who are very people-oriented and really likable, it's always the case I find they're dishonest. So one example, I had a CEO who spent nine years overseas in Asia. So he told me that he's going to retire in Hong Kong, knowing that I'm Chinese from Beijing. One day, very consistent over many years.
- 8.84** So one day, the Chief Financial Officer and the head of clinical and I have a conversation. I said the CEO will retire in Hong Kong. The Chief Financial said, no, he's going to retire in Miami. Then the other person said, no, he's going to retire in Guadalajara, in Mexico. Three of us, we all heard one version of the story. We're very,

very convinced our story is right. All of a sudden, I realized we should not argue. We were told a different story because of our ethnic background. You know the leader you can trust or not. Then I start to put everything he ever said to me in question because he's someone I really like, very enjoyable, very pleasant. He's a people person. What he said that's actually true?

- 8.85** I'm retiring in Hong Kong [laughter]. I think that one of the best ways to tell a really simple way is what happens when they're highly stressed, when something goes wrong. You can see patterns of behavior that they start blaming other people. They come apart at the seams, and it becomes pretty obvious. You never really know who someone is until they face real challenges. And then when they face challenges, they will tell you really, really explicitly. You'll find out real quickly.
- 8.86** You guys have been fantastic. Thank you for all the great questions. It was wonderful.
- 8.87** Yeah, thank you.
- 8.88** It made it exciting for us to be able to have the opportunity to have a genuine dialogue. It changes the dynamic completely. So thank you very much for that. And we've enjoyed it. It's been wonderful.
- 8.89** Thank you.